

NEURAL BELLMAN OPERATORS: PRESERVATION SUPPLEMENT TO REVISION R11

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This supplement preserves the complete preceding R10 paper and its entire preservation supplement without deleting equations, proofs, numerical tables, or unsuccessful calculations. The current manuscript is ECTA_R11. Earlier precision and comparative-performance statements are dated results, not new claims. The R11 revision adds a frozen method-specific tuning study, independent holdout evidence, exact continuous reference calculations, implemented-output error identities, and executed absolute-accuracy frontiers. All historical source files remain unchanged.

READING MAP

The R11 manuscript contains the original economy, all R10 main results and proofs, the added method-specific tuning study, the full absolute-accuracy calculations, and their verification details. A separate point-by-point response identifies the changes against every finding of the latest R9 referee report. The immutable source, result, build, and review identities are recorded in the revision directory.

The following pages reproduce the complete R10 manuscript. They are followed by the complete R10 supplement, which already preserves R9, R8, and earlier development material. In particular, recursive utility, sophisticated temporal selves, games, stochastic traces, action safeguards, unsuccessful earlier certificates, and all retained external experiments remain available. These earlier documents must be read with their original version labels. R11 does not turn an earlier failed tolerance or an exploratory numerical result into a current certificate merely by reproducing it.

PART I: COMPLETE R10 MANUSCRIPT

We study neural policy evaluation, improvement, and independent verification in a continuous economy with costly preference adjustment. An affine-preference market-deflator dual, combined with validated policy evaluation, certifies regret below .01 at the central state for three adjustment-cost coefficients under the original utility, continuous controls, and first-exit contract. The dual retains adapted adjustment through an explicit conditional-variance allowance. At cost coefficient two, access to adjustment raises optimal welfare by at least .04096859; increasing that coefficient to eight lowers optimal welfare by at least .02318257. Each effect exceeds the width of its corresponding certified interval. We provide an executed absolute-accuracy verification frontier and a quantitative allowance for implemented control errors. A structured action oracle and an absolute lower reference accompany the complete frozen external comparison. Proofs and computations distinguish policy evaluation, optimality, sampling uncertainty, and the economic inequalities actually established.

KEYWORDS: Stochastic control, Neural policy iteration, Endogenous preferences, Recursive utility, Verification, Computational economics.

1. INTRODUCTION

numerical decision rule is useful for economic analysis only when its error is small relative to the comparison being made. This requirement is distinct from a small training loss. It is also distinct from outperforming another learning algorithm. When a decision maker can change a reference state, an apparent welfare gain may reflect either the adjustment technology or an accurate evaluation of the policy without that technology. When termination is an economic contract, an inaccurate treatment of first exit can dominate both effects.

This paper studies Neural Bellman Operators (NBO): separated policy evaluation and feasible policy improvement, implemented by neural networks and assessed by independent verification. A motivating economy combines consumption, portfolio choice, stochastic risk aversion, costly preference adjustment. The analysis retains its original running utility, continuous controls, correlated Brownian shocks, and first-exit liquidation covenant. Neither reflecting the error at a training boundary nor altering the running payoff is used to establish the economic results below.

The main numerical result is a certified .01-optimal policy in the original continuous economy at the central initial state, for adjustment-cost coefficients .5, 2, and 8. The respective regret upper bounds are .00987440, .00971727, and .00996896. Each lower endpoint evaluates a feasible policy; each upper endpoint covers all original adaptive controls. The computation does not replace the first-exit contract, restrict the original optimum to time controls, or use a finite action grid to claim a continuous guarantee.

The economic consequences are quantitative. Raising the adjustment-cost coefficient from 2 to 8 lowers optimal welfare by an amount in $[-.02318257, .04286880]$. The width of this interval is smaller than its lower endpoint. Access to adjustment at coefficient two raises optimal welfare by an amount in $[\text{.04096859}, \text{.05973218}]$ relative to the *optimal* no-adjustment economy, rather than relative to a selected no-adjustment rule. Discounted expenditures of the deployed policies are also enclosed independently. The optimal expenditure itself is bounded by valid bounds, not identified by relabeling a learned policy as an optimizer.

polished within the feasible output class and evaluated by Gaussian polynomial moments exact first exit. An affine-preference market-deflator witness then bounds the flexible n. Portfolio terms cancel for every adapted original portfolio rule. A supporting source ty retains flexible adjustment, and a conditional-value preservation pays explicitly for m marginal value. Outward weighted Taylor boxes enclose the remaining one-Gaussian s. Explicit continuation potentials cover artificial verification faces without altering the c domain.

constructions improve on the preceding wealth-independent witness, whose $k = 2$ regret as .10634. That witness and its complete proofs remain available; the new certificate is labeling of its evaluation precision as optimality precision. The restricted economy's market-deflator bound is retained, together with its independently evaluated feasible t gives an initial restricted-value interval narrower than .00905. Combining the two es' bounds establishes the welfare inequalities above.

ond contribution addresses the inner optimization needed for verification. In a cou- nonconvex control family, all nonconvex coupling depends on the sum of the actions. imensional branch-and-bound calculation, together with strongly convex scalar dual lems, certifies the full continuous action box without a tensor action grid. Executed tes reach gaps below 10^{-5} through 128 action dimensions. At the eight- and sixteen- onal learned-critic test states, interval differentiation of the actual network weights also s the error in the floating-point critic gradient. This is an implemented bridge between parameters and an action certificate, not an assumption that an optimizer's stationary globally greedy.

ame nonconvex family is given an absolute accuracy axis. A pointwise running-cost on and removal of control bounds admit a Cole–Hopf representation that retains the nonquadratic terminal payoff. Orthogonal Gaussian coordinates integrate analytically, a positive scalar integral with an outward enclosure. The resulting number is a lower or the unchanged original minimization problem. Relative performance is evaluated ly in a frozen holdout experiment: three dimensions, two training budgets, twelve seed r panel, equal learning-rate search budgets, warmed initialization, randomized method d a third, independently specified feedback rule. The pinned author implementation of artNet is not modified.

ical verification, monotone approximation, and primal–dual numerical methods supply nt foundations. The stopped Itô argument is not claimed as a new general verification e. Nor is alternating neural evaluation and improvement claimed as a new algorithm- a. The contribution is the constructive combination of original-contract verification, dent economic upper and lower bounds, computable neural-jet allowances, and a non- ative global action audit, with each error quantity matched to an executed calculation. 8 compares these objects theorem by theorem with neighboring methods. Recursive util- isticated temporal selves, dynamic games, stochastic-trace identities, counterexamples, complete preceding manuscripts are preserved in the supplement rather than being used ge the empirical claims of the present application.

2. THE ECONOMIC ENVIRONMENT

state is $s = (u, x)$, where u denotes relative risk aversion and x denotes wealth. An ble action is $a = (c, \theta, p)$ in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

backlogs and time controls actually evaluated below are bounded and satisfy this requirement. More termination,

$$du_t = \theta_t dt + .05 dZ_t^u, \quad (1)$$

$$dx_t = [(.02 + .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

horizon is $T = 1$, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times (0, 2)$. Write $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u - 2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1 - t). \quad (4)$$

policy payoff and optimal value are

$$J_k^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

no-adjustment value V_R restricts θ to zero and retains *all* admissible consumption and portfolio choices. It is independent of k . The cost experiment holds the controls, transition law, stopping rule, settlement, utility normalization, and consumption units fixed. A state-dependent transformation of utility would generally define another economy.

The early settlement fee is a liquidation covenant. There is no continuation consumption or wealth injection following exit. At the lower wealth boundary the largest admissible wealth is negative. At the upper boundary, $p = 0$ eliminates the wealth diffusion and permits an upward drift. Thus neither a silently viable rectangle nor a theorem requiring uniform controlled-boundary regularity can be substituted for this contract. The verification below requires only local regularities for smooth test functions on the trace at which the process is actually stopped.

For a smooth function v , put

$$\mathcal{L}^a v = \theta v_u + [(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$R_a v = v_t + \mathcal{L}^a v + \ell_k - \rho v. \quad (7)$$

The mixed derivative is retained. Its sign is not inferred solely from the negative shock correlation.

PROPOSITION 1—Complete selection at a fixed jet: *For the generator in (7), a global maximizing action can be selected componentwise. Consumption is $\text{proj}_{[.05,.8]}(v_x^{-1/u})$ when $h_1 > 0$ and is $.8$ otherwise. Adjustment is $\text{proj}_{[-.2,.2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ and $h_2 = .04x^2v_{xx}$. If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5,.8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint maximizing $h_1p + h_2p^2/2$ is selected, with a fixed tie rule.*

The statement includes convex and zero-curvature portfolio objectives and nonpositive marginal wealth value. It applies to the continuous generator, not to the interpolated, killed-Euler approximation. It says nothing about global optimization over shared neural parameters.

fixed policy, exact evaluation solves $R_\pi v = 0$ with the stopping contract. Exact improvement maximizes the Hamiltonian of that fixed evaluation. Denote these operators by $\mathcal{E}$ and $\mathcal{I}$; policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. A neural implementation approximates both operators, independent evaluator measures the payoff of the policy that will actually be used.

In raw NBO implementation, critic updates hold the actor fixed. Actor updates hold parameters fixed, but retain differentiation through the controlled successor state. A parameters, when present in a game, are also frozen during unilateral improvement. This prevents an actor loss from lowering itself by changing the object being evaluated. It makes the joint network optimization convex. The ideal policy-iteration compactness and bounding arguments are retained in the supplement; no such argument is invoked to certify the learned Adam trajectory.

Inherited finite safeguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive submodular P_n^a , and $\delta = e^{-\rho\Delta}$, define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and candidate improvement gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error is at most b , backward comparison yields

$$\|V_{h,n}^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

The ρ -dependent envelope and threshold-correction proof are reproduced in the supplement. The audit can reduce the fraction of decisions changed while leaving the cost of a complete audit essentially unchanged. In particular, the retained 125-action tensor audit is not an enumerative certifier of Section 6. Its floating-point envelope is not relabeled as an continuous-time bound.

3.1. A quantitative bridge from a fitted network

$\rho(h) = (1 - e^{-\rho h})/\rho$, with $C_0(h) = h$.

ASSUMPTION 1—Admissible witnesses: *Witnesses are bounded, continuous on the stopped trace and $C^{1,2}$ in the interior of each time slab. Localized Itô identities pass to the stopping time. Uniform integrability. At slab interfaces an upper witness has nonpositive forward jumps and a lower witness has nonnegative forward jumps. The implemented policy has a well-defined state equation.*

LEMMA 1—Stopped one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and b_L on the stopped trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances are negative. Then*

$$J_k^\pi \geq L - b_L - C_\rho(T-t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T-t)\epsilon_U, \quad (9)$$

$$0 \leq V_k - J_k^\pi \leq U - L + b_U + b_L + C_\rho(T-t)(\epsilon_U + \epsilon_L). \quad (10)$$

These inequalities do not require an attained optimizer or a smooth optimal value.

LEMMA 2—Cover-and-oracle certificate for neural output: *Let a continuous neural witness and a feasible neural policy π satisfy Assumption 1, with stopped-trace error at most b . Let*

ulation proves, throughout cell j ,

$$|R_\pi v(z) - R_\pi v(z_j)| \leq \omega_j, \quad |\Gamma(z) - \Gamma(z_j)| \leq \nu_j, \quad \Gamma = \sup_a R_a v - R_\pi v.$$

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pose the representative evaluation intervals imply $|R_\pi v(z_j)| \leq e_j$ and a global action oracle satisfies $\Gamma(z_j) \leq q_j$, including jet, inner-solve, and arithmetic allowances. Set $e = \max_j (e_j + \omega_j)$ and $q = \max_j (q_j + \nu_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

particular, certified derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and $\nu_j = N_j h_j$. Any sequence of learned outputs for which these verified allowances vanish has vanishing regret, irrespective of whether its parameters converge.

This is a quantitative a posteriori connection to network errors, not a claim that an empirical mean-square loss controls a supremum. Interval differentiation or weight-based derivative bounds can supply the cell moduli. They may be conservative, and covering a high-dimensional domain remains costly. Section 6 executes the jet-to-action part for actual learned critics. The economic calculation instead fits an independent polynomial witness; it does not falsely claim to have certified a whole-state neural critic merely because its action audit passed at selected states.

4. A CONSTRUCTIVE CERTIFICATE FOR THE FLEXIBLE ECONOMY

The principal numerical target is a regret bound of .01 for the original problem (5), not for its finite approximation. The certificate below attains this target at $(0, 2, 1.25)$ for each of the three adjustment prices. It uses a feasible policy to bound value from below and a separate, preference-market-deflator construction to bound the optimum from above. The latter consists of arbitrary adapted consumption, portfolio choice, and preference adjustment. Neither a reference lattice nor an optimal-value label enters its construction.

4.1. An affine source that preserves the original opportunities

Let $\phi(u) = -.02(u - 2)^2$, $h = 1 - t$, and let Y satisfy (28), with its initial level $y_0 > 0$ now chosen as a witness parameter. Define

$$S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad (12)$$

$$\beta(u, y) = \partial_u \Psi(u, y), \quad (13)$$

$$m(t) = 2 + \int_0^t \bar{\theta}_s ds, \quad 0 \leq \bar{\theta}_s \leq .2, \quad (14)$$

where Ψ is the original consumption conjugate in (30). The reference drift $\bar{\theta}$ is deterministic. It is a parameter of an upper witness, not a restriction on the decision maker's adjustment policy. Let $\hat{y} = \text{proj}_{[.2, 1.2]} y$ and

$$Q(u) = \phi(m(1)) + \phi'(m(1))(u - m(1)), \quad b_T = \phi'(m(1)).$$

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (15)$$

one-sided inequality on the whole rectangle, not a fitted range. An alternative proof uses derivative identities and interval inequalities with explicit margins; see Appendix A.1. Concavity of S_0 outside this rectangle is not assumed.

$S_0(u, y) = S_0(m(r), \hat{y}) + \beta(m(r), \hat{y})(u - m(r))$. Consider the full-line auxiliary control

$$A_k(t, u, y) = \sup_{|\theta| \leq .2} \mathbb{E}_{t, u, y} \left[\int_t^1 e^{-\rho(r-t)} \{ \hat{S}_r(U_r, Y_r) - k\theta_r^2/2 \} dr + e^{-\rho(1-t)} Q(U_1) \right]. \quad (16)$$

$dZ_r = \theta_r dr + .05 dZ_r^u$, and the auxiliary control remains adapted. The affine source makes initial value of u an explicitly defined conditional expectation, which can be bounded solving another neural control problem. Put

$$Z_u(h, u) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (17)$$

$$Z_y(h, y) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (18)$$

THEOREM 3—Affine-preference dual and a computable initial upper bound: *For $k \geq .5$, $2, 2.4]$, and any deterministic $\bar{\theta} \in [0, .2]$, the original flexible optimum satisfies*

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (19)$$

ate the right side, put $\beta_t = \beta(m(t), \hat{Y}_t)$ and

$$b(s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E} \beta_t dt + e^{-\rho(1-s)} b_T, \quad (20)$$

$$g_k(q, a) = H_k(q) - qa + ka^2/2, \quad (21)$$

$$C = .00375 \int_0^1 t e^{-\rho t} \mathbb{E} \left[\partial_t \beta(m(t), \hat{Y}_t) \right] dt, \quad (22)$$

the l derivative includes the clipping map and $H_k(q) = \max_{|a| \leq .2} (qa - ka^2/2)$. Then

$$A_k(0, 2, y_0) \leq e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \left\{ \mathbb{E} S_0(m(t), \hat{Y}_t) - k\bar{\theta}_t^2/2 \right\} dt + C + \int_0^1 e^{-\rho s} g_k(b(s), \bar{\theta}_s) ds + \frac{.09D^2}{24k}, \quad (23)$$

explicit bound

$$D^2 = \frac{e^{aM + .045a^2}}{16} \{ (M + .09a)^2 + .09 \}, \quad a = 12/11, \quad M = \log y_0. \quad (24)$$

or policy.

The cancellation of the portfolio is the same accounting identity used in the restricted dual: $(x) = (.04x - Yc)dt +$ a local martingale. The new step is to retain flexible adjustment by replacing its source by a supporting affine function. The variance allowance in (23) pays explicitly for the random conditional marginal value of adjustment. Replacing that marginal by its unconditional mean without this allowance would generally produce an invalid lower bound. The covariance term C also matters: it retains the original correlation between preference and market shocks and is nonpositive here. It is evaluated, not silently discarded or replaced with a change of measure.

The source bound $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ imply $A_k - Q \geq -7.925272h$ on the verification angle, by choosing zero adjustment in the auxiliary problem. Hence the factors $8h$ in (18) for continuation at artificial faces. On actual wealth exits, the uncorrected affine witness already dominates the liquidation settlement. The global fallback $V_k \leq G$ covers continuation on an artificial exit. These facts connect (23) to the original stopped economy rather than to a surrogate full-line economy.

4.2. Policy generation, verified refinement, and the stopping rule

The lower-bound policies start from the retained one-input, twelve-unit neural time-control pairs. A constrained L-BFGS-B step polishes their sixteen consumption and adjustment outputs against the original payoff and the same exact budget projection. This is an explicit enlargement of a fixed network’s image to the feasible time-control outputs; it is not described as additional Adam iterations or as a theorem that the neural optimizer converges. Consumption and adjustment constants are rounded inward relative to the *exact decimal* action bounds, and feasibility is checked using their exact binary rational values. The finite-node training objective is a loss mechanism, not an asserted untruncated Gaussian expectation of singular CRRA utility. An independent polynomial-moment evaluator in Appendix G then encloses each deployed continuous-time policy under exact first preference exit. The full network histories, original outputs, polished outputs, and output-optimization records remain separate.

The dual reference is fitted with Gaussian quadrature, but its fitted objective is never used for a bound. Verification uses $t = v^2$, subdivides each of the sixteen time slabs, and covers the standard Gaussian coordinate on $[-6, 6]$. Constant and linear Taylor terms are integrated with interval-enclosed weighted moments. Hessian ranges enclose the entire remainder, including cases crossing the consumption constraint. Gaussian tails, the terminal tangent, the conditional variance allowance, localization, and outward arithmetic are separately included. The small dimensional interval tests establishing (15) precede every certificate execution.

The stopping rule is simple. Given a policy lower bound L and an upper bound U from Theorem 3, return the policy with tolerance ε only when $U - L \leq \varepsilon$. Otherwise refine the quadrature or return the explicit unachieved target at the declared work limit. Soundness does not require optimizer convergence. Nor does soundness imply that an arbitrary tolerance can be achieved with a fixed affine relaxation. For the three declared .01 instances, the finite executions now supply the termination certificate.

The finest upper bounds are approximately -1.26279687 , -1.28530724 , and -1.31820708 . Combined with the feasible-policy lower endpoints, they give regret upper bounds .00987440, .0071727, and .00996896 after outward display rounding. The central target is attained without any candidate-action grid, an Euler-error assumption, or an assumption that the policy is optimal within its neural class. The preceding wealth-independent witness and its larger gaps

k	Feasible policy payoff	Optimal-value upper	Regret upper
0.5	$[-1.27267126, -1.27265474]$	-1.26279686	0.00987440
2	$[-1.29502451, -1.29501124]$	-1.28530723	0.00971727
8	$[-1.32817604, -1.32816919]$	-1.31820708	0.00996896

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regret bounds are below .01. Each upper bound covers the full original continuous control set and stopping. Each lower bound evaluates the specified feasible time-control policy. Endpoints are rounded outward; gaps unrounded endpoints. These are central-state certificates, not uniform claims over every initial condition.

TABLE II

VERIFIED WORK NEEDED TO CROSS THE ABSOLUTE-ERROR TARGET

k	Source boxes	Regret upper	Verification seconds	.01 met
0.5	4,096	0.01027899	5.46	No
0.5	16,384	0.00995838	11.73	Yes
0.5	65,536	0.00987440	27.54	Yes
2	4,096	0.01013162	5.25	No
2	16,384	0.00980424	11.30	Yes
2	65,536	0.00971727	27.06	Yes
8	4,096	0.01037189	5.29	No
8	16,384	0.01005494	11.48	No
8	65,536	0.00996896	26.79	Yes

policy is independently evaluated once at the finest stated precision. Time includes that evaluation, dual on, and cumulative dual verification through each row, on one CPU thread. It excludes inherited network witness fitting, imports, and setup. This is an absolute-accuracy *verification* frontier, not an all-in training comparison or an external matched-accuracy superiority claim. Raw stage clocks are retained.

TABLE III

SEPARATE UPPER-BOUND ALLOWANCES AT $k = 2$

Boxes	Source remainder	Control gap	Variance allowance	Localization
4,096	0.000339131	0.000063959	0.000081080	0.000000247
16,384	0.000082675	0.000041416	0.000081080	0.000000247
65,536	0.000020398	0.000032100	0.000081080	0.000000247

the remainder is the Taylor allowance on the covered Gaussian region. Analytic Gaussian tails, interval weight minimal evaluation, and the covariance interval are included in the full records. The variance allowance does use merely by subdividing quadrature boxes. No row is obtained by treating a mesh difference as a rigorous error estimate.

ned in Appendix C, where their loss of wealth information can be compared with the instruction.

4.3. A quantitative bridge from implemented controls to the certificate

ound applies to the stored deployment constants themselves. A further perturbation antifies its robustness to control evaluation error, rather than inferring robustness from a training loss.

policies with $p = 0$, $c, \tilde{c} \in [.7, .8]$, $\theta, \tilde{\theta} \in [0, .2]$, and strictly feasible budgets as in (55). Let

$$10 \quad \delta_c = \int_0^1 |c_t - \tilde{c}_t| dt, \quad \delta_\theta = \int_0^1 |\theta_t - \tilde{\theta}_t| dt.$$

or original stopped payoffs satisfy

$$|J_k(c, \theta) - J_k(\tilde{c}, \tilde{\theta})| \leq (3 + .2e^{.02})\delta_c + (25 + .2k + .032)\delta_\theta + 8(14.1 + .02k)e^{-72}. \quad (25)$$

sequently, a validated output error bounded by the right side is an additive allowance the returned policy's regret certificate. Feasibility must still be checked separately; small unconstrained control error alone does not rule out early wealth exit.

for the stored policies the output allowance is zero: their binary64 constants are treated as exact by the evaluator. Proposition 2 instead gives an explicit interface for a budget-preserving implementation with perturbed outputs. Together with the neural-jet bound of Theorem 2, it reflects observable implementation errors to certified quantities without attributing an unproved convergence rate to Adam, RMSprop, or L-BFGS-B.

5. CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

for each feasible policy write $J_k(\pi) = A(\pi) - kB(\pi)$, where

$$B(\pi) = \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 dr, \quad 0 \leq B(\pi) \leq .02C_\rho(T - t).$$

policy set is common to every k .

THEOREM 4—Accuracy-calibrated cost comparisons: *The optimal value is convex and increasing in k . If optimizers exist, their adjustment budgets are nonincreasing across distinct prices. More generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (26)$$

$(k) \in [\underline{V}(k), \overline{V}(k)]$, the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive jet range intersected with

$$\left[\frac{\underline{V}(k) - \overline{V}(k_+)}{k_+ - k}, \frac{\overline{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \quad (27)$$

the theorem concerns discounted adjustment expenditures, not a pointwise ordering of controls. A re-solved V_u/k is not a comparative-static proof. The new full-value enclosures in Table I now resolve strict welfare losses across the cost grid. At $k = 2$, (27) also gives the full-budget enclosure $[\text{.00386376}, \text{.01960529}]$, conditional on existence of an optimizer. Its primitive lower endpoint is informative, while its upper endpoint is still the primitive bound. The newly enclosed expenditures in Table V describe the specified learned-and-polished policies, and an identified optimal budget.

ter comparison becomes possible by exploiting the restricted economy's structure.
e the auxiliary process

$$dY_t = .02Y_t dt - .3Y_t dZ_t, \quad \text{NBO: R11 PRESERVATION SUPPLEMENT} \quad (28)$$

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st-process constructed from the original shocks. It does not restrict portfolio information,
wealth, or add a traded asset. For

$$W_0(u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2,$$

ms in the generator of Yx cancel exactly. With $\theta = 0$, define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (29)$$

$$S(u, y) = \Psi(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (30)$$

1.5, 2.6], $y \in [.2, 12]$, one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$
ing its two arguments at these verification faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u,y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (31)$$

$U_r = .05 dZ_r^u$ and Y follows (28). Thus $-8h \leq F_R \leq 0$.

REM 5—A restricted-economy upper bound: *Let Z_R be the sum of the following
ative localization potentials:*

$$Z_{R,u} = 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (32)$$

$$Z_{R,y} = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (33)$$

$$V_R(0, 2, 1.25) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (34)$$

er bound covers all adaptive original consumption and portfolio controls with $\theta = 0$.
ently, for any feasible flexible policy with certified payoff lower bound $\underline{J}_k$ and any
ound $\bar{V}_R$ from (34),

$$V_k(0, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (35)$$

expectation in (31) has an explicit two-Gaussian representation. We transform time by
evaluate constant and linear Taylor terms with interval-enclosed weighted moments,
and the Hessian remainder on each box. Exponential Taylor remainders, Gaussian tails,
alization are included. This produces the executed refinement in Table IV; it does not
o a mesh difference as an error estimate. The final restricted-optimum upper bound
3599309 after outward display rounding. The evaluated restricted policy has payoff
4503942, -1.34503585], so the restricted problem itself has an initial optimal-value
e narrower than .00905.

	Weighted boxes	Dual witness-value enclosure	Width	Seconds
12	16384	$[-1.33876765, -1.33309541]$	0.00567	2.67
	131072	$[-1.33676857, -1.33544680]$	0.00132	15.13
	1048576	$[-1.33631272, -1.33599309]$	0.00032	110.24

se intervals enclose the *dual witness value*, not the restricted optimum. Only their upper endpoints are used as upper bounds for that optimum. A separately evaluated restricted policy gives the primal lower bound. Original stopping and all adaptive consumption and portfolio opportunities are retained.

TABLE V

CERTIFIED CURRENT-POLICY EXPENDITURES AND OPTIMAL VALUE OF ACCESS

k	Policy adjustment budget	Optimal access-value interval	Regret target
0.5	[0.01757696, 0.01757697]	[0.06332184, 0.08224255]	< .01
2	[0.01241866, 0.01241867]	[0.04096859, 0.05973218]	< .01
8	[0.00192681, 0.00192682]	[0.00781706, 0.02683234]	< .01

expenditure column evaluates the displayed learned policy. The access columns bound the difference between the *o* optimal values. The lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper endpoint pairs the flexible upper with a feasible restricted policy. All endpoints are rounded outward from precision records. The cost-two access interval has width below its entire positive gain; at cost eight, the interval still resolves the sign but not a comparably precise magnitude.

TABLE VI

ACCURACY-CALIBRATED OPTIMAL WELFARE LOSSES FROM HIGHER ADJUSTMENT COSTS

Cost change	Optimal welfare loss	Interval width	Width below loss
0.5 $\rightarrow$ 2	[0.01263598, 0.03222764]	0.01959166	Sign resolved
2 $\rightarrow$ 8	[0.02318257, 0.04286880]	0.01968623	Yes
0.5 $\rightarrow$ 8	[0.04553582, 0.06537918]	0.01984335	Yes

interval encloses $V_{k_1} - V_{k_2}$ in the unchanged original economy. Bounds use independent value endpoints, not fitted derivatives in k . The last column compares the width with the lower endpoint of the effect interval.

the welfare lower bounds remain strictly positive at all three prices. At $k = 2$ the optimal ss gain lies in $[\text{.04096859}, \text{.05973218}]$; its width is smaller than the lower endpoint. Table VI arately compares the flexible optima at different adjustment prices. The $2 \rightarrow 8$ and $.5 \rightarrow 8$ cts exceed their entire interval widths. The $.5 \rightarrow 2$ comparison resolves the sign but has a er relative uncertainty. No comparison assumes that a displayed policy is an optimizer or pprets the value-ordering theorem as a pointwise ordering of adjustment controls.

6. GLOBAL ACTION CERTIFICATION WITHOUT A TENSOR GRID

onsider the following minimization problem, which is also used for the external comparison:

$$dX_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (36)$$

$$f_d(a) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (37)$$

$$q_d(x) = \frac{1}{d} \sum_{i=1}^d (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \quad (38)$$

$$V_d^* = \inf_a \mathbb{E} \left[\int_0^1 f_d(a_t) dt + q_d(X_1) \right] \quad \text{PRESERVATION SUPPLEMENT}$$

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ective has genuinely coupled nonconvex actions. No analytic optimal-action or optimal-models are supplied to training. At a fixed critic gradient p , the action-dependent Hamiltonian

$$H_p(a) = \sum_i f_i(a_i) + .3 \sin z, \quad f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

is strongly convex with modulus at least $m = 1.8/d$.

LEM 6—Scalar global partition and convex dual bounds: *For any multiplier λ and d , put*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

On the interval $z \in [z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$D(\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \quad (39)$$

Each inner convex minimum can itself be bounded by $q(s_0) - e(s_0)^2/(2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible direction, or the full derivative magnitude at an interior point. Outward versions of these bounds, followed by scalar subdivision and pruning, give a global gap on $[-1, 1]^d$. The stored gradient $\hat{p}$ differs from the true neural gradient by componentwise bounds δ_i , the action’s true-gradient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

Interval-point root finding only proposes multipliers and actions. It is not trusted to solve subproblems exactly. Even an inaccurate proposal remains valid after the strong-tolerance allowance. This structure removes tensor enumeration; it does not establish a non-free complexity theorem or solve a generic nonseparable nonconvex Hamiltonian. The 28-dimensional 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex elements. It replaces neither raw actor training nor the preceding finite tensor safeguard in practice. At the learned eight- and sixteen-dimensional jets, interval forward differentiation of stored critic weights bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and 10^{-17} , respectively. These allowances are propagated into the action gap. Theorem 2 provides the additional state-cover and residual quantities that would be needed for a whole-system guarantee.

7. AN ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

7.1. A rigorous lower bound for the unchanged nonconvex family

Let $\sigma^2 = 1 + .1 \sin^2(1)$, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1) a^2$ for $|a| \leq 1$, bounding the running cost by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation. The final terminal function q_d is retained.

Action dim.	Tolerance	Verified gap	Scalar boxes	Convex subproblems	Seconds
8	1e-03	0.000258	33	264	0.299
16	1e-05	6.72e-06	47	376	0.493
16	1e-03	0.000931	37	592	0.476
16	1e-05	4.42e-06	57	912	0.751
32	1e-03	0.00068	85	2720	1.498
32	1e-05	3.52e-06	103	3296	1.991
64	1e-03	0.000647	163	10432	6.041
64	1e-05	3.06e-06	185	11840	5.955
128	1e-03	0.000948	317	40576	19.229
128	1e-05	4.68e-06	339	43392	20.651

rozen jet is used per dimension, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at recorded states; displayed gaps include interval neural-derivative allowances. At larger dimensions they are independently generated jets for the same Hamiltonian family. The experiment measures certificate work for these instances, not an average-case scaling law or a whole-state neural PDE residual.

TABLE VIII

ABSOLUTE LOWER BOUNDS FOR THE ORIGINAL NONCONVEX CONTROL PROBLEM

d	Quadrature cells	Relaxed-value enclosure	Original-value lower	Seconds
8	1024	$[-0.08599082, -0.08384626]$	-0.08599082	0.260
8	4096	$[-0.08518701, -0.08465086]$	-0.08518701	1.081
8	16384	$[-0.08498598, -0.08485194]$	-0.08498598	4.506
16	1024	$[-0.20520651, -0.20360335]$	-0.20520651	0.271
16	4096	$[-0.20460665, -0.20420585]$	-0.20460665	1.067
16	16384	$[-0.20445644, -0.20435623]$	-0.20445644	4.212
32	1024	$[-0.28757809, -0.28684340]$	-0.28757809	0.245
32	4096	$[-0.28730394, -0.28712026]$	-0.28730394	0.988
32	16384	$[-0.28723515, -0.28718923]$	-0.28723515	4.172

v encloses the relaxed value. Its lower endpoint is also a lower bound for the original bounded-control value. The width shown by successive enclosures is quadrature uncertainty; it does not include or estimate the structural relaxation gap.

PROPOSITION 3—Scalar-integral absolute lower bound: *Let Z be normal with mean $\tilde{\mu}/(r_0 + 4)$ and variance $\sigma^2 r_0/(r_0 + 4)$. Then*

$$V_d^* \geq \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (40)$$

outward enclosure of the positive expectation yields a rigorous numerical lower bound for original objective (38).

the relaxation is an upper/lower-bound device, not a new training problem with a known manufactured value. The terminal cosine, which couples all state coordinates, remains in the far expectation. Its tails are bounded analytically and its interior is integrated by interval angles.

d	Budget	NBO cost	SOC cost	LQ cost	Median difference	Adjusted p
8	10	0.5812	1.0842	0.5121	-0.4783	0.001465
8	30	0.5785	1.0557	0.5121	-0.4783	0.001465
16	10	0.4663	0.7893	0.1740	-0.1672	1
16	30	0.4630	0.8456	0.1740	-0.2878	0.438
32	10	0.3813	1.2011	0.0854	-0.8691	0.001465
32	30	0.3799	1.0218	0.0854	-0.6962	0.001465

realized cost is better. The median difference is NBO minus SOC-MartNet. The last column is a one-sided t -test p -value adjusted across the six predeclared dimension–budget panels. The sign test concerns the joint training-and-audit pipeline under independent seed draws, not an outward numerical error guarantee.

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TABLE X

ABSOLUTE REFERENCE EXCESS AND ACTUAL TRAINING CLOCKS

d	Budget	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
8	10	-0.08499	0.6662	1.1692	10.002	10.006
8	30	-0.08499	0.6635	1.1407	30.004	30.008
16	10	-0.20446	0.6707	0.9938	10.002	10.005
16	30	-0.20446	0.6674	1.0500	30.004	30.006
32	10	-0.28724	0.6686	1.4883	10.003	10.007
32	30	-0.28724	0.6672	1.3090	30.003	30.007

is the estimated mean policy cost minus the rigorous original-value lower bound. It provides an absolute axis but is not itself a deterministic regret enclosure: sampling error and floating-point rollout evaluation are not certified. Training clocks include completed-update overshoot and exclude separately recorded warmup and setup.

7.2. Holdout design and exact deployment interpretation

comparison retains the original actor and critic architectures: two tanh hidden layers of 128 units each, a bounded actor output, and the same terminal-value embedding. NBO uses a separate evaluation and improvement with Adam, three critic steps per actor step, 256 states, and four successors. SOC-MartNet uses its pinned author solver, RMSprop, a shared output test network, and the inherited $J = 2$, $K = 1$ multiplier setup. These different implementation systems are disclosed rather than described as identical implementations.

The committed protocol freezes dimensions 8, 16, 32 and training budgets of 10 and 30. At each dimension, both methods receive the same six validation trials: two seeds and three learning-rate multipliers. Selection minimizes mean independent validation cost, with a random tie rule. Twelve different holdout seeds, 720–731, are then evaluated at each dimension–budget pair. Within each pair, actor and critic starting weights are identical and their gradients are checked. Framework warmup precedes both clocks, and method order is seed-randomized. Training-only and setup-plus-training clocks are retained. The baseline $a_i(t, x) = \frac{1}{1 + 4(1 - t)}[-2(x_i - .5)]$ is specified independently from the associated quadratic objective and is evaluated on the unchanged nonquadratic objective.

The trained network is deployed as a sample-and-hold control on 128 dates and evaluated on 96 independent, paired Gaussian paths. In (38), drift under a held action and volatility are constant on a slab; the Gaussian transition and running-cost integral are therefore exact in expectation for *that deployed continuous-time policy*. A 64-date evaluation uses aggregated increments and changes the deployment policy. Its difference from the 128-date result is bounded by an Euler-error bound. Floating-point network evaluation and Monte Carlo uncertainty are distinct from the outward deterministic lower bound in Table VIII.

their interpretation is not uniform. In dimensions 8 and 32, NBO has lower paired cost than all twelve seeds at both budgets; each multiplicity-adjusted sign-test value is .001465. In dimension 16 the adjusted values are 1 and .438, so these panels do not establish a robust seed-level ordering under the declared test. Tripling the budget gives only small improvements in NBO mean cost, from .5812 to .5785, .4663 to .4630, and .3813 to .3799 respectively. Most importantly, the independently specified quadratic-feedback baseline has lower mean cost than all learned methods in every panel. Its mean costs are .5121, .1740, and .08544 in dimensions 8, 16, and 32. These retained results support the stated NBO–SOC–MartNet comparison at the three budget points, not a claim that NBO is the best available control rule or that additional training would already have reached an accurate optimum.

The raw records retain every seed difference, the median and interquartile range, approximate Student intervals, costs for all three rules, path standard errors, training curves, optimizer clocks, and checkpoint counts. Approximate Student intervals are descriptive; the exact sign calculation avoids assuming Gaussian seed-level differences. The source protocol predates holdout execution, while the benchmark family and adapters themselves were developed in earlier revisions. Holdout results do not turn one problem family into evidence of general superiority across economic environments. Two budget points are not a matched-accuracy frontier, and a finite absolute lower bound need not be tight enough to rank methods at a small prescribed regret.

8. RELATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

The strongest claims should be compared with methods that establish bounds, not only with methods that train networks. Theorem 1 is an application of classical stopped verification to both upper and lower witnesses. Its role here is to handle the actual liquidation trace without supposing a smooth or boundary-regular optimal value. Theorem 2 supplies a computation-coverage guarantee, derivative, and oracle allowance. Neither theorem claims that classical verification is absent from the literature.

Monotone, stable, consistent approximations with comparison provide convergence to viscosity solutions in the framework of Barles and Souganidis (1991). Controlled Markov-chain approximation is developed systematically by Kushner and Dupuis (2001); adaptive sparse-grid methods address high-dimensional dynamic models in Brumm and Scheidegger (2017). Such results concern specified approximation schemes and their hypotheses. The finite telescoping regret in the preserved manuscript names the operator discrepancies needed for a continuous limit; an observed mesh difference does not supply them. The original-payoff certificates instead evaluate continuous generators or exact Gaussian moments directly, with separately bounded analytic and arithmetic remainders. They do not assert a convergence rate for the truncated killed-Euler scheme.

Primal–dual numerical control methods also pair feasible decisions with bounds. Brown, Smith, and Sun (2010) relax nonanticipativity with penalties and establish weak and strong optimality for stochastic dynamic programs. Brown and Smith (2011) use related bounds in portfolio optimization with transaction costs. Our restricted and flexible constructions are martingale-deflator witnesses with explicit stopped localization; they are not claims that policy-optimality or upper-bound analysis is new. Theorem 3 adds a verified affine-preference source and a conditional-coefficient variance allowance. Its distinction from an information-relaxation bound is that an uncomputed penalty is that every source integral, variance constant, and face allowance are evaluated here. Lasserre, Henrion, Prieur, and Trélat (2008) develop occupation-measure and moment relaxations for polynomial nonlinear optimal control under their stated assumptions. They do not invoke their deterministic convergence theorem for the present logarithmic, stochastic,

trace inequalities are verified here instead.

Proposition 3, Proposition 4, and Theorem 5 identify the current and preceding model-specific conditions behind the economic bounds. The affine-preference theorem retains costly adaptation through an adapted conditional coefficient in the boundaries and moments quantitatively, and the declared central-state target. The fitted generator proposition couples a polynomial to a small localization potential; the restricted theorem cancels every portfolio term computing a bounded-source expectation. Theorem 6 exploits rank-one action coupling using convexity of the remaining scalar problems. It is a structured convex-duality and bound-and-bound construction, not a generic replacement for global nonconvex optimization. Proposition 3 preserves the nonquadratic terminal payoff in a computable relaxation of the exact problem.

Policy iteration and neural PDE solvers are important computational antecedents. Kim and Mijatović (2015) analyze continuous-time policy improvement. Kim, Cho, Kim, and Kim (2026) separate iteration, policy-network, and residual errors in entropy-regularized learning. Kim, Kim, Kim, and Cho (2026) study interior bounds and the mismatch between open-loop and closed-loop performance. Lee and Kim (2025) combine policy iteration and reinforcement learning. Cai, Fang, and Zhou (2025) impose martingale conditions without an explicit terminal infimum. NBO's raw training comparison is with a pinned implementation of the method, not a locally recreated approximation described as author code. Our certificates audit a policy produced by another solver. This portability is a property of independent training, while the training results remain evidence about the specifically recorded NBO and CartNet adapters.

Extensive utility, temporal-self equilibria, and games require their own evaluation operators and optimality notions. Their complete equations, assumptions, finite gain bounds, failures, and numerical results are preserved in the supplement. No new large-scale recursive or game-theoretic result is inferred from the additive-utility certificates above. The stochastic-trace bias and local-optimizer counterexamples are preserved for the same reason: they identify conditions that a neural training loss cannot replace.

9. CONCLUSION

The original continuous preference-adjustment economy now has accepted central-state targets at the declared .01 regret tolerance for all three cost coefficients. A feasible learned-and-constructed policy supplies each lower bound, while an independent affine-preference dual supplies an upper bound over the original continuous adaptive controls. This closes the numerical target gap by deleting the original stopping contract or replacing it by a manufactured-payoff problem. The value of access and the welfare cost of more expensive adjustment are resolved at a level calibrated to the reported effects. At coefficient two the optimal access gain is at least .096859; raising the coefficient to eight lowers optimal welfare by at least .02318257. The results guarantee exceed their corresponding interval widths. Optimal adjustment budgets obey the natural ordering and a valid secant enclosure, but are not equated with the budgets of the baseline policies.

The constructive tools also retain their distinct roles: fitted Bernstein witnesses, stopped martingale moment evaluation, market-deflator localization, quantitative output and neural-network allowances, and a structured continuous-action oracle. The unchanged coupled comparison problem has an absolute lower reference and a complete frozen holdout record, including the baseline and sign-test outcomes. Those experiments do not establish a universally applicable neural solver or an external matched-accuracy dominance theorem. All preceding

APPENDIX A: PROOF OF THE AFFINE-PREFERENCE DUAL

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This appendix gives the analytic inequalities behind the new upper bound and identifies exactly which finite interval calculations enter the proof. The frozen fitting objectives are not of inputs. The only fitted objects subsequently used are the exact stored values of y_0 and the even reference drifts, whose admissible ranges are checked independently.

A.1. The supporting-plane inequality

Let $r = u - 1$, $l = \log y$, and

$$L = \text{proj}_{[-\log(.8), -\log(.05)]}(l/u), \quad c = e^{-L}, \quad E = e^{Lr}.$$

The consumption maximizer is c , including both constrained branches. Write $\chi = 1$ on the interior branch and zero on a constrained branch; on a box intersecting a switching surface its value enclosure is $[0, 1]$. The derivatives required by the certifier are

$$\begin{aligned} \Psi &= -E/r - yc, & \beta &= E(1 - Lr)/r^2, \\ (S_0)_l &= -yc + .01y, \\ (S_0)_{uu} &= -\frac{E\{(Lr - 1)^2 + 1\}}{r^3} + \chi \frac{ycL^2}{u}, \\ (S_0)_{ul} &= -\chi \frac{ycL}{u}, \\ (S_0)_u &= -yc + \chi \frac{yc}{u} + .01y. \end{aligned} \tag{41}$$

On the interior branch $yc = E$. The envelope is continuously differentiable across the consumption constraint; the displayed one-sided second derivatives enclose its almost-everywhere Hessian. This is sufficient both for supporting inequalities and for the integral Taylor remainder below.

The lower-consumption constraint is active on $u \in [1.5, 2.5]$, $y \in [.2, 12]$, because $(12)/1.5 < -\log(.05)$. Moreover $L < 2$ throughout that rectangle. On its interior consumption branch,

$$\partial_l(S_0)_{uu} = \partial_u \beta_l = \frac{EL(2 - L)}{u^2} \geq 0. \tag{42}$$

On the upper-consumption branch $(S_0)_{uu}$ is independent of l and $\beta_l = 0$. The jump in $(S_0)_{uu}$ on increasing l enters the interior branch is nonnegative. When increasing u leaves that branch, β_l jumps upward to zero. Thus $(S_0)_{uu}$ is nondecreasing in l , and β_l is nondecreasing in l including the switching boundary.

The following enclosures are computed by the interval program on complete interval covers, by point sampling:

$$\sup_{1.5 \leq u \leq 2.3} (S_0)_{uu}(u, 12) < -.0862, \tag{43}$$

$$\sup_{2.3 \leq u \leq 2.5} \beta(u, 12) < -.9687, \tag{44}$$

cover has 1,600 intervals and the second has 2,000. All decimal endpoints and analytic bounds are enclosed at 70-digit interval precision; all subsequent binary operations and hulls are rounded outward. Full endpoints and the finite cover prescription are stored in `int_primitive_checks.json` and its generating program.

2), the first inequality proves concavity on $[1.5, 2.3]$ for every y in the verification set. For $u \in [2.3, 2.5]$ and $m \in [2, 2.2]$, the difference $\beta(u, y) - \beta(m, y)$ is nondecreasing in y . For $y \in [1.5, 2.3]$, β_i is nondecreasing in u . Its maximum therefore occurs at $y = 12$. At that endpoint, the inequality gives $\beta(m, 12) \geq \beta(2.2, 12)$, and the last two inequalities in (45) make the difference nonnegative. Integrating the derivative between m and u now proves (15) on both sides of the interval. This argument does not infer a two-dimensional supporting plane from concavity at a single point without controlling the cross derivative.

The same program covers $m \in [2, 2.2]$ by 1,000 intervals and uses $(S_0)_i < 0$, $\beta_i \leq 0$ to check β at y endpoints. It obtains

$$-6.989 < S_0(m, y) < -1.2443, \quad |\beta(m, y)| < .9712. \quad (46)$$

Coarsely simpler bounds $-7.1 \leq S_0 \leq 0$ and $|\beta| \leq 1.1$ are used for tails and localization. The inequalities continue to hold after clipping y .

A.2. An adapted solution of the affine auxiliary problem

This subsection starts at $(0, 2, y_0)$ and uses the full-line auxiliary processes. All sources are in U , with bounded coefficients, and all controls are bounded. Fubini's theorem, optional expectation, and optional stopping on bounded rectangles are legitimate by Gaussian moment bounds. With

$$B_s = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\beta_t \mid \mathcal{F}_s] dt + e^{-\rho(1-s)} b_T,$$

the independence of future Brownian increments gives $\mathbb{E}[\beta_t \mid \mathcal{F}_s] = \mathbb{E}[\beta_t \mid Y_s]$, even when the control observes the full correlated Brownian history. In particular, B_s is an adapted function of Y and $\mathbb{E}B_s = b(s)$.

$$U_t - m(t) = \int_0^t (\theta_s - \bar{\theta}_s) ds + .05 Z_t^u.$$

The argument in (16), followed by conditional Fubini, shows that its control-dependent part is

$$\mathbb{E} \int_0^1 e^{-\rho s} \{(\theta_s - \bar{\theta}_s) B_s - k \theta_s^2 / 2\} ds.$$

The pointwise maximizer $\theta_s = \text{proj}_{[-.2, .2]}(B_s/k)$ is adapted and bounded. It gives a well-defined auxiliary U because it is a function of the exogenous process Y . Thus this maximization does not require perfect foresight nor a new assumption about the optimizer of the original problem. Its exact value yields

$$A_k(0, 2, y_0) = e^{-\rho} \phi(m(1)) + \int_0^1 e^{-\rho t} \{\mathbb{E}S_0(m(t), \hat{Y}_t) - k \bar{\theta}_t^2 / 2\} dt + C$$

$$+ \int_0^t e^{-\rho(t-s)} \mathbb{E} g_k(B_s, \bar{\theta}_s) ds. \quad (47)$$

the stochastic source term, the joint Gaussian covariance is

$$20 \quad \text{Cov}(.05Z_t^u, \log Y_t) = (.05)(-.3)(-.25)t = .00375t.$$

Gaussian integration by parts consequently gives exactly the term C in (22). The clipping consumption switches are continuous Lipschitz maps; their weak derivatives, rather than existent classical derivatives at isolated switches, suffice for this identity. Since $\beta_l \leq 0$, $\beta_l = 0$. In particular, treating the two noises as independent would discard a real term of the final economy.

A.3. The conditional-variance allowance

The function H_k is continuously differentiable, with $H'_k(q) = \text{proj}_{[-.2, .2]}(q/k)$. Its derivative is $1/k$ -Lipschitz. The same bound applies to the derivative of $q \mapsto g_k(q, a)$ for fixed a . Therefore

$$\mathbb{E} g_k(B_s, \bar{\theta}_s) \leq g_k(b(s), \bar{\theta}_s) + \frac{\text{Var}(B_s)}{2k}. \quad (48)$$

Let $l_s = \log Y_s$ and write $B_s = f_s(l_s)$. The law of l_s is Gaussian with mean $M - .025s$ and variance $.09s$. The Gaussian Poincare inequality gives

$$\text{Var}(B_s) \leq .09s \mathbb{E}|f'_s(l_s)|^2.$$

By completeness, with an orthonormal Hermite expansion $f = \sum_{n \geq 0} c_n H_n$ under a Gaussian measure of variance v , one has $\text{Var} f = \sum_{n \geq 1} c_n^2$ and $v \mathbb{E}|f'|^2 = \sum_{n \geq 1} n c_n^2$. Polynomial approximation bounds the inequality to the bounded Lipschitz functions used here. No unknown mixing or concentration constant enters the calculation.

Differentiation of the conditional Gaussian transition is a translation derivative, so

$$f'_s(l_s) = \int_s^1 e^{-\rho(t-s)} \mathbb{E}[\partial_l \beta_t \mid Y_s] dt.$$

Additional Jensen and Minkowski give

$$\text{Var}(B_s) \leq .09s \left\{ \int_s^1 e^{-\rho(t-s)} \sqrt{\mathbb{E}|\partial_l \beta_t|^2} dt \right\}^2 \leq .09s D^2 C_\rho (1-s)^2. \quad (49)$$

Next verify the constant in this inequality. On the interior branch and before clipping, for $l \in [2, 2.2]$,

$$|\partial_l \beta(m, e^l)|^2 = e^{2(1-1/m)l} \frac{l^2}{m^4} \leq \frac{1}{16} e^{(12/11)l} l^2 \quad (l > 0).$$

For $l \leq 0$, the upper-consumption constraint makes the derivative zero. At a clipped y face the derivative is also zero almost everywhere. The positive right-hand side can therefore be integrated over the entire untruncated Gaussian as an upper bound. Its exact exponential moment

$$\mathbb{E}[e^{al} l^2] = e^{a\mu + a^2 v/2} \{(\mu + av)^2 + v\}.$$

on the relevant range. Using $\mu \leq M$ and $v \leq .09$ gives (24). Finally,

$$\int_0^1 e^{-\rho s} s C_\rho (1-s)^2 ds \leq \int_0^1 s(1-s)^2 ds = 1/12.$$

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ing (47), (48), and (49) proves (23). The allowance is positive and remains present at quadrature resolution.

A.4. Transfer from the auxiliary problem to the actual contract

$$W(t, u, x, y) = y(x - .5) + .1 \log(.5) + A_k(t, u, y).$$

ounting identity for Yx cancels every original adaptive portfolio choice. The maximiza-
 ing Ψ bounds every original consumption. Inequality (15) bounds the remaining source
 as (u, y) lies in the verification rectangle. Dynamic programming for the explicitly
 affine auxiliary problem then makes discounted W plus accumulated original payoff a
 martingale up to the first relevant face, for any original policy. One can establish the same
 at directly from conditional Fubini and the adapted selector above; no differentiability of
 own original value is needed. Smooth approximations of the bounded Lipschitz source,
 form source-error corrections, give the equivalent classical verification argument.
 terminal tangent obeys $Q \geq \phi$ everywhere because ϕ is concave. Write $F = A_k - Q$.
 the feasible auxiliary choice $\theta = 0$, discounted Ito applied to the affine function Q gives

$$, u, y) \geq \mathbb{E} \int_t^1 e^{-\rho(r-t)} \{S_0(m(r), \hat{Y}_r) + \beta(m(r), \hat{Y}_r)(U_r - m(r)) - \rho Q(U_r)\} dr.$$

starting $u \in [1.5, 2.5]$, one has

$$\mathbb{E}|U_r - m(r)| \leq .7 + .05 = .75, \quad \mathbb{E}|Q(U_r)| \leq .0008 + .008(.75) = .0068.$$

ently (46) implies

$$y) \geq \{-7.1 - 1.1(.75) - .04(.0068)\}(1-t) = -7.925272(1-t) > -8(1-t). \quad (50)$$

a uniform bound for the required starting rectangle; it is not a claim that the affine
 l is bounded on the entire u line.

$\in [.5, 2]$ and $y \geq .2$,

$$y(x - .5) + .1 \log(.5) - .1 \log x \geq 0,$$

ifferentiation in x , with equality at $x = .5$. Thus $W \geq G - 8h$ on actual wealth exits
 $\geq G$ at $h = 0$. For the preference localization, $.2(120) + .00125(120)^2 = 42$ bounds
 d generator for either sign and every original θ . For log-deflator localization, $dl =$
 $t - .3dZ^x$ and $22.33 = .025(22) + .045(22)^2$ bounds either sign. Each term in (18) is
 gative supermartingale potential that contributes at least $8h$ on its artificial face. Their
 therefore raises the continuation value to at least G at every artificial face. The already
 ned global upper $V_k \leq G$, for $k \geq .5$, applies beyond that face. Stopped verification
 and this fallback after it prove (19). All constants correspond to the original discount,
 n correlation, action bounds, and liquidation fee.

B.1. Interval quadrature of the dual

the frozen references have sixteen time slabs. A slab $t \in [j/16, (j+1)/16]$ is transformed $\in [\sqrt{j/16}, \sqrt{(j+1)/16}]$ and subdivided into 4, 8, or 16 intervals. The standard Gaussian is covered by 64, 128, or 256 intervals on $[-6, 6]$, respectively. Thus the three meshes use 6, 16,384, and 65,536 source boxes. All square-root endpoints are enclosed before binary64 version.

In a box the reference mean and log-deflator are

$$m(v^2) = 2 + \mu_j + \bar{\theta}_j(v^2 - j/16), \quad l(v, z) = M - .025v^2 - .3vz,$$

where $\mu_j = \sum_{i < j} \bar{\theta}_i/16$ is an interval-enclosed sum of exact deployment constants. The source Gaussian in (v, z) follows from (41) and the chain rule. The entire image box is enclosed, with a null of all possible consumption branches. The program asserts that the covered Gaussian stays inside the y clipping faces for each of the fitted references.

With $w_v = 2ve^{-.04v^2}$ and $w_z = (2\pi)^{-1/2}e^{-z^2/2}$, the source integral uses the exact interval-enclosed zeroth, first-centered, and second-centered moments of both weights. The exponential series for the time weight retains terms through degree twelve in $-.04v^2$ and bounds the remaining terms analytically. The Gaussian weight uses its degree-110 exponential series with analytic absolute remainder on $[-6, 6]$. Antiderivatives of each polynomial term are evaluated at interval endpoints. No floating-point Gaussian quadrature weight is assumed exact in the deflation.

For r_v, r_z are box radii, a valid second-order remainder bound is

$$\frac{1}{2} \|S_{vv}\|_{\infty} M_{v,2} M_{z,0} + \|S_{vz}\|_{\infty} r_v r_z M_{v,0} M_{z,0} + \frac{1}{2} \|S_{zz}\|_{\infty} M_{v,0} M_{z,2}.$$

Interval first-centered moments retain their signs; constant and linear terms are not discarded though all weighted boxes were symmetric. The source is C^1 with locally bounded one-sided Hessians across an active-consumption switch, so the integral Taylor formula with the Hessian remains valid there.

Gaussian Chernoff bound gives $\Pr(|Z| > 6) \leq 2e^{-18}$. The source tail lies between $1.1 \Pr(|Z| > 6)$ and zero at each time; the coefficient tail has magnitude at most $1.1 \Pr(|Z| > 6)$ for the covariance tail,

$$|\partial_l \beta| \leq 12^{6/11} \log(12)/4 < 2.5$$

the nonzero-derivative region, so its omitted contribution lies in $[-.00375(2.5) \Pr(|Z| > 6), 0]$. The signs are retained because $\beta_l \leq 0$. These corrections account for the full unbounded Hessian, not just the sampled central region.

For $b(s)$, interval source-coefficient integrals are accumulated backwards. The remaining portion of the current time cell ranges independently from zero to its entire discounted mass. Multiplying by the interval for $e^{\rho s}$ encloses $b(s)$ throughout that cell. The convexity of $g_k(q, \bar{\theta})$ ensures the maximum over its resulting q interval occur at an endpoint. At an uncertain saturation threshold the unconstrained quadratic conjugate supplies a safe upper bound. This calculation supplies the control-gap integral in (23).

Every elementary binary64 addition, subtraction, multiplication, division, and pairwise sum is rounded outward. Analytic constants and weighted moments use 70-digit interval arithmetic. The log objective, the source-quadrature enclosure, the policy-payoff enclosure, and the optimality test have different meanings and are recorded in different files. The acceptance test subtracts

B.2. Proof of the feasible-output allowance

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OF PROPOSITION 2: Couple the two policies to the same preference Brownian Wealth is deterministic, decreasing, and remains in $(.5, 1.25]$ because of the stated and consumption constraints. The difference of preference paths at time t is at most $|\tilde{\theta}_s|ds$. The terminal wealth difference is at most $e^{.02}\delta_c$.

the event that neither policy exits through preference before time one, $u \in [1.2, 2.8]$. For $8]$, put $r = u - 1$ and $L = -\log c$. Since $Lr < 1$, the running utility satisfies

$$|\partial_c \ell| = c^{-u} < 3, \quad 0 \leq \partial_u \{c^{1-u}/(1-u)\} = e^{Lr}(1-Lr)/r^2 \leq 1/r^2 \leq 25.$$

terminal preference derivative has magnitude at most .032, while the terminal wealth difference is at most .2. Finally, $|\frac{k}{2}|\theta^2 - \tilde{\theta}^2| \leq .2k|\theta - \tilde{\theta}|$. Integrating these inequalities and using factors at most one gives the first two terms of (25).

policy's preference exit has probability at most $2e^{-72}$ by the same maximal inequality of the independent evaluator. The union thus has probability at most $4e^{-72}$. On any stopped policy has magnitude below 6 before the adjustment cost, that cost contributes at most 8.1. Each payoff therefore has absolute value at $1 + .02k$. Bounding the payoff difference on the exit union by twice this quantity gives term of (25). The argument never replaces the stopped economy with a reflected or payoff. *Q.E.D.*

B.3. What the finite execution establishes

three cost instances pass the .01 acceptance test. The interval upper bounds apply to original adaptive controls; the lower bounds apply to the stored polished time policies. distinction is essential to an optimality certificate. The computations do not identify a optimal feedback surface, establish a rate for the neural optimizer, or prove that a fixed relaxation converges to exact value as its quadrature mesh tends to zero. Its nonvanishing cases are displayed explicitly. The finite target and the economic inequalities reported in text follow from the executed upper and lower bounds alone.

APPENDIX C: THE PRECEDING FITTED CONSTRUCTION, RETAINED FOR COMPARISON

Following section is preserved verbatim from the completed R9 paper. Its policies, witnesses, tables, and statements that the .01 target was not met describe that preceding construction. The current R10 certificate is Theorem 3 and Table I. This dated preservation is alternative current claim.

APPENDIX D: CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

Initial numerical bounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state certificate is not advertised as a uniform certificate for every state in D . The policy used here is trained in a time-control neural subclass. The retained lattice-trained, state-feedback policies, their finite corrections, and their previous continuous bounds remain separately in the supplement.

at $h = 1 - t$ and restrict the *verification domain*, not the economic domain, to $u \in [1.5, 2.6]$.

$$24 \quad F(h, u) = h \sum_{i=0}^{12} \sum_{j=0}^{14} \beta_{ij} T_i(2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

coefficients are obtained from a method-of-lines initializer and residual least squares. The initializer's collocation residual is not the certificate. For $w = G + F$, the consumption and folio maximizers are .8 and .8. Maximizing the remaining wealth term over $x \in [.5, 2]$ gives a scalar residual majorant

$$\mathcal{R}_k F = -F_h + .00125 F_{uu} - .04 F + H_k \{F_u - .04(u - 2)\} + \frac{.8^{1-u}}{1-u} \quad (51)$$

$$+ .0008(u - 2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (52)$$

$$H_k(q) = \max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

source is the original utility, not a payoff manufactured to make F exact.

Lower coefficients are enclosed at 70-digit interval precision and converted to Bernstein coefficients. The utility term is replaced by a polynomial with a proved uniform remainder below 7×10^{-9} . De Casteljau subdivision uses outward rounding at every elementary step. The positive-saturation, negative-saturation, and interior branches of H_k are all included whenever a certified q range permits them. Outward thresholds also cover a cell touching a switching face. The same calculation proves $F/h \geq -8$.

PROPOSITION 4—Localized fitted upper witness: *Suppose $F(0, u) = 0$, $F/h \geq -8$, and $F' \leq \epsilon$ on the verification rectangle. For $k \geq .5$, let*

$$Z(h, u) = 8h \{e^{-120(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (53)$$

$$U(t, u, x) = G(u, x) + F(1 - t, u) + Z(1 - t, u) + \epsilon_+ C_\rho(1 - t). \quad (54)$$

$U V_k(0, 2, 1.25) \leq U(0, 2, 1.25)$. No reflecting boundary or truncation of the original policy introduced at $u = 1.5$ or 2.6.

The exponential terms bound continuation when an artificial verification face is reached. The locally valid upper witness G is used beyond that face. At the central state the localization error is less than 1.219×10^{-7} , rather than an eight-unit initial gap. Uniform residual selection and localization are different terms and are reported separately.

D.2. A learned feasible policy and independent stopped evaluation

The actor has one time input, a twelve-unit tanh layer, and two outputs. At sixteen fixed nodes, a sigmoid constrains adjustment to $[0, .2]$, and a differentiable budget projection constrains consumption to $[\text{.05}, \text{.8}]$. The portfolio is zero. Training uses the original expected payoff, Gaussian quadrature, and no optimal-value or greedy-action labels. The deployed controls are the resulting frozen binary64 constants on each time slab; there is no claim that this control class exhausts the state-feedback opportunities in (5).

k	Policy payoff enclosure	Optimal-value upper	Regret upper
0.5	$[-1.27324522, -1.27322724]$	-1.16618072	0.10706450
2	$[-1.29504316, -1.29502986]$	-1.18870455	0.10633861
8	$[-1.32818098, -1.32817412]$	-1.22254633	0.10563464

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Endpoints and upper bounds are rounded outward. Each regret upper is the independent optimal-value upper at the lower endpoint of the learned policy’s payoff interval. All controls in the optimal-value bound are continuous. The .01 full-policy target is not met in any row.

TABLE D.II
EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE

Time boxes	Payoff width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
1024	0.000209	0.363	36	0.00477	0.079	0.110237
4096	5.25e-05	0.614	57	0.00195	0.116	0.107396
16384	1.33e-05	1.618	86	0.000894	0.186	0.106339

These refine the independent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two columns measure the respective integration and subdivision stages; complete timings, including coefficient construction, are in the raw records. A shrinking evaluation or residual allowance need not eliminate the approximation gap between the chosen upper witness and policy.

Budget projection enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-.02r} dr < 1.25 - .5e^{-.02}. \tag{55}$$

$= e^{-.02t} \{1.25 - \int_0^t e^{-.02r} c_r dr\}$ is deterministic. Since consumption exceeds .02(1.25), wealth decreases and its interval-enclosed terminal value remains strictly above .5. Wealth never reaches the remaining preference exit is controlled by a Brownian maximal inequality, not by a boundary check.

The evaluator independent of the training quadrature approximates utility by a degree-90 polynomial in $u - 2$, with a uniform analytic remainder on the original preference interval. Free moments are evaluated by recurrence, integrated over outward time rectangles, and used for the stopped event using a maximal probability bound and Cauchy–Schwarz. The wealth, adjustment expenditure, polynomial approximation, time integration, exact-exit condition, and arithmetic are all enclosed. There is no Euler, action-grid, or spatial-interpolation on this route. Those omissions are consequences of the deployed policy and analytic computation, not uncomputed entries assigned zero.

Table D.I reports the actual original-economy bounds. Table D.II makes the constructive content explicit. The last row gives substantially more precise policy evaluation than optimization. The fitted witness is independent of the policy and valid over the full original action and action ranges. Its loss of wealth-specific information is economically consequential. The late wealth-sensitive linear-programming pilot is retained with its computation log and candidate diagnostics; an unvalidated collocation objective is not used in either table.

E.1. Selection, stopped comparison, and the neural bridge

PROOF OF PROPOSITION 1: For a fixed jet the three controls enter additively. The continuation objective has derivative $c^{-u} - v_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it is increasing; otherwise its unconstrained maximizer is $v_x^{-1/u}$. The adjustment objective is $-k\theta^2/2$, which is strictly concave for $k > 0$. The portfolio objective is $h_1p + h_2p^2/2$. A nonnegative h_2 gives the projected vertex. A nonnegative h_2 gives a maximum at an endpoint, including the linear and constant cases. Compactness and the stated tie rule give a feasible selection. These are control-space statements at a fixed jet, not parameter-space convergence statements. *Q.E.D.*

PROOF OF THEOREM 1: Localize the process on interior compact sets and apply discounted Bellman's formula to an upper witness on each time slab. The discounted terminal witness plus accumulated reward equals the initial witness, the expected integral of $R_a U$, and the expected sum of discounted forward jumps. The latter sum is nonpositive. Boundary comparison gives $U + b_U$. Nonnegativity of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$J_k^a(t, s) \leq U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

Form integrability permits removal of localization. Taking the supremum over all admissible controls proves the optimal-value bound; attainment is unnecessary. Apply the same identity to L using the implemented policy. Its forward jumps are nonnegative, $g \geq L - b_L$, and $R_\pi L \geq -\epsilon_L$, proving the lower bound. Subtraction proves the regret statement. The process is stopped at the economic exit; there is no use of an unproved interior trace of the optimal value. *Q.E.D.*

PROOF OF THEOREM 2: The cover and representative intervals imply $|R_\pi v| \leq e$ and $0 \leq q$ everywhere. Hence $\sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $L = v$, $b_U = b_L = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen norm gives the two moduli by integration along segments, or along a specified path inside the cell. In a nonconvex cell, a verified path-length radius must replace the Euclidean radius. The overshooting-allowance assertion follows directly from the bound and does not require a compactness argument of network parameters. *Q.E.D.*

The finite counterpart follows from positivity of the continuation operators: propagate the valuation error with one factor δ , propagate the optimality error with one factor δ , and add the candidate gain. Backward induction from terminal error b gives (8). The full state-dependent selection, correction rule, and recursive-operator versions are reproduced in the preservation theorem. Unlike this finite identity, a continuous interpretation additionally needs either operator discrepancy terms or an independent continuous verification route. The present terminal-payoff calculation uses the latter.

E.2. Arithmetic convention

The mathematical constants specifying the economy are exact decimals. Stored network weights, policy constants, and fitted coefficients are interpreted as exact binary64 numbers. Constant formation and polynomial coefficient conversion use 70-digit interval arithmetic; linear optimization bounds use at least 50 digits. Vectorized interval addition, subtraction, multiplication, division by a strictly positive interval, and pairwise summation round both

interval multiplication; this is conservative even when its lower endpoint is negative. The interval library approximation is treated as a rigorous interval merely because its value was subsequently rounded once. Exponentials in vectorized routines are replaced by polynomials with analytic remainders; internal logarithms are rigorous constants, and square roots are formed in the multiprecision layer. The derivative and residual claims concern the mathematical function represented by the stored constants, not the real-time floating-point implementation of an arbitrary neural policy. The time controls are themselves stored constants, so their continuous implementation is rigorous. External-network rollouts are explicitly floating-point performance estimates. The function is retained in every numerical table.

APPENDIX F: ORIGINAL-ECONOMY FITTED WITNESSES

F.1. *Global fallback and localized upper bound*

$= -.02(u - 2)^2 + .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. Investment gain is bounded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part is at most $.000512$. Consumption utility is maximized at the upper endpoint $u = 2.8$ of the stated domain. The portfolio drift and diffusion contributions at $p = .8$ are $.00352$. Maximizing $-.08/x - .004 \log x$ over $[.5, 2]$ gives $-.04 - .004 \log 2$. Consequently

$$G \leq \frac{.8^{-1.8}}{-1.8} + .000512 + .001024 + .00195 + .00352 - .04 - .004 \log 2 < -.8659.$$

$\geq g$. Stopped verification gives $V_k \leq G$ globally on the original domain, including as a verification bound at an artificial verification face.

PROOF OF PROPOSITION 4: The derivatives of $G + F$ in x coincide with those of G . Thus the c and p selection applies. The residual after maximizing over wealth is precisely (52). $8h \exp\{\lambda(u - u_0) + \kappa h\}$, where $h = 1 - t$,

$$(\partial_t + \mathcal{L}^a - \rho)z = 8e^{\lambda(u - u_0) + \kappa h} \{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

Choices $(|\lambda|, \kappa) = (120, 42), (160, 64)$ satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore are nonpositive killed generators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) = 0$ follows that U has nonpositive residual inside the verification domain.

$F = 1$, $F = Z = C_\rho = 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G - 8h$. At either artificial preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first at an economic exit or an artificial face. If the artificial face occurs first, the remaining original-payoff is bounded by G under every admissible continuation, by the global fallback bound. Concatenated stopped verification gives the asserted bound without modifying the dynamics or opportunity set. *Q.E.D.*

F.2. *Polynomial source enclosure and refinement*

$u - 1 = A + Bz$, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and $(A + D) < 1$. The uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

by e^{LBz} times the degree-eight Taylor polynomial of e^{LBz} . Its remainder is at most $(LB)^9/9!$. Multiplying the two enclosures, with the relevant sup-norm bounds, gives stored uniform error below 2.117×10^{-9} for $-.8^{-(u-1)}/(u-1)$. This error is added to the dual upper bound, not fitted away.

The H_k branches are $-.2q - .02k$ for $q \leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for $q \geq .2k$. On each box, interval Bernstein coefficients enclose q and every potentially applicable dual polynomial. A box touching a switching threshold includes both adjoining branches. Comparison uses outward threshold intervals; binary64 representation of $.2k$ cannot exclude the branch. A tensor polynomial lies between its minimum and maximum Bernstein coefficients. De Casteljau subdivision preserves this enclosure and is repeated on boxes whose upper coefficient exceeds the prescribed ceiling. The independent check of $F/h \geq -8$ uses the same reach-hull property.

For a fixed polynomial whose true residual lies strictly below a ceiling, Bernstein refinement actually verifies that ceiling, up to the explicitly accounted source and arithmetic allowances. This is a certification statement about a fixed candidate with slack. It is not a theorem that dual fitting, a wealth-independent ansatz, or arbitrary Adam training produces approximants converging to the optimal value. The executed ceiling sequence and all visited-cell counts are stored. The separate wealth-sensitive linear-programming pilot has not received this validation and contributes no upper endpoint to the reported results.

APPENDIX G: INDEPENDENT EVALUATION OF THE LEARNED TIME POLICY

The projected policy has constant c_j, θ_j on each of sixteen slabs, $p = 0$, and $0 \leq \theta_j \leq .2$. Training chooses these constants, but the subsequent evaluator reads only the frozen policy. The exact wealth formula in the main text and the outward budget check show $x_1 > .5$. Since $x_1 = .025$ and $x_0 = 1.25$, wealth is decreasing throughout the horizon and never reaches the lower boundary. Neither Brownian correlation nor a wealth discretization affects this conclusion for $p = 0$.

Without preference stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An upper-boundary exit requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is contained in $\{\sup_{t \leq 1} |W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each gives

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

The original preference diffusion is still stopped at exact first exit. Its rare-exit correction is not zero.

At $r = u - 2$ and $L_j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r}/(1+r)$. Replace the reciprocal by $\sum_{m=0}^{80} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The metric remainder is at most $.8^{81}/.2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11}/11!$. The product bound, including the reciprocal and exponential suprema, yields a uniform error for degree-90 polynomial P_j . This bound holds on the whole original preference interval, not just at training quadrature nodes.

For a Gaussian random variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$, $M_1 = \mu$, and

$$M_n = \mu M_{n-1} + (n-1)v M_{n-2}.$$

Since $\mathbb{E}P_j(u_t - 2)$ is evaluated by this recurrence with $v = .0025t$, and the terminal quadratic preference settlement has an exact Gaussian expectation. On each time rectangle the intervals

interval range times interval length therefore enclose its integral. These ranges are
drature-error estimates inferred from two discretizations. The three resolutions are
dent enclosures of the same integral.

mpare this free-Gaussian polynomial calculation with the stopped surplus payoff,
the event of preference exit. Off E the two paths agree through T . On E , the omitted
ial tail is bounded using

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$$\mathbb{E}[\mathbf{1}_E |P_j(u_t - 2)|] \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

Gaussian moments are bounded uniformly by evaluating their nonnegative coefficient
s at $\mu = .2$ and $v = .0025$. The same argument bounds the free terminal settlement,
tual exit settlement has magnitude below 8.1. Discounted adjustment cost omitted after
most $.02k\mathbf{1}_E$. One valid total stopping allowance is thus

$$\sqrt{p_{\text{exit}}} \left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}} (.02k + 8.1).$$

form utility approximation allowance is added separately, multiplied by $C_\rho(1)$. At
e stopping allowance is below 5.388×10^{-16} ; the utility approximation allowance is
 151×10^{-7} before multiplying by $C_\rho(1)$. The larger remaining width comes from the
y enclosed time integration, not from ignoring a boundary event.

ree adjustment budget is exactly $\sum_j (\theta_j^2/2) \int_{j/16}^{(j+1)/16} e^{-.04t} dt$. Stopping can reduce this
by at most $.02p_{\text{exit}}$. Its interval, and the strictly positive wealth-feasibility margin, are
at full precision. Tight budget intervals describe these deployed policies, not optimal-
dentification.

APPENDIX H: ECONOMIC COMPARISON AND THE RESTRICTED DUAL

H.1. Cost ordering and value enclosures

OF THEOREM 4: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreas-
ause $B(\pi) \geq 0$. The supremum is convex and nonincreasing. The two approximate
ty inequalities give

$$A_1 - k_1 B_1 \geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

and dividing by $k_2 - k_1 > 0$ gives (26). Zero allowances give budget ordering for
imizers at distinct prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq$
 $(k_+ - k)B(\pi_k)$; feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute
ppropriate value-enclosure endpoints and intersect with the primitive budget range to obtain
Q.E.D.

ccess lower bound follows from $V_k \geq J_k$ and $V_R \leq \bar{V}_R$. Its nonnegativity follows
the restricted policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq$
 $\bar{V}_R$. These inequalities compare two optimal values even though the feasible policies
g the lower bounds need not be optimal.

PROOF OF THEOREM 5: For Y as specified, quadratic covariation gives

$$30 \quad d(Y_t x_t) = (.04Y_t x_t - Y_t c_t)dt + \text{local martingale.}$$

$.06pYx$ portfolio drift cancels the $-.06pYx$ covariation term for every adaptive p . Including Y , the discount term, and the preference quadratic yields

$$R_{c,p}W_0 = \frac{c^{1-u}}{1-u} - yc + .01y - .00005 - .004\log(.5) + .0008(u-2)^2 \leq S(u, y).$$

There is no W_0 mixed u - x or u - y derivative. Correlation is retained in the transition law of (Y, x) used to evaluate the source potential.

For a source lower bound use the feasible consumption $.35$: its utility on $u \in [1.5, 2.6]$ is bounded below by $-2/\sqrt{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative quantity on this rectangle. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize utility at $c = .8, u = 2.6$, use $.01y \leq .12$, and bound the remaining terms by $.0033$. This gives $-.7698 < 0$. The clipped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

The discounted Feynman–Kac potential satisfies the conditional expectation identity for the remaining source integral. Consequently $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is a martingale. Stopping this identity on a bounded verification rectangle is legitimate because the source is bounded. Inside the rectangle $\tilde{S} = S$, so adding the identity to discounted Itô verification for W_0 cancels its residual majorant. This argument does not require an assumed globally smooth optimal value, or differentiability of the clipped source at its outer faces.

For reference localization with $\theta = 0$ uses $\kappa = .00125\lambda^2$, giving 50 and 72 for $|\lambda| = 200$ and $|\lambda| = 22$. For $l = \log Y$, $dl = -.025dt - .3dZ^x$. Each exponential localization term with $|\lambda| = 22$ is nonpositive killed generator because $22.33 = .025(22) + .045(22)^2$ dominates either sign of the log drift. As before, the factor $8h$ supplies at least $8h$ at its associated artificial face.

For $x \in [.5, 2]$ and $y \geq .2$,

$$W_0 - G = y(x - .5) + .1\log(.5/x) \geq 0 :$$

The expression is zero at $.5$ and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source is at least $-8h$, so $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial y face a localization term pays for the entire possible source deficit, giving continuation value at least G . Actual preference exit cannot precede these inner preference faces. Invoke the global fallback $V_R \leq G$ after an artificial exit and apply the stopped identity before it. This verifies (34) over all original adaptive consumption and portfolio controls. The value of Y_0 is an auxiliary witness parameter, not an imposed wealth price or policy restriction. *Q.E.D.*

H.3. Validated evaluation of the restricted witness

Let z_1, z_2 be independent standard Gaussians and put $r = v^2$. At the central state,

$$u = 2 + .05v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

The source expectation is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box $z_2 \in [-6, 6]$ lies strictly inside the stated u, y verification rectangle under this transformation. The omitted probability is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission

completeness, the Hessian used by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$, $\log c, y = e^l$, and let χ be one for an interior consumption solution and zero for a strictly bounded solution. Across a switching box use $\chi \in [0, 1]$ with the following equations (IMPLEMENTATION)

$$\begin{aligned} Q_u &= \frac{e^{L(u-1)}\{1 - L(u-1)\}}{(u-1)^2}, & Q_{uu} &= -\frac{e^{L(u-1)}[\{L(u-1) - 1\}^2 + 1]}{(u-1)^3}, \\ S_u &= Q_u + .0016(u-2), & S_l &= -yc + .01y, \\ Q_{uu} &= Q_{uu} + \chi yc L^2/u + .0016, & S_{ul} &= -\chi yc L/u, \\ S_{ll} &= -yc + \chi yc/u + .01y. \end{aligned}$$

velope is continuously differentiable across clipping surfaces; its gradient is locally constant and these piecewise second derivatives give a valid integral Taylor bound across such a box. Differentiate the displayed (u, l) transformation by the chain rule to obtain a Hessian in (u, l) . Every intermediate expression is enclosed on the entire box. Bounds on the absolute values of the entries give the usual one-half diagonal and full mixed-term Taylor remainder. Constant and linear Taylor terms are integrated exactly for their polynomial approximation. Gaussian weighted moments through order two use 111 terms of $e^{-z^2/2}$ and are bounded on $|z| \leq 6$ by $e^{18} 18^{111}/111!$. Time moments use thirteen terms of $e^{-.04v^2}$ and are bounded on $|v| \leq 3$ by $e^{3} 3^{41}/41!$. All antiderivatives, remainders, additions, and sums are rounded outward. The two diagonal second moments are nonnegative; intersecting their enclosures with the nonnegative half-line is valid. Mixed moments are conservatively bounded by cell-radius products times probability mass. Successive box counts are 16,384, 131,072, and 1,048,576. Their intervals enclose the initial *dual witness*. Only an upper endpoint is used to bound the restricted optimum. Integrating the final upper endpoint with the independently certified restricted-policy lower bound gives the true optimal-value interval of width less than .009047. The distinct width below .000320 is integration of the witness, not the primal–dual gap.

APPENDIX I: CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

I.1. The scalar global action bound and interval neural jets

PROOF OF THEOREM 6: For any action with $\sum_i a_i = z$ and any real λ ,

$$\sum_i f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

Second derivative of $.3 \sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic minorant gives

$$\lambda z + .3 \sin z \geq \lambda z_0 + .3 \sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

implies (39). Each scalar f_i has second derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus λs is m -strongly convex. Its tangent quadratic at any feasible s_0 gives a lower bound $q'(s_0)^2/(2m)$. At an endpoint, only a feasible descent direction is relevant: an outward-minimization direction contributes zero, while an inward-pointing direction contributes

line or directly from the sign of the linear term. The stated lower bound remains valid if the quadratic minimizer lies beyond the opposite endpoint, because relaxing the displacement can lower it.

The sum of these scalar lower bounds is no larger than $D(\lambda)$. A feasible action gives a rough upper bound by direct interval evaluation. The scalar interval $[-d, d]$ covers every feasible action sum. Subdivision, pruning only a box whose lower bound exceeds a feasible upper bound, and taking the minimum surviving lower bound preserve a global enclosure. Root finding affects efficiency but not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box. The true candidate upper and true minimum lower can each worsen by at most this amount, adding the $4 \sum_i \delta_i$ gap correction. *Q.E.D.*

For an affine layer with stored weights, interval automatic differentiation propagates both evaluations and spatial derivatives. A tanh layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a strictly positive denominator and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The final embedding is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as a previously rounded autograd constant. This gives a componentwise interval for the derivative of the represented network. Subtracting the stored floating-point gradient gives the δ_i . The two automated tests use the inherited eight- and sixteen-dimensional learned checkpoints and the same as the action certificates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17} respectively. These are local jet checks, not whole-state residual bounds.

1.2. A lower bound retaining the nonquadratic terminal payoff

PROOF OF PROPOSITION 3: For $|a| \leq 1$, $\sin(a)/a \geq \sin 1$, with the continuous value at 1. Consequently $a^2 + .1 \sin^2 a \geq r_0 a^2$, where $r_0 = 1 + .1 \sin^2 1$. Also $.3 \sin(\sum_i a_i) \geq -.3$. Replacing running cost by $r_0 \|a\|^2/d - .3$ decreases every original control's cost; allowing all $\mathbb{R}^d$ decreases the infimum further. The original terminal payoff q_d is unchanged. The relaxed HJB equation is

$$v_t + \frac{\sigma^2}{2} \Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

$\lambda = \sigma^2 r_0 / (2d)$ and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$v(0, 0) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

Bounded cosine and coercive quadratic terminal terms give finite positive expectations. The Gaussian heat solution is smooth before the terminal date and has at most quadratic growth. Standard localization of its verification identity, or truncation followed by monotone Gaussian approximation, validates the relaxed lower bound without assuming a smooth solution of the original unded-control problem.

Complete the square for the quadratic terminal component. Its Gaussian normalization constants

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

Under the tilted Gaussian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and variance $s^2 = \sigma^2 r_0 / (r_0 + 4)$. Orthogonal coordinates disappear, leaving

$$v(0, 0) = \frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

gives a lower enclosure for v , because $-\lambda \log$ is decreasing. That lower endpoint is $Q.E.D.$

the scalar integral let $\eta = .2/\lambda$ and $R = \lceil \sqrt{2(\eta+50)} \rceil$. The omitted Gaussian contribution is no larger than $2e^{\eta-R^2/2}$. Interval ranges of the positive integrand times cell lengths the integral on $[-R, R]$. The successive 1,024, 4,096, and 16,384-cell calculations add analytic tail allowance and then apply the decreasing logarithm with outward rounding. Widths measure integration uncertainty in the relaxed value. Relaxing the action bound running sine can leave a large structural gap even when this quadrature width is small.

APPENDIX J: REPLICATION AND SCOPE LEDGER

entral economic certificate is independent of the inherited killed-Euler discretization. The upper has a verified continuous-generator residual, an explicit source-polynomial remainder, a stopped-trace inequality, and exponentially bounded artificial-face continuation. The lower has exact deterministic wealth, interval Gaussian polynomial moments, an interval utility approximation remainder, outward time integration, and an exact-exit probability bound. The restricted upper cancels continuous portfolio controls algebraically and evaluates the restricted-source stochastic potential with Taylor, Gaussian-tail, and localization allowances. The results do not establish the still-uncomputed operator discrepancy budget for the old finite

external experiment has a different status. Each declared sample-and-hold policy is a continuous control in the stated additive-noise dynamics. Real-arithmetic transitions are stored for holding slab, but the stored path estimates use floating-point network evaluations and a Monte Carlo sample. The rigorous lower bound supplies an absolute reference; subtracting a sampled mean does not create an outward policy-regret certificate. Paired path errors, path variation, completed-update clock overshoot, setup, and warmup are recorded separately. The exact sign test concerns the frozen joint training-and-audit pipeline under independent runs. It does not certify an underlying PDE or imply that the six panels represent six different problem families.

A new table is generated from raw JSON or path records; display endpoints are rounded in the conservative direction. The frozen external protocol, all tuning trials, all 72 holdout pairs, endpoints, and all path arrays are retained. The validation program checks coverage of the six panels, matching initialization hashes, pinned author revision, finite outputs, refinement of the primitive feasibility, the positive access lower bounds, and the explicitly false flexible decision flags. The complete R8, R4, and delivered R5 textual and numerical material is included in the preservation supplement, and earlier source files remain byte-for-byte unchanged. A failed pilot nor a historical manufactured-payoff test is used as new original-economy evidence.

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PART II: COMPLETE R10 PRESERVATION SUPPLEMENT

The current paper and proofs are in ECTA_R10. This supplement preserves the complete completed R9 paper and the entire R9 preservation supplement without deleting equations, numerical tables, counterexamples, or unsuccessful calculations. Earlier precision statements are dated results, not current R10 claims. All underlying historical sources remain unchanged in the revision branch.

READING MAP

The R10 paper contains the affine-preference dual, its complete verification proof, the final-model certificates below .01, the quantitative output-error allowance, the economic comparisons, and every main R9 theorem or its explicitly preserved appendix counterpart. A separate point-by-point response to the R9 referee accompanies the paper in the R10 revision history.

The following pages reproduce the completed R9 paper, including its older .10634 flexibility bound. They are followed by the complete R9 supplement, which preserves R8, R4, delivered R5 numerical material, recursive utility, sophisticated temporal selves, games, stochastic-trace identities, finite safeguards, and historical diagnostics. Historical claims must read with their original version labels. The R10 current status is determined by its own then, result files, and source/result/build manifest, not by the older precision flags reproduced.

PART I: COMPLETED R9 PAPER, PRESERVED IN FULL

possibly preferred adjustment, create neural policies and market equilibria. A positive value of access to adjustment under the original utility and first-exit central state, the certified gain is at least .04094994 when the adjustment-cost two. Constructive interval refinement encloses the no-adjustment optimum. The flexible policy is evaluated much more precisely than its current .10634. For coupled nonconvex controls, a scalar dual branch-and-bound procedure provides lower bounds through 128 dimensions, and a Cole–Hopf relaxation supplies lower bounds. A frozen, symmetrically tuned, multi-budget comparison refutes outcomes, including a stronger independently specified feedback baseline. We compare separate policy evaluation, optimality, computational cost, and the economic calculation establishes.

Keywords: Stochastic control, Neural policy iteration, Endogenous preferences, Recursion, Certification, Computational economics.

1. INTRODUCTION

Neural Bellman Operators (NBO): separated policy evaluation and feasible policy implemented by neural networks and assessed by independent verification. The economy combines consumption, portfolio choice, stochastic risk aversion, and adjustment. The analysis retains its original running utility, continuous Brownian shocks, and first-exit liquidation covenant. Neither reflecting the boundary nor altering the running payoff is used to establish the economic result.

The result is a strictly positive, verified lower bound on the value of access to the economy. At the central initial state and adjustment-cost coefficient $k = 2$, access is certified by at least .04094994 in the model's fixed utility units. The corresponding value is .074788 at $k = .5$ and .00781212 at $k = 8$. These are comparisons with the restricted economy, not merely with an arbitrarily chosen no-adjustment policy. The flexible policy supplies the flexible economy's lower bound. An auxiliary market policy supplies the restricted economy's upper bound while allowing every original policy and portfolio policy. A scalar state used to construct the witness does not alter the opportunity set.

The argument has three parts. First, smooth one-sided witnesses bound the economy without assuming a smooth optimal value or regularity of the boundary. Second, fitting and certification are separate operations. A polynomial fitted to the boundary equation is converted to outward Bernstein coefficients and refined

ot .0000133. This replaces the previous 8.33956 bound for a different
stantially smaller original-model certificate; it does not establish the
arget. The positive access-welfare result is valid nevertheless because
ble policy³⁸ with a substantially tighter upper bound on the restricted
fore establishes a specific economic inequality rather than interpreting
tion as an optimal policy surface.

addresses the inner optimization needed for verification. In a cou-
family, all nonconvex coupling depends on the sum of the actions.
ch-and-bound calculation, together with strongly convex scalar dual
e full continuous action box without a tensor action grid. Executed
low 10^{-5} through 128 action dimensions. At the eight- and sixteen-
test states, interval differentiation of the actual network weights also
floating-point critic gradient. This is an implemented bridge between
n action certificate, not an assumption that an optimizer’s stationary

family is given an absolute accuracy axis. A pointwise running-cost
of control bounds admit a Cole–Hopf representation that retains the
nimal payoff. Orthogonal Gaussian coordinates integrate analytically,
integral with an outward enclosure. The resulting number is a lower
original minimization problem. Relative performance is evaluated
out experiment: three dimensions, two training budgets, twelve seed
ning-rate search budgets, warmed initialization, randomized method
ndently specified feedback rule. The pinned author implementation of
ified.

monotone approximation, and primal–dual numerical methods supply
he stopped Itô argument is not claimed as a new general verification
ing neural evaluation and improvement claimed as a new algorithm
on is the constructive combination of original-contract verification,
per and lower bounds, computable neural-jet allowances, and a non-
n audit, with each error quantity matched to an executed calculation.
objects theorem by theorem with neighboring methods. Recursive util-
l selves, dynamic games, stochastic-trace identities, counterexamples,
ng manuscripts are preserved in the supplement rather than being used
claims of the present application.

2. THE ECONOMIC ENVIRONMENT

, where u denotes relative risk aversion and x denotes wealth. An
 c, θ, p) in the compact set

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

, discounting is $\rho = .04$, and the open state domain is $D = (1.2, 2.8) \times \inf\{r \geq t : S_r \notin D\}$. For $k > 0$, running utility and settlement are

$$= \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$= -.02(u-2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1-t). \quad (4)$$

and optimal value are

$$\int_t^\tau e^{-\rho(r-t)} \ell_k(S_r, a_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \Big], \quad V_k = \sup_\pi J_k^\pi. \quad (5)$$

value V_R restricts θ to zero and retains *all* admissible consumption and is independent of k . The cost experiment holds the controls, transition law, event, utility normalization, and consumption units fixed. A state-dependent utility would generally define another economy.

ent fee is a liquidation covenant. There is no continuation consumption or allowing exit. At the lower wealth boundary the largest admissible wealth at the upper boundary, $p = 0$ eliminates the wealth diffusion and permits another a silently viable rectangle nor a theorem requiring uniform controlled- can be substituted for this contract. The verification below requires only with test functions on the trace at which the process is actually stopped.

tion v , put

$$[(.02 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$^av + \ell_k - \rho v. \quad (7)$$

is retained. Its sign is not inferred solely from the negative shock correla-

-Complete selection at a fixed jet: *For the generator in (7), a global can be selected componentwise. Consumption is $\text{proj}_{[.05, .8]}(v_x^{-1/u})$ when otherwise. Adjustment is $\text{proj}_{[-.2, .2]}(v_u/k)$. Let $h_1 = .06xv_x - .0025xv_{ux}$ If $h_2 < 0$, the portfolio is $\text{proj}_{[-.5, .8]}(-h_1/h_2)$; if $h_2 \geq 0$, an endpoint $p^2/2$ is selected, with a fixed tie rule.*

cludes convex and zero-curvature portfolio objectives and nonpositive . It applies to the continuous generator, not to the interpolated, killed-Euler . It says nothing about global optimization over shared neural parameters.

to retain differentiation through the controlled successor state. A present in a game, are also frozen during unilateral improvement. This or loss from lowering itself by changing the object being evaluated. It network optimization convex. The ideal policy-iteration compactness are retained in the supplement; no such argument is invoked to certify ory.

reguard is a distinct algorithm. For $\mathcal{T}_n^a w = r_n^a + \delta P_n^a w$, positive sub- ρ^Δ , define the policy-evaluation residual $e_n = \|v_n - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$ and element gap $\eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\pi_n} v_{n+1}\|_\infty$. If the terminal error comparison yields

$$\|v_n^* - V_{h,n}^\pi\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

ope and threshold-correction proof are reproduced in the supplement. The fraction of decisions changed while leaving the cost of a complete y unchanged. In particular, the retained 125-action tensor audit is not ifier of Section 6. Its floating-point envelope is not relabeled as an bound.

1. A quantitative bridge from a fitted network

)/ ρ , with $C_0(h) = h$.

missible witnesses: *Witnesses are bounded, continuous on the stopped erior of each time slab. Localized Itô identities pass to the stopping ity. At slab interfaces an upper witness has nonpositive forward jumps onnegative forward jumps. The implemented policy has a well-defined*

one-sided comparison: *Suppose Assumption 1 holds, $U \geq g - b_U$ and d trace, $\sup_a R_a U \leq \epsilon_U$, and $R_\pi L \geq -\epsilon_L$, where all four allowances*

$$L - b_L - C_\rho(T - t)\epsilon_L, \quad V_k \leq U + b_U + C_\rho(T - t)\epsilon_U, \quad (9)$$

$$U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (10)$$

quire an attained optimizer or a smooth optimal value.

nd-oracle certificate for neural output: *Let a continuous neural witness policy π satisfy Assumption 1, with stopped-trace error at most b . Let*

cluding jet, inner solve, and arithmetic allowances. Set $e = \max_j (e_j + \omega_j)$. Then

$$0 \leq V_k - J_k^\pi \leq 2b + C_\rho(T - t)(2e + q). \quad (11)$$

NBO: R11 PRESERVATION SUPPLEMENT

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derivative bounds M_j, N_j and cell radii h_j permit $\omega_j = M_j h_j$ and
ence of learned outputs for which these verified allowances vanish has
spective of whether its parameters converge.

ve a posteriori connection to network errors, not a claim that an empirical
ontrols a supremum. Interval differentiation or weight-based derivative
e cell moduli. They may be conservative, and covering a high-dimensional
costly. Section 6 executes the jet-to-action part for actual learned critics.
tation instead fits an independent polynomial witness; it does not falsely
ed a whole-state neural critic merely because its action audit passed at

CONSTRUCTIVE CERTIFICATES IN THE ORIGINAL ECONOMY

al bounds in this section concern $(t, u, x) = (0, 2, 1.25)$. A central-state
rtised as a uniform certificate for every state in D . The policy used here
ime-control neural subclass. The retained lattice-trained, state-feedback
nite corrections, and their previous continuous bounds remain separately
ement.

ttting an upper witness without importing an optimal value

restrict the *verification domain*, not the economic domain, to $u \in [1.5, 2.6]$.

$$(h, u) = h \sum_{i=0}^{12} \sum_{j=0}^{14} \beta_{ij} T_i(2h - 1) T_j \left(2 \frac{u - 1.5}{1.1} - 1 \right).$$

obtained from a method-of-lines initializer and residual least squares. The
on residual is not the certificate. For $w = G + F$, the consumption and
are .8 and .8. Maximizing the remaining wealth term over $x \in [.5, 2]$ gives
jorant

$$-F_h + .00125F_{uu} - .04F + H_k\{F_u - .04(u - 2)\} + \frac{.8^{1-u}}{1-u} \quad (12)$$

$$+ .0008(u - 2)^2 + .00195 + .00352 - .04 - .004 \log 2, \quad (13)$$

$$\max_{|\theta| \leq .2} (q\theta - k\theta^2/2).$$

calized fitted upper witness: Suppose $F(0, u) = 0$, $F/h \geq -8$, and on rectangle. For $k \geq .5$, let

$$= 8h\{e^{-126(u-1.5)+42h} + e^{-160(2.6-u)+64h}\}, \quad (14)$$

$$= G(u, x) + F(1 - t, u) + Z(1 - t, u) + \epsilon_+ C_\rho(1 - t). \quad (15)$$

$(0, 2, 1.25)$. No reflecting boundary or truncation of the original policy for 2.6.

bound continuation when an artificial verification face is reached. The guess G is used beyond that face. At the central state the localization 19×10^{-7} , rather than an eight-unit initial gap. Uniform residual on are different terms and are reported separately.

ned feasible policy and independent stopped evaluation

the input, a twelve-unit tanh layer, and two outputs. At sixteen fixed constrains adjustment to $[0, .2]$, and a differentiable budget projection to $[.05, .8]$. The portfolio is zero. Training uses the original expected return, and no optimal-value or greedy-action labels. The deployed frozen binary64 constants on each time slab; there is no claim that this tests the state-feedback opportunities in (5).

enforces

$$\sum_{j=0}^{15} c_j \int_{j/16}^{(j+1)/16} e^{-.02r} dr < 1.25 - .5e^{-.02}. \quad (16)$$

$\int_0^t e^{-.02r} c_r dr\}$ is deterministic. Since consumption exceeds $.02(1.25)$, interval-enclosed terminal value remains strictly above $.5$. Wealth never reference exit is controlled by a Brownian maximal inequality, not by a check.

ment of the training quadrature approximates utility by a degree-90 a uniform analytic remainder on the original preference interval. Free evaluated by recurrence, integrated over outward time rectangles, and event using a maximal probability bound and Cauchy–Schwarz. The ent expenditure, polynomial approximation, time integration, exact-exit are all enclosed. There is no Euler, action-grid, or spatial-interpolation e omissions are consequences of the deployed policy and analytic ed entries assigned zero.

al original-economy bounds. Table II makes the constructive refinement gives substantially more precise policy evaluation than optimality

TABLE II
 NBO: R11 PRESERVATION SUPPLEMENT

EXECUTED REFINEMENT FOR THE $k = 2$ CERTIFICATE					
Payoff width	Sec.	Bernstein boxes	Residual upper	Sec.	Regret upper
0.000209	0.363	36	0.00477	0.079	0.110237
5.25e-05	0.614	57	0.00195	0.116	0.107396
1.33e-05	1.618	86	0.000894	0.186	0.106339

pendent policy evaluator. Bernstein boxes refine the already fitted upper witness. The two
 the respective integration and subdivision stages; complete timings, including coefficient
 the raw records. A shrinking evaluation or residual allowance need not eliminate the
 approximation gap between the chosen upper witness and policy.

ed witness is independent of the policy and valid over the full original
 ges. Its loss of wealth-specific information is economically consequential.
 ositive linear-programming pilot is retained with its computation log and
 gnostics; an unvalidated collocation objective is not used in either table.

CERTIFYING THE VALUE OF PREFERENCE ADJUSTMENT

blicly write $J_k(\pi) = A(\pi) - k B(\pi)$, where

$$= \mathbb{E}^\pi \int_t^\tau e^{-\rho(r-t)} \theta_r^2 / 2 \, dr, \quad 0 \leq B(\pi) \leq .02 C_\rho (T - t).$$

mon to every k .

curacy-calibrated cost comparisons: *The optimal value is convex and
 If optimizers exist, their adjustment budgets are nonincreasing across
 generally, let π_i be ε_i -optimal at k_i , with $k_2 > k_1$. Then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \tag{17}$$

)), the budget of an optimizer at k , for $k_- < k < k_+$, lies in the primitive
 ted with

$$\left[\frac{V(k) - \bar{V}(k_+)}{k_+ - k}, \frac{\bar{V}(k_-) - \underline{V}(k)}{k - k_-} \right]. \tag{18}$$

erns discounted adjustment expenditures, not a pointwise ordering of
 V_u/k is not a comparative-static proof. The central full-value enclosures

$$dY_t = .02Y_t dt - .3Y_t dZ_t^x, \quad Y_0 = 1.55. \quad (19)$$

ected from the original shocks. It does not restrict portfolio information, 44
aded asset. For

$$u, x, y) = y(x - .5) + .1 \log(.5) - .02(u - 2)^2,$$

or of Yx cancel exactly. With $\theta = 0$, define

$$\max_{.5 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad (20)$$

$$(u, y) + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2. \quad (21)$$

12], one has $-8 < S < 0$. Extend S to a bounded function $\tilde{S} \in [-8, 0]$
ents at these verification faces. Let

$$F_R(h, u, y) = \mathbb{E}_{u,y} \int_0^h e^{-.04r} \tilde{S}(U_r, Y_r) dr, \quad (22)$$

and Y follows (19). Thus $-8h \leq F_R \leq 0$.

stricted-economy upper bound: *Let Z_R be the sum of the following
potentials:*

$$= 8h \{ e^{-200(u-1.5)+50h} + e^{-240(2.6-u)+72h} \}, \quad (23)$$

$$= 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (24)$$

$$) \leq W_0(2, 1.25, 1.55) + F_R(1, 2, 1.55) + Z_R(1, 2, 1.55). \quad (25)$$

all adaptive original consumption and portfolio controls with $\theta = 0$.
ossible flexible policy with certified payoff lower bound $\underline{J}_k$ and any
5),

$$, 2, 1.25) - V_R(0, 2, 1.25) \geq \max\{0, \underline{J}_k - \bar{V}_R\}. \quad (26)$$

) has an explicit two-Gaussian representation. We transform time by
and linear Taylor terms with interval-enclosed weighted moments,
remainder on each box. Exponential Taylor remainders, Gaussian tails,
aded. This produces the executed refinement in Table III; it does not
nce as an error estimate. The final restricted-optimum upper bound

TABLE IV

STIFIED ORIGINAL-MODEL EXPENDITURES AND VALUE OF ACCESS

Policy adjustment budget	Access gain: lower	Access gain: enclosure
[83159556, 0.01960483159557]	0.06274788	[0.06274788, 0.17885870]
[15505497, 0.01242015505498]	0.04094994	[0.04094994, 0.15633486]
[18262027, 0.00192618262028]	0.00781212	[0.00781212, 0.12249308]

evaluates the displayed learned policy. The access columns bound the difference between the lower endpoint pairs a feasible flexible policy with the restricted dual upper; the upper endpoint upper with a feasible restricted policy. The positive lower endpoints are informative though the two-sided access intervals are not narrow estimates of the welfare magnitude.

outward display rounding. The evaluated restricted policy has payoff [0.34503585], so the restricted problem itself has an initial optimal-value an .00905.

bounds remain strictly positive at all three prices. At $k = 2$, the quadrature restricted witness is more than two orders of magnitude narrower than the flexible learned policy is optimal. This comparison establishes a minimum economically meaningful gain the flexible learned policy is optimal. It does not identify the exact access expenditure-budget magnitude, or an optimal pointwise policy ordering.

GLOBAL ACTION CERTIFICATION WITHOUT A TENSOR GRID

ing minimization problem, which is also used for the external comparison:

$$dX_t = 2a_t dt + \sqrt{.32} dW_t, \quad X_0 = 0, \quad a_t \in [-1, 1]^d, \quad (27)$$

$$f_d(a) = \frac{1}{d} \sum_{i=1}^d \{a_i^2 + .1 \sin^2 a_i\} + .3 \sin \left(\sum_i a_i \right), \quad (28)$$

$$q_d(x) = \frac{1}{d} \sum_{i=1}^d (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right), \quad (29)$$

$$J^* = \inf_a \mathbb{E} \left[\int_0^1 f_d(a_t) dt + q_d(X_1) \right].$$

quinely coupled nonconvex actions. No analytic optimal-action or optimal-policy is available to training. At a fixed critic gradient p , the action-dependent Hamilto-

03	0.00068	85	2720	1.498
05	3.52e-06	103	3296	1.991
03	0.000647	163	10432	6.041
05	3.46e-06	185	11840	5.955
03	0.000948	317	40576	19.229
05	4.68e-06	339	43392	20.651

vision, at both tolerances. At $d = 8, 16$, these are gradients of retained learned critics at gaps include interval neural-derivative allowances. At larger dimensions they are for the same Hamiltonian family. The experiment measures certificate work for these t an average-case scaling law or a whole-state neural PDE residual.

$$) + .3 \sin z, \quad f_i(s) = \frac{s^2 + .1 \sin^2 s}{d} + 2p_i s, \quad z = \sum_i a_i.$$

x with modulus at least $m = 1.8/d$.

global partition and convex dual bounds: *For any multiplier λ and*

$$D(\lambda) = \sum_i \min_{|s| \leq 1} \{f_i(s) - \lambda s\}.$$

$z_0 - r, z_0 + r]$, a lower bound for the full Hamiltonian is

$$\lambda) + \lambda z_0 + .3 \sin z_0 - |\lambda + .3 \cos z_0| r - .15 r^2. \tag{30}$$

es an upper bound. Each inner convex minimum can itself be bounded $2m)$, where $q = f_i - \lambda s$ and e is the derivative magnitude in a feasible full derivative magnitude at an interior point. Outward versions of v scalar subdivision and pruning, give a global gap on $[-1, 1]^d$. differs from the true neural gradient by componentwise bounds δ_i , the dient gap is at most its stored-gradient gap plus $4 \sum_i \delta_i$.

ding only proposes multipliers and actions. It is not trusted to solve exactly. Even an inaccurate proposal remains valid after the strong- is structure removes tensor enumeration; it does not establish a y theorem or solve a generic nonseparable nonconvex Hamiltonian. 10^{-5} calculation uses 339 scalar boxes and 43,392 scalar convex neither raw actor training nor the preceding finite tensor safeguard in eight- and sixteen-dimensional jets, interval forward differentiation nts bounds the ℓ^1 gradient discrepancies below 4.584×10^{-17} and

16384	[-0.20445644, -0.20435623]	-0.20445644	4.212
1024	[-0.28757809, -0.28684340]	-0.28757809	0.245
4096	[-0.28730394, -0.28712026]	-0.28730394	0.988
16384	[-0.28723515, -0.28718923]	-0.28723515	4.172

value. Its lower endpoint is also a lower bound for the original bounded-control value. The
 isive enclosures is quadrature uncertainty; it does not include or estimate the structural
 relaxation gap.

ctively. These allowances are propagated into the action gap. Theorem 2
 al state-cover and residual quantities that would be needed for a whole-

ABSOLUTE BOUND AND A FROZEN EXTERNAL COMPARISON

rigorous lower bound for the unchanged nonconvex family

$^2(1)$, $\sigma^2 = .32$, and $\lambda_d = \sigma^2 r_0 / (2d)$. Since $\sin^2 a \geq \sin^2(1)a^2$ for $|a| \leq 1$,
 cost by $r_0 \|a\|^2 / d - .3$ and allowing unbounded controls gives a relaxation.
 function q_d is retained.

-Scalar-integral absolute lower bound: *Let Z be normal with mean
 riance $\sigma^2 r_0 / (r_0 + 4)$. Then*

$$\frac{\sigma r_0}{+4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda_d \log \mathbb{E} \exp \left[-\frac{.2}{\lambda_d} \cos Z \right]. \quad (31)$$

*e of the positive expectation yields a rigorous numerical lower bound for
 (29).*

n upper/lower-bound device, not a new training problem with a known
 The terminal cosine, which couples all state coordinates, remains in the
 s tails are bounded analytically and its interior is integrated by interval

. Holdout design and exact deployment interpretation

ains the original actor and critic architectures: two tanh hidden layers
 bounded actor output, and the same terminal-value embedding. NBO
 tion and improvement with Adam, three critic steps per actor step, 256
 our successors. SOC-MartNet uses its pinned author solver, RMSprop,
 etwork, and the inherited $J = 2$, $K = 1$ multiplier setup. These different
 are disclosed rather than described as identical implementations.
 otocol freezes dimensions 8, 16, 32 and training budgets of 10 and 30
 ension, both methods receive the same six validation trials: two seeds

TABLE VIII

MUTE REFERENCE EXCESS AND ACTUAL TRAINING CLOCKS

	Lower bound	NBO excess	SOC excess	NBO sec.	SOC sec.
	-0.08499	0.6662	1.1692	10.002	10.006
	-0.08499	0.6635	1.1407	30.004	30.008
	-0.20446	0.6707	0.9938	10.002	10.005
	-0.20446	0.6674	1.0500	30.004	30.006
	-0.28724	0.6686	1.4883	10.003	10.007
	-0.28724	0.6672	1.3090	30.003	30.007

policy cost minus the rigorous original-value lower bound. It provides an absolute
deterministic regret enclosure: sampling error and floating-point rollout evaluation are
clocks include completed-update overshoot and exclude separately recorded warmup
and setup.

multipliers. Selection minimizes mean independent validation cost,
elve different holdout seeds, 720–731, are then evaluated at each
Within each pair, actor and critic starting weights are identical and
Framework warmup precedes both clocks, and method order is seed-
y and setup-plus-training clocks are retained. The baseline $a_i(t, x) =$
 $l + 4(1 - t)\}$ is specified independently from the associated quadratic
on the unchanged nonquadratic objective.

s deployed as a sample-and-hold control on 128 dates and evaluated
paired Gaussian paths. In (29), drift under a held action and volatility
e Gaussian transition and running-cost integral are therefore exact in
ployed *continuous-time policy*. A 64-date evaluation uses aggregated
changes the deployment policy. Its difference from the 128-date result is
bound. Floating-point network evaluation and Monte Carlo uncertainty
outward deterministic lower bound in Table VI.

or NBO over the author implementation in mean cost in all six panels,
not uniform. In dimensions 8 and 32, NBO has lower paired cost
th budgets; each multiplicity-adjusted sign-test value is .001465. In
ed values are 1 and .438, so these panels do not establish a robust
the declared test. Tripling the budget gives only small improvements
5812 to .5785, .4663 to .4630, and .3813 to .3799 respectively. Most
ently specified quadratic-feedback baseline has lower mean cost than
very panel. Its mean costs are .5121, .1740, and .08544 in dimensions
ned results support the stated NBO–SOC-MartNet comparison at the

the problem family into evidence of general superiority across economic budget points are not a matched-accuracy frontier, and a finite absolute lower bound is not enough to rank methods at a small prescribed regret.

FROM VERIFICATION TO RIGOROUS CONTROL AND NEURAL COMPUTATION

Our methods should be compared with methods that establish bounds, not only with classical methods but also with modern methods. Theorem 1 is an application of classical stopped verification to the problem of finding a lower witness. Its role here is to handle the actual liquidation trace without the need for a boundary-regular optimal value. Theorem 2 supplies a computation-tractable, and oracle allowance. Neither theorem claims that classical verification is sufficient for the problem.

Our consistent approximations with comparison provide convergence to vis-à-vis the framework of Barles and Souganidis (1991). Controlled Markov-chain models developed systematically by Kushner and Dupuis (2001); adaptive sparse-grid methods for high-dimensional dynamic models in Brumm and Scheidegger (2017). Such methods provide approximation schemes and their hypotheses. The finite telescoping of the approximation names the operator discrepancies needed for a continuous approximation and mesh difference does not supply them. The original-payoff certificates for continuous generators or exact Gaussian moments directly, with separately verified arithmetic remainders. They do not assert a convergence rate for the scheme.

Classical control methods also pair feasible decisions with bounds. Brown (2010) relax nonanticipativity with penalties and establish weak and strong duality for dynamic programs. Brown and Smith (2011) use related bounds in portfolio optimization with transaction costs. Our no-adjustment construction is a market-deflator witness and explicit stopped localization; it is not a claim that policy-plus-upper-bound is sufficient. Lasserre, Henrion, Prieur, and Trélat (2008) develop occupation-measure methods for polynomial nonlinear optimal control under their stated assumptions. The deterministic convergence theorem for the present logarithmic, stochastic, and explicit source bounds, interval polynomial conversions, Gaussian moment bounds are verified here instead.

Theorem 4 identify the model-specific constructions responsible for the bounds. The former couples a fitted generator witness to a small localization bound on every portfolio term before computing a bounded-source expectation. The rank-one action coupling and strong convexity of the remaining scalar problem are captured by convex-duality and branch-and-bound construction, not a generic method for nonconvex optimization. Proposition 3 preserves the nonquadratic relaxation of the exact comparison problem.

Verification and neural PDE solvers are important computational antecedents. Kushner (2015) analyze continuous-time policy improvement. Kim, Cho, Kim, and Song (2015) analyze rate iteration, policy-network, and residual errors in entropy-regularized

al-self equilibria, and games require their own evaluation operators
 their complete equations, assumptions, finite gain bounds, failures, and
 are preserved in the supplement. No new large-scale recursive or game
 from the additive-utility certificates above. The stochastic-trace bias
 izer counterexamples are preserved for the same reason: they identify
 aining loss cannot replace.

9. CONCLUSION

olish a positive minimum value of access to preference adjustment in
 onomy. They do so by combining a feasible learned policy with an
 upper bound on the restricted optimum, rather than by assuming either
 the restricted economy is enclosed within .00905 at the central state.
 uation is much more accurate than its present .10634 regret bound at
 e-policy objective remains a distinct numerical target.
 sults give operational ways to improve and audit these calculations:
 , stopped Gaussian-moment evaluation, market-deflator localization,
 , and a non-enumerative continuous-action oracle. The coupled com-
 as a rigorous absolute lower bound and a frozen multi-budget, multi-
 ord. Together these results sharpen the connection between neural
 ic conclusions while leaving every inherited derivation, experiment,
 e available for further review.

APPENDIX A: VERIFICATION AND COMPUTATIONAL ERROR

selection, stopped comparison, and the neural bridge

ON 1: For a fixed jet the three controls enter additively. The con-
 erivative $c^{-u} - v_x$ and second derivative $-uc^{-u-1} < 0$. If $v_x \leq 0$ it
 its unconstrained maximizer is $v_x^{-1/u}$. The adjustment objective is
 ictly concave for $k > 0$. The portfolio objective is $h_1p + h_2p^2/2$. A
 ojected vertex. A nonnegative h_2 gives a maximum at an endpoint,
 constant cases. Compactness and the stated tie rule give a feasible
 ese are control-space statements at a fixed jet, not parameter-space
Q.E.D.

1: Localize the process on interior compact sets and apply discounted
 r witness on each time slab. The discounted terminal witness plus
 ls the initial witness, the expected integral of $R_a U$, and the expected
 rd jumps. The latter sum is nonpositive. Boundary comparison gives
 ty of discounting and the deterministic bound $\tau \leq T$ therefore imply

$$+ b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T - t)\epsilon_U.$$

Lemma 2. The cover and representative intervals imply $|R_\pi v| \leq e$ and $v \leq e$. Since $\sup_a R_a v = R_\pi v + \Gamma \leq e + q$ and $R_\pi v \geq -e$. Apply Theorem 1 with $b = b$, $\epsilon_U = e + q$, and $\epsilon_L = e$. A derivative bound with respect to the chosen v moduli by integration along segments, or along a specified path inside the cell, a verified path-length radius must replace the Euclidean radius. The assertion follows directly from the bound and does not require a compact set. *Q.E.D.*

Part follows from positivity of the continuation operators: propagate the one factor δ , propagate the optimality error with one factor δ , and add the forward induction from terminal error b gives (8). The full state-dependent rule, and recursive-operator versions are reproduced in the preservation of this finite identity, a continuous interpretation additionally needs either compactness terms or an independent continuous verification route. The present work uses the latter.

A.2. Arithmetic convention

Constants specifying the economy are exact decimals. Stored network constants, and fitted coefficients are interpreted as exact binary64 numbers. Polynomial coefficient conversion use 70-digit interval arithmetic; bounds use at least 50 digits. Vectorized interval addition, subtraction, multiplication by a strictly positive interval, and pairwise summation round both with the adjacent binary64 number. An interval square may be enclosed by multiplication; this is conservative even when its lower endpoint is negative. Arbitrary approximation is treated as a rigorous interval merely because its result is rounded once. Exponentials in vectorized routines are replaced by analytic remainders; interval logarithms, trigonometric constants, and functions are handled in the multiprecision layer.

Residual claims concern the mathematical function represented by the stored floating-point implementation of an arbitrary neural policy. The constants are themselves stored constants, so their continuous implementation is exact. Neural-network rollouts are explicitly floating-point performance estimates. These are included in every numerical table.

APPENDIX B: ORIGINAL-ECONOMY FITTED WITNESSES

B.1. Global fallback and localized upper bound

$2)^2 + .1 \log x$, consumption and portfolio selection give $c = .8$ and $p = .8$. s is bounded above by $.000512/k \leq .001024$ for $k \geq .5$. The preference part is $.000512$. Consumption utility is maximized at the upper endpoint $u = 2.8$

artificial verification face.

ON 2: The derivatives of $G + F$ in x coincide with those of G . Thus
 n applies. The residual after maximizing over wealth is precisely (13).
) $+ \kappa h\}$, where $h = 1 - t$,

$$z = 8e^{\lambda(u-u_0)+\kappa h} \{-1 + h(-\kappa + \lambda\theta + .00125\lambda^2 - \rho)\}.$$

(20, 42), (160, 64) satisfy $\kappa = .2|\lambda| + .00125\lambda^2$. Both terms therefore
 nerators for every admissible control. Moreover $(\partial_t - \rho)C_\rho(1 - t) =$
 nonpositive residual inside the verification domain.

$= 0$, and $U = G$. At an actual wealth exit, $F \geq -8h$ implies $U \geq G -$
 preference face, one summand of Z is at least $8h$, so $U \geq G$. Stop first
 artificial face. If the artificial face occurs first, the remaining original-
 l by G under every admissible continuation, by the global fallback
 l stopped verification gives the asserted bound without modifying the
 opportunity set. *Q.E.D.*

2. Polynomial source enclosure and refinement

, where $A = 1.05$, $B = .55$, and $|z| \leq 1$. Let $D = (A^2 - B^2)^{1/2}$ and
 uniformly convergent expansion

$$\frac{1}{A + Bz} = \frac{1}{D} \left[1 + 2 \sum_{n=1}^{\infty} (-q)^n T_n(z) \right]$$

gree 16 bounded by $2q^{17}/\{D(1 - q)\}$. If $L = -\log(.8)$, approximate
 ne degree-eight Taylor polynomial of e^{LBz} . Its remainder is at most
 plying the two enclosures, with the relevant sup-norm bounds, gives
 elow 2.117×10^{-9} for $-.8^{-(u-1)}/(u - 1)$. This error is added to the
 fitted away.

$-.2q - .02k$ for $q \leq -.2k$, $q^2/(2k)$ for $|q| \leq .2k$, and $.2q - .02k$ for
 erval Bernstein coefficients enclose q and every potentially applicable
 ox touching a switching threshold includes both adjoining branches.
 ward threshold intervals; binary64 representation of $.2k$ cannot exclude
 ynomial lies between its minimum and maximum Bernstein coeffi-
 vision preserves this enclosure and is repeated on boxes whose upper
 escribed ceiling. The independent check of $F/h \geq -8$ uses the same

whose true residual lies strictly below a ceiling, Bernstein refinement
 iling, up to the explicitly accounted source and arithmetic allowances.
 tement about a fixed candidate with slack. It is not a theorem that
 independent ansatz, or arbitrary Adam training produces approximants

e stopping, $u_t = 2 + \mu(t) + .05W_t^u$, with $\mu(t) = \int_0^t \theta_r dr \in [0, .2]$. An requires $W_t^u \geq 12$ at some $t \leq 1$; a lower exit requires $W_t^u \leq -16$ and is $\mathcal{D}_{t \leq 1} \{|W_t^u| \geq 12\}$. The exponential martingale maximal inequality on each

$$p_{\text{exit}} \leq 2e^{-72} < 1.077 \times 10^{-31}.$$

ee diffusion is still stopped at exact first exit. Its rare-exit correction is not

$j = -\log c_j$. Before exit, $|r| \leq .8$ and utility is $-e^{L_j} e^{L_j r} / (1+r)$. Replace $\sum_{m=0}^{\infty} (-r)^m$ and the exponential by its degree-ten Taylor polynomial. The s at most $.8^{81}/.2$; the exponential remainder is at most $e^{.8L_j} (.8L_j)^{11}/11!$. cluding the reciprocal and exponential suprema, yields a uniform error for mial P_j . This bound holds on the whole original preference interval, not ture nodes.

lom variable with mean μ and variance v , its raw moments satisfy $M_0 = 1$,

$$M_n = \mu M_{n-1} + (n-1)v M_{n-2}.$$

s evaluated by this recurrence with $v = .0025t$, and the terminal quadratic has an exact Gaussian expectation. On each time rectangle the intervals , and the polynomial coefficients enclose the entire integrand. Outward e times interval length therefore enclose its integral. These ranges are estimates inferred from two discretizations. The three resolutions are es of the same integral.

ee-Gaussian polynomial calculation with the stopped economic payoff, preference exit. Off E the two paths agree through T . On E , the omitted nded using

$$-2)\| \leq \sqrt{p_{\text{exit}}} \|P_j(u_t - 2)\|_2, \quad \|P_j\|_2 \leq \sum_m |b_{jm}| \sqrt{M_{2m}}.$$

ents are bounded uniformly by evaluating their nonnegative coefficient and $v = .0025$. The same argument bounds the free terminal settlement, ment has magnitude below 8.1. Discounted adjustment cost omitted after . One valid total stopping allowance is thus

$$\overline{p_{\text{exit}}} \left\{ \max_j \|P_j\|_2 + \|G(u_1, x_1)\|_2 \right\} + p_{\text{exit}} (.02k + 8.1).$$

D.1. Cost ordering and value enclosures

3: For each policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing. The supremum is convex and nonincreasing. The two approximate

$$\geq A_2 - k_1 B_2 - \varepsilon_1, \quad A_2 - k_2 B_2 \geq A_1 - k_2 B_1 - \varepsilon_2.$$

$k_2 - k_1 > 0$ gives (17). Zero allowances give budget ordering for net prices. If π_k is optimal at k , feasibility at k_+ implies $V(k_+) \geq$ feasibility at k_- implies $V(k_-) \geq V(k) + (k - k_-)B(\pi_k)$. Substitute closure endpoints and intersect with the primitive budget range to obtain *Q.E.D.*

and follows from $V_k \geq J_k$ and $V_R \leq \overline{V}_R$. Its nonnegativity follows. Policy set is contained in the flexible policy set. Likewise $V_k - V_R \leq$ utilities compare two optimal values even though the feasible policies need not be optimal.

Cancellation of the portfolio and stopped localization

4: For Y as specified, quadratic covariation gives

$$\langle \tilde{z}_t x_t \rangle = (.04Y_t x_t - Y_t c_t)dt + \text{local martingale}.$$

It cancels the $-.06pYx$ covariation term for every adaptive p . Including and the preference quadratic yields

$$yc + .01y - .00005 - .004 \log(.5) + .0008(u - 2)^2 \leq S(u, y).$$

$-x$ or $u-y$ derivative. Correlation is retained in the transition law of the source potential.

and use the feasible consumption .35: its utility on $u \in [1.5, 2.6]$ is $\overline{.35}$, while $-.35y \geq -4.2$. The remaining terms sum to a nonnegative. Thus $S > -7.581 > -8$. For an upper bound discard $-yc$, maximize use $.01y \leq .12$, and bound the remaining terms by .0033. This gives stopped extension is bounded in $[-8, 0]$, so $-8h \leq F_R \leq 0$.

mean-Kac potential satisfies the conditional expectation identity for gral. Consequently $e^{-\rho r} F_R(h - r, U_r, Y_r) + \int_0^r e^{-\rho s} \tilde{S}(U_s, Y_s) ds$ is

fore, the factor $8h$ supplies at least $8h$ at its associated artificial face.
 $y \geq .2$,

$$W_0 - G = y(x - .5) + .1 \log(.5/x) \geq 0 \quad \text{NEO, R11 PRESERVATION SUPPLEMENT}$$

55

at $.5$ and has derivative $y - .1/x \geq 0$. At an actual wealth exit the source
 to $W_0 + F_R + Z_R \geq g$. At a terminal time it is at least G . At any artificial
 tion term pays for the entire possible source deficit, giving continuation
 ual preference exit cannot precede these inner preference faces. Invoke
 $r \leq G$ after an artificial exit and apply the stopped identity before it. This
 original adaptive consumption and portfolio controls. The value of Y_0 is an
 umenter, not an imposed wealth price or policy restriction. *Q.E.D.*

D.3. Validated evaluation of the restricted witness

endent standard Gaussians and put $r = v^2$. At the central state,

$$v\{-.25z_1 + \sqrt{.9375}z_2\}, \quad l = \log(1.55) - .025v^2 - .3vz_1.$$

on is integrated with weight $2ve^{-.04v^2}\varphi(z_1)\varphi(z_2)$ on $v \in [0, 1]$. The box
 istrictly inside the stated u, y verification rectangle under this transformation.
 ity is at most $4e^{-18}$. Since the extended source lies in $[-8, 0]$, omission
 $[-8C_\rho(1)4e^{-18}, 0]$, whose width is less than 4.778×10^{-7} . The initial
 s less than 2.200×10^{-9} .

the Hessian used by the evaluator is explicit. Let $c = \text{proj}_{[.05, .8]} e^{-l/u}$,
 and let χ be one for an interior consumption solution and zero for a strictly
 Across a switching box use $\chi \in [0, 1]$. With $Q = c^{1-u}/(1-u)$,

$$\begin{aligned} \frac{\{1 - L(u-1)\}}{(u-1)^2}, \quad Q_{uu} &= -\frac{e^{L(u-1)}[\{L(u-1)-1\}^2+1]}{(u-1)^3}, \\ .0016(u-2), \quad S_l &= -yc + .01y, \\ \chi ycL^2/u + .0016, \quad S_{ul} &= -\chi ycL/u, \\ \chi yc/u + .01y. \end{aligned}$$

inuously differentiable across clipping surfaces; its gradient is locally
 eewise second derivatives give a valid integral Taylor bound across such a
 the displayed (u, l) transformation by the chain rule to obtain a Hessian in
 mediate expression is enclosed on the entire box. Bounds on the absolute
 he usual one-half diagonal and full mixed-term Taylor remainder.

near Taylor terms are integrated exactly for their polynomial approximation
 ian weighted moments through order two use 111 terms of $e^{-z^2/2}$ and

. Only an upper endpoint is used to bound the restricted optimum. The upper endpoint with the independently certified restricted-policy lower bound gives an interval of width less than .009047. The distinct width below .000320 is not the primal–dual gap.

CONTINUOUS ACTION AND ABSOLUTE-VALUE CERTIFICATION

the scalar global action bound and interval neural jets

5: For any action with $\sum_i a_i = z$ and any real λ ,

$$f_i(a_i) = \sum_i \{f_i(a_i) - \lambda a_i\} + \lambda z \geq D(\lambda) + \lambda z.$$

$.3 \sin z$ is bounded below by $-.3$. For $z = z_0 + e$, $|e| \leq r$, its quadratic

$$.3 \sin z \geq \lambda z_0 + .3 \sin z_0 + (\lambda + .3 \cos z_0)e - .15e^2,$$

scalar f_i has second derivative $(2 + .2 \cos 2s)/d \geq 1.8/d = m$. Thus f_i is convex. Its tangent quadratic at any feasible s_0 gives a lower bound. At an endpoint, only a feasible descent direction is relevant: an outward-pointing direction contributes zero, while an inward-pointing direction contributes $-m$. This follows either by minimizing the tangent quadratic over the displacement, or by the sign of the linear term. The stated lower bound remains valid if the displacement goes beyond the opposite endpoint, because relaxing the displacement can

only improve the lower bounds is no larger than $D(\lambda)$. A feasible action gives a direct interval evaluation. The scalar interval $[-d, d]$ covers every division, pruning only a box whose lower bound exceeds a feasible minimum surviving lower bound preserve a global enclosure. Root finding is not validity. Finally $|H_p(a) - H_{\hat{p}}(a)| \leq 2 \sum_i \delta_i$ on the action box. The certified and true minimum lower can each worsen by at most this amount, $Q.E.D.$

With stored weights, interval automatic differentiation propagates both derivatives. A $\tanh$ layer is evaluated as $(e^{2z} - 1)/(e^{2z} + 1)$ with a value and derivative $1 - \tanh^2 z$, all in multiprecision intervals. The derivative is differentiated as its mathematical formula, including the exact $\sqrt{d}$, not as a constant. This gives a componentwise interval for the derivative. Subtracting the stored floating-point gradient gives the δ_i . The two certified eight- and sixteen-dimensional learned checkpoints and the same error estimates. The summed errors are below 4.584×10^{-17} and 7.877×10^{-17} for local jet checks, not whole-state residual bounds.

$$\Delta v - \frac{d}{r_0} |\nabla v|^2 - .3 = 0, \quad v(1, x) = q_d(x), \quad \sigma^2 = .32.$$

and use the logarithmic heat representation. At $X_0 = 0$ it gives

$$) = -.3 - \lambda \log \mathbb{E} \exp\{-q_d(\sigma Z)/\lambda\}, \quad Z \sim N(0, I_d).$$

and coercive quadratic terminal terms give finite positive expectations. The on is smooth before the terminal date and has at most quadratic growth. of its verification identity, or truncation followed by monotone Gaussian the relaxed lower bound without assuming a smooth solution of the original lem.

re for the quadratic terminal component. Its Gaussian normalization con-

$$\frac{.25r_0}{r_0 + 4} + \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3.$$

sian law the scalar sum divided by $\sqrt{d}$ has mean $m = 2\sqrt{d}/(r_0 + 4)$ and $r_0 + 4$). Orthogonal coordinates disappear, leaving

$$+ \frac{\sigma^2 r_0}{4} \log(1 + 4/r_0) - .3 - \lambda \log \mathbb{E} \exp\{-(.2/\lambda) \cos(m + sZ_1)\}.$$

ation in the proposition. An upper enclosure for the positive scalar ex-
er enclosure for v , because $-\lambda \log$ is decreasing. That lower endpoint is
and for the original value. *Q.E.D.*

gral let $\eta = .2/\lambda$ and $R = \lceil \sqrt{2(\eta + 50)} \rceil$. The omitted Gaussian contri-
an $2e^{\eta - R^2/2}$. Interval ranges of the positive integrand times cell lengths
in $[-R, R]$. The successive 1,024, 4,096, and 16,384-cell calculations add
wance and then apply the decreasing logarithm with outward rounding.
e integration uncertainty in the relaxed value. Relaxing the action bound
can leave a large structural gap even when this quadrature width is small.

APPENDIX F: REPLICATION AND SCOPE LEDGER

mic certificate is independent of the inherited killed-Euler discretization.
verified continuous-generator residual, an explicit source-polynomial
-trace inequality, and exponentially bounded artificial-face continuation.
exact deterministic wealth, interval Gaussian polynomial moments, an
imation remainder, outward time integration, and an exact-exit probability
ted upper cancels continuous portfolio controls algebraically and evaluates

does not create an outward policy-regret certificate. Paired path errors, -update clock overshoot, setup, and warmup are recorded separately. erns the frozen joint training-and-audit pipeline under independent certify an underlying PDE or imply that the six panels represent six ilies.

erated from raw JSON or path records; display endpoints are rounded on. The frozen external protocol, all tuning trials, all 72 holdout pairs, th arrays are retained. The validation program checks coverage of the initialization hashes, pinned author revision, finite outputs, refinement ility, the positive access lower bounds, and the explicitly false flexible complete R8, R4, and delivered R5 textual and numerical material is on supplement, and earlier source files remain byte-for-byte unchanged. a historical manufactured-payoff test is used as new original-economy

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ement preserves the complete R8 manuscript and proofs, the complete R4
d derivations, and all six delivered R5 numerical tables without altering their
also specifies the R9 replication and evidence map. Recursive utility, temporal
ic games, Merton benchmarks, stochastic-trace identities, finite safeguards,
payoff tests, and unsuccessful numerical attempts remain available for review
inal-economy welfare bounds, constructive certificates, and frozen external
e developed in the R9 main paper and its mathematical appendix.

PART I: READING AND REPLICATION MAP

cript is `ECTA_R9.tex`; its Appendix proves the new results. The accom-
resses the September 22, 2026 R8 numerical-methods report at commit
work is isolated on the R9 constructive-certification branch. Earlier main
have not been advanced by this revision.

directory contains the original-payoff actor and moment evaluator, fitted-
d Bernstein verifier, restricted market-deflator verifier, scalar global action
al-jet evaluator, absolute-value relaxation, and frozen external-comparison
les are generated from their retained JSON records. Every new trained
all 4,096-path samples for each of the 72 holdout pairs are retained. Source
aes, exact commands, scope checks, and failed-pilot status are in the R9
tion records.

erial is historical but preserved in full. References to a preceding revision, its
urrent target, or its conclusions refer to that historical record. In particular,
payoff result is not evidence for original-payoff precision, and its six-seed
is superseded for current inference by the frozen R9 holdout. Neither
ilently rewritten. Historical finite safeguards remain distinct from the new
cle. The R4 recursive, temporal-self, and game derivations retain their own
ilibrium concepts; no new large-scale extension result is inferred from the
utations.

cost effect from an approximation error. The difficulty is compounded by the fact that the boundary is an actual termination event rather than an artificial edge of a

Neural Bellman Operators (NBO) as a model-based framework for policy learning, feasible policy improvement, and verification of the implemented policy. This application combines stochastic risk aversion, consumption, portfolio choice, and preference adjustment. Its stopping and settlement contract is specified as a martingale. Our mathematical results identify three distinct objects: an exact policy-optimal policy, a neural implementation, and an independently evaluated policy. None of these objects lie within the error bounds.

We construct a continuous-time verification pair. A supersolution controls the value of the policy, whereas a subsolution controls the payoff of the particular policy. The difference, together with explicitly signed residual and boundary conditions, forms the argument. This argument uses smooth test functions, not smoothness of the optimal policy. We do not require continuity of the optimal value at an irregular controlled boundary. We provide both a direct continuous-generator implementation and an operator-splitting approach for a discretized implementation. The latter separates time, action, and state, and handles exit, inner-solve, and arithmetic errors; a refinement difference is

related to computational accuracy to an economic comparative static. Raising the stopping tolerance lowers the optimal discounted adjustment budget. For policies that are nearly optimal, the possible reversal in this ordering is bounded by the change in price. Value enclosures also bound optimal policies. Value enclosures and bound the welfare gain from access to the adjustment budget. These results make the required numerical precision depend on the economic parameters and on a common uncalibrated stopping tolerance.

Our analysis has complementary components. We re-evaluate the retained components of the full finite-action safeguard, and separate time, state, and action. We provide outward-interval verification pairs for their deployment in the original problem. These pairs are valid but too conservative to resolve the stated economic problem. Instead, explicitly modified-payoff experiment trains a bounded actor on the problem, covariance, and exit geometry and obtains an all-state continuous-time solution to a coupled nonconvex control problem in eight and sixteen dimensions. We compare the budget with the pinned author implementation of SOC-MartNet. The results show that neural controls, no analytical greedy-action labels, paired independent evaluation, and a six-seed design. Its outcome measures realized cost at that budget, and the distance from the optimum.

Our PDE evaluation, and their combination have important antecedents. We study continuous-time policy improvement. Kim, Cho, Kim, and Lee (2020) study evaluation and improvement for entropy-regularized stochastic control problems. We study policy-network, and residual errors. The updated interior analysis of Kim et al. (2026) treats stationary second-order equations with bounded-domain

sophisticated temporal selves, and dynamic games remain part of the not share an identical evaluation equation or an identical optimality criterion. Equilibrium conventions, proofs, and computational records, and gain bound changes when the evaluation operator is nonlinear or when rivals elementary material preserves the complete preceding R4 manuscript and the current manuscript integrates the R5 evidence and the new continuous al-comparison results.

AN ECONOMY WITH COSTLY PREFERENCE ADJUSTMENT

(u, x) , where u is relative risk aversion and x is financial wealth. An a progressively measurable process $a = (c, \theta, p)$ taking values in

$$A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

the state satisfies

$$+ .05 dZ_t^u, \quad (1)$$

$$+ .06p_t)x_t - c_t]dt + .2p_t x_t dZ_t^x, \quad d\langle Z^u, Z^x \rangle_t = -.25 dt. \quad (2)$$

, the open state domain is $D = (1.2, 2.8) \times (.5, 2)$, and the discount rate ess is stopped at $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$. Its running payoff and

$$= \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad k > 0, \quad (3)$$

$$= -.02(u-2)^2 + .1 \log x, \quad g(t, u, x) = G(u, x) - 8(1-t). \quad (4)$$

$$\left[e^{-\rho(\tau-t)} \ell_k(S_\tau, a_\tau) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V_k(t, s) = \sup_{\pi} J_k^\pi(t, s). \quad (5)$$

state dynamics, stopping rule, and settlement do not depend on k . Bounded Lipschitz state coefficients give a well-defined stopped state equation for low. Bounded payoffs on the stated domain give finite values. The cardinal consumption units are held fixed across comparisons. A transformation of the controlled state u is generally a different economy.

ion fee is a specified liquidation covenant, not an estimated preference tion there is no subsequent wealth injection or continuation consumption. sential: a constant preference diffusion cannot be confined to the stated

evaluation pair in Section 4 uses inequalities on the actual stopped trace in this setting.

tion v , define

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$$2 + .06p)x - c]v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux}, \quad (6)$$

$$+ \ell_k - \rho v. \quad (7)$$

is $\sup_{a \in A} R_a V_k = 0$ in the appropriate solution sense. All covariance and derivative, are retained. A negative shock correlation by itself does the optimal portfolio response, and dividing a re-solved V_u by k does ordering of adjustment policies.

complete control selection for a fixed jet: *Let v have the derivatives in $R_a v$ is obtained separately in the three control components. If $v_x > 0$, otherwise $c^* = .8$. Also $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. Write*

$$h_1 = .06xv_x - .0025xv_{ux}, \quad h_2 = .04x^2v_{xx}.$$

$\text{proj}_{[-.5, .8]}(-h_1/h_2)$. If $h_2 \geq 0$, choose an endpoint maximizing $h_1p +$ rule.

cludes nonpositive marginal wealth value, zero curvature, convex port- strained optima. It removes action discretization from a continuous- particular model. It is not a closed-form optimizer of the killed-Euler, Bellman target, and it is not a global convergence theorem for shared underlying derivative rule was present in the retained derivations; the tests every curvature branch explicitly.

EVALUATION, IMPROVEMENT, AND FINITE CORRECTION

policy evaluation solves $R_\pi v = 0$ with the economic stopping contract. feasible control maximizing the Hamiltonian of that fixed evaluation. denoted $\mathcal{E}$ and $\mathcal{I}$, and exact policy iteration is $\pi_{j+1} = \mathcal{I}(\mathcal{E}\pi_j)$. If a , $R_{\pi_{j+1}}(\mathcal{E}\pi_j) \geq 0$. Verification then gives $J^{\pi_{j+1}} \geq J^{\pi_j}$. ment requires more than this monotonicity. Under compactness of ty of evaluation into a precompact $C^{1,2}$ jet class, closure under im- the chosen selector at cluster points, and comparison for the optimality pint value is optimal. These are hypotheses on the exact operators. successful stochastic-gradient training. Appendix B gives the limiting distinctions between parameter stationarity, policy convergence, and

3.2. A certificate for the declared finite economy

$P_n^a w$, where P_n^a is a positive substochastic continuation matrix and $\delta =$
 yoffs are included in r_n^a . Let $\hat{\pi}$ be a feasible policy and v_n a critic, with
 at most b . Define

$$\|v_n - \mathcal{T}_n^{\hat{\pi}^n} v_{n+1}\|_\infty, \quad \eta_n = \|\max_a \mathcal{T}_n^a v_{n+1} - \mathcal{T}_n^{\hat{\pi}^n} v_{n+1}\|_\infty.$$

ite gain accounting: *On this finite economy,*

$$\|V_{h,n}^* - V_{h,n}^{\hat{\pi}}\|_\infty \leq 2\delta^{N-n}b + \sum_{j=n}^{N-1} \delta^{j-n}(2e_j + \eta_j). \quad (8)$$

ently evaluated with the exact finite terminal data, $e_j = b = 0$. A state-
 btained by setting $B_N = 0$ and

$$g_n(s) + \delta \max_a (P_n^a B_{n+1})(s), \quad g_n = \max_a \mathcal{T}_n^a V_{h,n+1}^{\hat{\pi}} - V_{h,n}^{\hat{\pi}}.$$

$$_n \leq B_n.$$

must range over the entire declared candidate economy. For the retained
 125-point action grid together with the frozen continuous neural proposal
 e. The certificate does not cover all continuous controls merely because
 ted in a neural network. Floating-point computation of the finite envelope
 from the real-arithmetic statement of the theorem.

3.3. The safeguarded algorithm and its cost

gorithm first evaluates the raw policy, computes all candidate gains, and
 ain above a specified threshold. It re-evaluates the corrected policy and
 ported envelope meets the target. In the new re-audit the threshold is

$$\gamma = \frac{\varepsilon}{2 \sum_{n=0}^{N-1} \delta^n}. \quad (9)$$

sufficient condition is $M_j^{-1} \sum_m g_{j,m} = o(\gamma_j)$.

mechanism, not evidence that longer training has already made the
 66 gible. It distinguishes a useful limit statement from an unmeasured
 y audit still enumerates the candidate actions. With N dates, S lattice
 s, the implementation uses $NS(A + 2)$ value-action evaluations per
 valuation and the extra proposal, and $NS(A + 1)$ envelope-action
 ction uses four shock branches and bilinear interpolation. There is no
 n for this safeguard.

VERIFICATION OF THE CONTINUOUS ECONOMY

d in continuous time must be stated before it can be certified. For the
 we use their feasible bilinear interpolation on the exact uniform state
 erval the feedback function is frozen in time, but is evaluated at the
 y: $a_r = \pi_n(S_r)$. This is not a simulation that holds the initial action
 nes a bounded, piecewise-time, Lipschitz state-feedback rule for the
 ics and exact first-exit contract.
 $\rho^h)/\rho$, with $C_0(h) = h$.

missible verification witnesses: U and L are bounded, continuous
 and $C^{1,2}$ in the interior of each time slab. Their derivatives permit
 and the stopped expectations are uniformly integrable. The policy π
 ed state equation. At interior slab interfaces either the witnesses are
 $U(t^-, s) \geq U(t^+, s)$ and $L(t^-, s) \leq L(t^+, s)$.

desired continuous verification pair: Under Assumption 1, suppose
 ain

$$\sup_{a \in A} R_a U \leq \epsilon_U, \quad R_\pi L \geq -\epsilon_L,$$

il and exit trace $U \geq g - b_U$ and $L \leq g + b_L$, where all four constants

$$U(t, s) \leq U(t, s) + b_U + C_\rho(T - t)\epsilon_U, \quad (10)$$

$$U(t, s) \geq L(t, s) - b_L - C_\rho(T - t)\epsilon_L, \quad (11)$$

$$-J^\pi \leq U - L + b_U + b_L + C_\rho(T - t)(\epsilon_U + \epsilon_L). \quad (12)$$

elong to $C^{1,2}$ or be continuous at every controlled boundary point.

d not be neural networks, and need not have the same approximation
 can propose either witness, but verification requires an enclosure

4.1. An explicit original-payoff construction

osition 1 gives $c = .8$, $p = .8$, and $\theta = -.02(u - 2)$ when $k = 2$. A 160-by-
 partition bounds $\sup_{s,a} R_a G$ above by $-\alpha$, with $\alpha > 0$. For each frozen
 und $R_\pi g$ below by $-\beta$ on every state cell and time slab. Consequently,

$$U = G - \alpha C_\rho(1 - t), \quad L = g - \beta C_\rho(1 - t) \quad (13)$$

witnesses. Since $\alpha < 8$, $U \geq g$ on the stopped trace, while $L \leq g$. Their
 s

$$8 + (\beta - \alpha)C_\rho(1). \quad (14)$$

encloses logarithms, powers, products, and all bilinear-action component
 outputs are frozen binary64 constants interpreted as exact real numbers;
 exact uniform rational grid. The returned bound is rounded outward again
 binary64. This certifies the specified real-arithmetic feedback, not every
 entation of the stochastic differential equation.

TABLE I

ORIGINAL-PAYOFF CONTINUOUS NDU WITNESS BOUNDS

Interval cells	α	β	Regret upper bound
4320	0.864832	7.078878	14.091405
8640	0.864832	1.211230	8.339561

odal policies, deployed as continuously reevaluated bilinear feedback on each time slab.
 l continuous Brownian model and exact first exit using outward interval arithmetic. Both
 bounds exceed the .01 precision target.

bounds in Table I are numerically wide. They replace an absent continuous
 cuted, valid certificate, but do not establish regret below .01 or .05. In
 a substitute for a sufficiently narrow welfare enclosure. The theorem and
 ke that distinction testable rather than hiding it in a missing discretization

4.2. A complete discretization accounting route

route to continuous verification. Let $\mathcal{B}_n^*$ be the exact continuous-time
 igroup over one decision interval, allowing all admissible within-interval
 e the exact semigroup for the declared continuously implemented policy.

$$\| \delta_0 \|_\infty \leq B_h + \delta^* (b^* + b^*) + \sum_{n=0} \delta^n (\kappa_n + \kappa_n^*), \quad (15)$$

and finite optimal–policy gap. Each κ can be enclosed by a declared chain separating time freezing (including within-step feedback), restriction of , shock quadrature, first exit and settlement timing, inner optimization, is valid when each adjacent operator difference is bounded on the function. A null term is unknown, not zero. A coarse–fine solution none of the uncomputed tails. Nor does a regular-domain scheme ply at an irregular controlled exit boundary. The direct generator pair the errors by auditing the continuous implemented policy itself; it does of the training scheme vanish.

NUMERIC COMPARISONS AT FINITE NUMERICAL ACCURACY

suppress it from the notation. For each feasible policy write

$$-t) \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \Big], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

$$) - k B(\pi)\}, \text{ and } 0 \leq B(\pi) \leq .02 C_\rho (T - t).$$

ment costs and approximate optimality: *If the feasible policies and independent of k , V is convex and nonincreasing. For $k_1 < k_2$, optimal $B(\pi_1)$. More generally, if $V(k_i) - J_{k_i}^{\pi_i} \leq \varepsilon_i$, then*

$$B(\pi_2) - B(\pi_1) \leq \frac{\varepsilon_1 + \varepsilon_2}{k_2 - k_1}. \quad (16)$$

y $|\widehat{B}_i - B(\pi_i)| \leq \xi_i$, their possible reversal is bounded by the right side differentiability point with an attained optimizer, $V'(k) = -B(\pi_k)$.

is sharp without further restrictions on the feasible policy set. The counted adjustment expenditure, not pointwise values of $\theta(t, u, x)$. The ment is an economic theorem for the original continuous contract; the st still establish sufficient accuracy to quantify its magnitude.

due intervals, budgets, and access to adjustment: *Suppose $L(k) \leq c < k_+$. Every attained optimal adjustment budget at k belongs to*

$$\frac{k) - U(k_+)}{k_+ - k} \Big\}, \min \left\{ .02 C_\rho (T - t), \frac{U(k_-) - L(k)}{k - k_-} \right\} \Big]. \quad (17)$$

enclose the flexible and the restricted $\theta \equiv 0$ values, the value of access $[L_F - U_R, U_F - L_R]$, intersected with $[0, \infty)$.

distinguish numerical validity, achieved precision, and comparative per-al-payoff NDU certificate in Table I is an outward interval bound for a The finite gain envelopes in this section are float64 calculations for their economies. The learned-geometry experiment has a rigorous continuous deliberately modified running payoff. The external comparison estimates npling uncertainty; it does not know the optimal value. These distinctions etation of the results, not alternative names for the same error. panel was executed from source commit 14ecbd48 in a clean Python 3.11 Torch 2.10.0, NumPy 2.3.5, SciPy 1.17.0, and mpmath 1.3.0. Numerical read. The ledger retains the processor description, exact versions, source e hashes, checkpoint weights, rollout arrays, all seed-level records, and generation reads those committed records and fails on missing cells. The are development checks, not extra independent confirmatory seeds.

preserving a learned policy through thresholded correction

vo retained policies without retraining or selecting a new policy using the he twelve-date seed-10 policy had a raw localized envelope of .0922420. sitive-gain safeguard replaced 54.6602 percent of interior state-date de- a .00544535 envelope. The threshold in (9), at target .05, replaces only tains a .00682279 envelope. At target .01, the correction share is 4.9350 pe is .00676060. Thus most of the earlier replacements were unnecessary red finite-economy tolerance in this sample.

TABLE II
THRESHOLDED CORRECTION OF RETAINED NDU POLICIES

t	Raw bound	Final bound	Override (%)	Audits	Seconds
5	0.042805	0.042805	0.000	1	0.472
1	0.042805	0.003278	8.796	2	0.849
5	0.092242	0.006823	0.875	2	1.641
1	0.092242	0.006761	4.935	2	1.630

velopes, not continuous-model certificates. Seconds include all listed audits and envelope t worker. Each audit still enumerates the full 125-action grid plus the proposal. Historical training time is not added across different environments.

addresses the role of correction rather than concealing it. The corrected raw rule, and every audit still enumerates 125 grid actions plus the neural elve-date runs, two audits use 2,362,200 value-action evaluations and ction evaluations, each with four transition branches. The new policy is not

ch correction shares vanish, but this two-policy exercise does not
 ose convergence curve. Neither its finite envelope nor its smaller
 e continuous NDU error budget.

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Separating time, state, and action approximation

roximation factor at a time. The time experiment fixes a 25×31 lattice
 component; the state experiment fixes 48 dates and five action points;
 es 48 dates and the 25×31 lattice. All retain the original four-sign
 t-segment liquidation treatment. The latter two approximations are not
 his panel.

TABLE III

ONE-FACTOR NDU APPROXIMATION DIAGNOSTICS

V	State nodes	n_a	Value	Exit prob.	h^2/Δ
4	25×31	5	-1.451486	0.104315	0.1067
8	25×31	5	-1.549787	0.193358	0.2133
6	25×31	5	-1.608530	0.227492	0.4267
8	25×31	5	-1.549787	0.193358	0.2133
8	37×46	5	-1.444048	0.151095	0.0948
8	49×61	5	-1.391646	0.102389	0.0533
8	25×31	3	-1.606541	0.245260	0.2133
8	25×31	5	-1.549787	0.193358	0.2133
8	25×31	9	-1.536711	0.182784	0.2133

uate $(u, x) = (2, 1.25)$. n_a is the number of nodes per action component. Four-sign
 at liquidation are unchanged; these differences do not bound continuous-model error.

ce is not evidence of convergence. Its central value moves from
 -1.60853 at 96 dates, while the exit probability rises substantially.
 fixed makes h^2/Δ increase, rather than shrink; the experiment ex-
 ead of fitting an unjustified time order. Conversely, refining the state
 the value from -1.54979 to -1.39165 . Action refinement improves
 the three approximation factors cannot be conflated. The operator-
 lose to machine precision; agreement of two implementations of the
 t an enclosure of its continuous-model bias.

5 dates, a 49×61 lattice, and nine points in each action component,
 stment-cost economy is re-solved. Table IV shows the same ordering
 e as Theorem 3, with an active preference margin. This is a stronger
 n the earlier five-point panel, but still not a verified continuous welfare
 and quadrature terms in (15) remain unbounded by these calculations.
 hardson remainder nor a continuous policy-convergence rate is inferred

learned actor on the full NDU control and exit geometry

continuous verification transfers from an analytic example to an actually
 n (2), the three-dimensional action box, correlated shocks, state domain,
 y. Change only the running payoff to

$$\tilde{\ell}(t, s, a) = \ell_2(s, a) - \sup_{b \in A} R_b g(t, s). \quad (18)$$

s independent of the chosen action, but depends on state and time. Thus
 optimal behavior relative to the original-payoff economy; it is not an
 e normalization. In the modified problem $\sup_a \tilde{R}_a g = 0$ and the explicit
 (.8, -.02(u - 2), .8). Itô verification proves that the modified value is g
 ping contract.

parameter actor with constant bounded sigmoid consumption and portfolio
 -tanh adjustment output. The actor maximizes the Hamiltonian of the
 no optimal-action labels. Six declared seeds each receive 4,000 Adam steps
 d 256 random states per step. This is a controlled verification experiment,
 apacity or high-dimensional scaling claim. The witness is known rather
 r and all of its achieved errors are learned.
 ghts, the control gap is bounded over the entire state domain by outward
 ne consumption gain is bounded using

$$\int_c^{.8} z^{-u} dz - .1(.8 - c)/x \leq (.8 - c)\{c^{-2.8} - .05\},$$

ne exact quadratic difference, and the adjustment gain equals $(\theta + .02(u -$
 3-cell interval partition in u encloses the latter. The full regret is bounded
 ummed action-gap enclosure. The time discretization of a rollout and a
 probability do not enter this certificate, because its proof is for the original

he .01 modified-problem target, with continuous regret bounds between
 2263. Table V separately reports componentwise control errors rather than
 cy as policy sup-norm accuracy. This is the desired transfer test with a
 original first-exit/control geometry. It does not identify the original NDU
 ghten Table I merely by sharing the state equation.

0112212	0.0006273	0.0000635	0.0006387	2.794
0099308	0.0005551	0.0000334	0.0007077	2.780

ward interval, arithmetic and the same state, action, covariance, and first-exit geometry the running payoff is manufactured. Four actor parameters are trained for 4,000 steps without optimal-action labels; the critic witness is known.

nonconvex control and an external author implementation

does not use the Cole–Hopf quadratic-control model or regress to an starting from $X_0 = 0$ in $\mathbb{R}^d$, minimize

$$\left[\begin{array}{c} \vdots \\ \tau_1 \end{array} \right], \quad dX_t = 2a_t dt + \sqrt{2}(.4)dW_t, \quad a_t \in [-1, 1]^d, \quad (19)$$

$$f(a) = d^{-1} \sum_{i=1}^d (a_i^2 + .1 \sin^2 a_i) + .3 \sin \left(\sum_i a_i \right), \quad (20)$$

$$q(x) = d^{-1} \sum_i (x_i - .5)^2 + .2 \cos \left(d^{-1/2} \sum_i x_i \right). \quad (21)$$

control components. For example, at an interior vector whose sum is running cost has a negative direction along the all-ones vector for both the problem is genuinely nonconvex in the action. No closed-form control labels are supplied to either method, and the quadratic exponential eliminate this Hamiltonian.

or is the unmodified `SOCMartNet.train` routine in the authors’ `SOCMartNet`, pinned at commit `991ea8dd`. Its complete module hashes manifest. The integration consists of model and network adapters and after a completed update at the time budget. The authors’ solver is like PINN comparator. The pinned implementation uses a Hessian-free; we therefore use constant diffusion in this comparison. This is a mented interface used here, not an assertion that the published SOC-odes volatility control.

identical initial actor and critic weights within each seed. Both use width 48, bounded actor outputs, a terminal-exact critic, 16 training CPU thread. NBO uses three evaluation updates per improvement s, four Gaussian successors, and Adam with learning rate .003. SOC-onal test network, RMSprop updates, and the declared martingale and or/value learning rate is .003/ $\sqrt{d}$, the test-network learning rate is .01, r settings are fixed in the protocol. The test network has 600 outputs; p costs are reported rather than excluded from the method description.

TABLE VI NONCONVEX-CONTROL COMPARISON AT A FIXED TRAINING BUDGET			
NBO cost	SOC cost	Difference	Paired 95% interval
0.59567	2.31352	-1.71785	$[-2.13823, -1.29747]$
0.48067	2.72799	-2.24732	$[-2.51288, -1.98176]$

SOC-MartNet realized cost at 128 audit steps; negative favors NBO. Each dimension has six audit seeds, 4,096 common-random-number paths per pair, and ten seconds of training per dimension. The interval describes across-seed uncertainty, not optimality loss. Maximum paired path standard errors are 0.01098, 0.00900, respectively.

TABLE VII MEASURED TRAINING AND SETUP RESOURCES			
Method	Train (s)	Setup + train (s)	Outer updates
NBO	10.003	10.692	1536.7
SOC-MartNet	10.012	10.015	465.3
NBO	10.004	10.689	1477.3
SOC-MartNet	10.012	10.014	458.0

over six seeds. Identical actor/critic parameter counts within each dimension: 3,224/2,881 at $d = 16$. SOC-MartNet also trains its 600-output test network. Paired policy audit means are 10 seconds. Outer-update definitions differ across algorithms and are not equal units of work.

At dimension sixteen, the paired mean NBO-minus-SOC cost is -1.71785 , with a six-seed 95% interval of $[-2.13823, -1.29747]$. At dimension sixteen it is -2.24732 , with interval $[-2.51288, -1.98176]$. Thus the new evidence favors NBO under the specified adapter and problem. This is a demonstrated benefit in a coupled nonconvex problem, in contrast to the published SOC-MartNet hardware or learning curves, or a statement that SOC-MartNet is near optimal.

4.3. Uncertainty, mesh diagnostics, and retained failures

The primary audit uses 4,096 paths with common Brownian increments for the two independent seed for each training seed. The primary outcome uses 128 time steps to audit aggregates adjacent increments, so its difference from the finer audit is a mesh diagnostic. Table VI reports the maximum paired path standard error over six seeds. The interval is based on six independent training/audit runs. This approximation rather than a distribution-free coverage claim. Paths are not independent across training runs.

three twelve-date raw policies failed. Its recursive raw policies missed eight/sixteen/thirty-two-dimensional quadratic-control NBO means the in-house PINN-PI comparator. Those results are neither relabeled d after observing the new comparison. The R4 Merton, nonlinear ursive, temporal-self, game, and coupled linear-quadratic calculations, incomplete runs, are reproduced in the preservation supplement. The changes the tested model and comparator; it does not retrospectively ults.

RELATION TO NEIGHBORING NUMERICAL METHODS

relevant methodological axes. Its entries summarize the cited analyses parison; they are not a ranking by paper title. In particular, an interior nknown outer data, a stopped-control regret theorem with a contractual didate-action gain envelope address different mathematical objects.

TABLE VIII
RELATED METHODS AND THE OBJECTS COMPARED

Principal mechanism or guarantee	Relation to the present evidence
Policy updates for entropy-linearized HJB equations; iteration, policy-network, and PDE residual errors	Neighboring policy-evaluation/improvement analysis; not the external implementation tested here
Stationary interior error control with dual and outer-information contributions under stated regularity conditions	Clarifies what interior collocation can control without supplying the stopped NDU boundary data
Policy iteration and reusable operator approximation across terminal data	Representation reuse is a different performance target from one trained policy at a fixed budget
Stingale training of value and control without an explicit Hamiltonian in- m	Pinned author routine tested on coupled constrained nonconvex controls in $d = 8, 16$
Separated evaluation, improvement, independent gain or witness verification	Continuous stopped-model theorem, explicit economic error calibration, finite correction costs, and paired author-code experiment

Kim (2026), Kim, Kim, Kim, and Cho (2026), Lee and Kim (2025), Cai, Fang, and benchmark uses constant diffusion and a ten-second training budget. Its scope does not general ranking for recursive utilities, games, or higher dimensions.

g a method, the distinction between state dimension and action dimension- e representations avoid a tensor state grid during the high-dimensional

$= 1 - \gamma$, and $bV > 0$. The retained Epstein–Zin normalization is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (22)$$

logarithmic limit when $a = 0$; see Epstein and Zin (1989), Duffie and
ation is $V_t + \mathcal{L}^{c,p}V + f(c, V) = 0$, not an additive-reward equation with
eters renamed. The transform $V = \exp(W)/b$ enforces the sign domain.
margin for bV , bounds on consumption, and a monotone implicit step are
or a finite Lipschitz bound on the evaluation operator.

extends beyond affine evaluation. If the action operators $\mathcal{T}_n^a$ are monotone
nitz with constants L_n on a verified continuation-value interval, then (8)
aced by $\prod_{m=n}^{j-1} L_m$, and the terminal factor by $\prod_{m=n}^{N-1} L_m$. Appendix D
A sign-enforcing architecture alone does not verify the required compact
ity.

record retains the stationary homothetic anchors $p = .3$ and $c/x = .0455$
eters, the executed implicit recursive reference, and the R5 trained recursive
losses around .0425 and .0502 against a .01 target were corrected using a
Those corrected results establish finite-model performance of the corrected
polish small correction cost or a scaled neural recursive-portfolio solution.
d their limitations are preserved rather than replaced by an additive-control

8.2. *Sophisticated temporal selves*

l acting self with present-bias parameter β and ordinary discount δ , the
luation equations differ:

$$c_n(s) \in \arg \max_c \{u(c)\Delta + \beta \delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (23)$$

$$V_n(s) = u(c_n(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n(s), \varepsilon))]. \quad (24)$$

carries the extra β . With two consumption dates, terminal log wealth, zero
e sophisticated first-date share is $1/(1 + 2\beta)$, hence .4166667 at $\beta = .7$.
valuation by geometric- β evaluation gives .4566210, a different behavioral
derivation, executed residual check, and feasible-deviation calculation
plement. Equilibrium verification concerns each acting self’s feasible
selves fixed, not a fictitious commitment planner who can change all future

has two firms, capacity investment, stochastic demand, and multiple verification remain small-scale. Neither experiment establishes generality or a competitive limit. The operator formulation, nonzero present value, and strategic state transitions are retained as substantive extensions; experiments on these objects remain a separate quantitative requirement, not an additive nonconvex benchmark.

9. COMPUTATION WITH STOCHASTIC TRACES

z_k are independent standard Gaussian probes, put $\bar{q}_K = K^{-1} \sum_k z'_k A z_k$.

$$\mathbb{E}(c - \bar{q}_K/2)^2 = (c - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (25)$$

Unpaired one-probe residuals retains a variance penalty that does not independent probe means give an unbiased cross-product estimator dual, but not a nonnegative sample objective or a uniform residual variance accounting, and the retained Monte Carlo regression are given in Appendix E.

With m columns, an exact contraction can use m directional second order contraction uses K probe operations and the loading operations. Differentiation, collocation coverage, optimization iterations, and the budget all enter total cost. Avoiding a dense Hessian is useful, but small dimensional crossover or eliminate the cost of verification. The quadratic tests with off-diagonal drift, rewards, and shocks are useful for their known quadratic value family and faster Riccati comparator performance conclusion.

10. CONCLUSION

Evaluation approximation, an improvement rule, and a verification spectable. In a stopped preference–portfolio economy, this separation retained Hamiltonian audit, one-sided continuous value witnesses, and for adjustment-cost comparisons at finite numerical accuracy. The new state and paired author-code comparison address different questions: method and realized performance on coupled nonconvex controls. The conservative original-payoff certificate or the substantial cost of a

continues to include recursive utility, sophisticated temporal selves, and their distinct evaluation and equilibrium conditions. The remaining steps: construct sufficiently narrow original-model witnesses, or enclose

$$\left[+ \int_t^\sigma e^{-\rho(r-t)} \ell(S_r, a_r) dr \right] = w(t, s) + \mathbb{E} \int_t^\sigma e^{-\rho(r-t)} R_{a_r} w(r, S_r) dr, \quad (26)$$

a compact interior localization or at the economic stopping time. Localization of the stochastic integral. The hypotheses on the witness and uniform integrability of the regularity of the optimal value is invoked.

$\leq \epsilon_U$ for every admissible action. The terminal inequality $g \leq U + b_U$ and

$$U(t, s) + b_U + \epsilon_U \mathbb{E} \int_t^\tau e^{-\rho(r-t)} dr \leq U(t, s) + b_U + C_\rho(T-t)\epsilon_U.$$

over policies; attainment is unnecessary. For $w = L$, use the specified policy $L \geq -\epsilon_L$ and $g \geq L - b_L$, and the same integration identity to obtain the regret bound, and nonnegativity follows from the definition

slabs, apply (26) on each slab and sum. Forward-time jumps contribute to the upper-witness identity. They are nonpositive under Assumption 1, and are preserved. Lower-witness jumps have the opposite sign. This explains why the approximation may not silently omit its interface errors. The actual policy is continuous in time because Itô's formula is applied to the witnesses, not the policy. *Q.E.D.*

A.2. Spatial residual potentials

residual satisfies $R_a U \leq q_U$ and a nonnegative potential obeys

$$z_{U,t} + \mathcal{L}^a z_U - \rho z_U \leq -q_U \quad \text{for every } a.$$

0 by the affine dependence of R_a on the witness. Its boundary inequality is satisfied by z_U . Similarly, if $R_\pi L \geq -q_L$ and $z_{L,t} + \mathcal{L}^\pi z_L - \rho z_L \leq -q_L$, then Theorem 2 applies to $U + z_U$ and $L - z_L$. A potential solving an occupation time inequality would weight residuals by reachable discounted occupation; an approximate construction after the displayed differential inequality is enclosed. This construction converts an empirical occupancy histogram into a theorem.

B. Verification of the explicit original-economy witnesses

$$= -e^{-\rho(1-t)} \text{ and } \rho C_\rho(1-t) = 1 - e^{-\rho(1-t)},$$

$$(\partial_t - \rho)[-aC_\rho(1-t)] = a$$

constant in its time index. The stopped coefficients are bounded and regularity, so the discounted stochastic integrals are integrable. The sign is a statement about the original continuously stopped policy, including minimal boundary regularity conditions of a different PDE theorem fail. In bounds all a in the upper witness through the exact maximizing lower witness on every time-state cell by the convex range of the mapped to a Cartesian box; enlarging the action range can only make the derivative. Interval dependency can likewise enlarge the bound, never change. No assertion of floating-point simulation conformance is needed: constants are defined by the stored constants and exact interpolation weights. A further allowance would be required for another implementation.

A.4. Proof of Proposition 1

part of $R_a v$ is the sum of

$$\frac{c^{1-u}}{1-u} - v_x c, \quad v_u \theta - \frac{k}{2} \theta^2, \quad h_1 p + \frac{h_2}{2} p^2.$$

changes h_1 but introduces no product between control components. It has derivative $c^{-u} - v_x$ and strictly negative second derivative. The first derivative is positive everywhere. Otherwise its unique unconstrained maximum and projection gives the constrained optimum. The adjustment objective is quadratic because $k > 0$. The portfolio objective is strictly concave when $h_2 = 0$, and convex when $h_2 > 0$; the stated vertex or endpoint rule is compact product feasibility and additivity make these separate maximizers *Q.E.D.*

A.5. A learned-geometry regret bound

$\sup_a R_a g(t, s)$ for the original running payoff. The modified residual is bounded with equality for the feasible explicit optimizer. Applying (26) first to the explicit optimizer proves $\tilde{V} = g$. For the learned policy,

$$- \tilde{J}^\pi(t, s) = \mathbb{E}_{t,s}^\pi \int_t^\tau e^{-\rho(r-t)} \{H_g^* - R_\pi g\}(r, S_r) dr.$$

terms cancel. At $k = 2$ the adjustment term completes the square to bound her two action differences are exactly those bounded in Section 6.3. The upper bound times $C_\rho(1 - t)$ gives the recorded continuous regret. Regret errors are separate interval consequences, not inferred by reversing *Q.E.D.*

ent, and Hessian. The function component of any such limit must equal $\bar{V}$, and its derivative must equal $\bar{V}'$. Its derivative components are the derivatives of that same function. If the value sequence converges in that strong norm: otherwise a subsequence would contradict compactness. If the sequence $\pi_{j_m} \rightarrow p$, continuity of evaluation gives $\mathcal{E}p = \bar{V}$. Continuity of evaluation at a cluster point gives $\pi_{j_m+1} \rightarrow \mathcal{I}(\bar{V})$, and continuity of evaluation gives $\mathcal{E}\mathcal{I}(\bar{V}) = \bar{V}$. The evaluation equation of the improved limiting policy therefore reads

$$R_{\mathcal{I}(\bar{V})}\bar{V} = 0.$$

The limiting property of that selector implies $\sup_a R_a \bar{V} = 0$. Boundary comparison of the proof establishes equality of limiting values, not equality of consecutive values. Compactness, selector closure, and strong evaluation continuity are hypotheses, not properties of arbitrary neural architecture or Adam trajectory. *Q.E.D.*

B.2. Proof of Theorem 1

Stochasticity give $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \delta \|v - w\|_\infty$. The maximized operator norm is constant. Since the candidate set includes the enacted action,

$$\|v_n - \max_a \mathcal{T}_n^a v_{n+1}\|_\infty \leq e_n + \eta_n.$$

Approximate equation with optimal backward induction yields

$$\|v_n - V_{h,n}^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

Exact policy evaluation gives the same inequality with only e_j in the sum. To prove (8).

Bound let $d_n = V_{h,n}^* - V_{h,n}^\pi$. Exact evaluation implies $g_n \geq 0$. For every maximum-difference inequality gives

$$d_n \leq g_n + \delta \max_a P_n^a d_{n+1}.$$

By induction and positivity give $d_n \leq B_n$. The reference optimal value is bounded by recursion: only policy evaluation, transition matrices, and feasible gains are bounded. *Q.E.D.*

B.3. Threshold correction and arithmetic scope

If the gain is at most γ , Theorem 1 gives a regret bound at most $\gamma \sum_n \delta^n$. Thus the margin when this uniform condition holds. Correcting a subset of decisions guarantees that condition, since the re-evaluated continuation changes. The

ported in the numerical tables are float64 quantities with independent regression tests. They are not outward-rounded enclosures of every η_n . To make a machine-rigorous finite certificate one must add roundoff errors to e_n and η_n , or evaluate their analogues with validated arithmetic. Interval and learned-geometry certificates do use outward interval arithmetic, but only for this reason.

Proof of the continuous semigroup error decomposition

Inside each step, the exact surviving-continuation map has Lipschitz survival probabilities are at most one and continuation is received at the premium over all admissible within-step controls preserves this bound. and $E_n^\pi = \|J_n^\pi - w_n^\pi\|_\infty$. Then

$$E_n^* \leq \kappa_n^* + \delta E_{n+1}^*, \quad E_n^\pi \leq \kappa_n^\pi + \delta E_{n+1}^\pi.$$

The positive norm-one lift of the finite gap prove (15). If $\mathcal{B}^{(0)}, \dots, \mathcal{B}^{(m)}$ operator chain from the exact semigroup to the lifted numerical operator, bounds its discrepancy on a specified continuation w by $\sum_i \|(\mathcal{B}^{(i)} - w)\|$. must have a verified bound; changing the tested continuation between steps without controlling that change invalidates the telescoping argument. *Q.E.D.*

APPENDIX C: PROOFS FOR ECONOMIC COMPARISONS

$A(\pi) - kB(\pi)$ is affine and nonincreasing in k . The supremum is therefore attained at k' . The budget is bounded by $.02C_\rho(T - t)$ because $|\theta| \leq .2$. If π attains the budget at price k' ,

$$A(k') \geq A(\pi) - k'B(\pi) = V(k) - (k' - k)B(\pi).$$

The gradient of V at k . At a differentiability point every attained optimizer has

C.1. Proof of Theorem 3

Feasibility and feasibility of the rival candidate policy give

$$A(\pi_1) - k_1B(\pi_1) \geq A(\pi_2) - k_1B(\pi_2) - \varepsilon_1,$$

$$A(\pi_2) - k_2B(\pi_2) \geq A(\pi_1) - k_2B(\pi_1) - \varepsilon_2.$$

Adding yields $(k_2 - k_1)\{B(\pi_2) - B(\pi_1)\} \leq \varepsilon_1 + \varepsilon_2$. The exact monotonicity case. Adding the two budget measurement discrepancies yields the inequality. Feasible-set independence of k is essential: otherwise a comparison might not be available.

inequality with the optimizer at k and prices k_+ and k_- . Rearrangement

$$\frac{V(k) - V(k_+)}{k_+ - k} \leq B(\pi_k) \leq \frac{V(k_-) - V(k)}{k - k_-}.$$

erator by its appropriate outer value enclosure gives (17), and intersecting budget range is valid. Inclusion of restricted policies in the flexible set. The interval subtraction rule gives the other endpoints. No optimizer on non selected policy, or pointwise action ordering is required. *Q.E.D.*

APPENDIX D: RECURSIVE AND EQUILIBRIUM GAIN BOUNDS

interval of continuation values is invariant under all action operators, each on it, and each has Lipschitz constant at most $L_n \geq 0$. The maximized Lipschitz. Repeating the two approximate-equation comparisons in the finite

$$\|V_n^* - v_n\|_\infty \leq e_n + \eta_n + L_n \|V_{n+1}^* - v_{n+1}\|_\infty,$$

$$\|V_n^\pi - v_n\|_\infty \leq e_n + L_n \|V_{n+1}^\pi - v_{n+1}\|_\infty.$$

product-weighted error formula stated in the text. These hypotheses must be recursive discretization. In particular, the sign condition $bV > 0$ does not require derivatives of the driver close to zero and does not supply a uniform L_n . players' strategies and regard player i 's unilateral deviations over all future problem with its own payoff and transition law. The preceding certificate $V_i^* - V_i^\pi$. Requiring this bound for every player gives an approximate Nash equilibrium state included in the norm. Checking only a single-date action deviation in one rollout does not replace the all-future-date best-response calculation. Temporal selves the acting self's criterion uses the β -weighted continuation continuation evaluation uses δ . A unilateral deviation certificate must use the δ -criterion, with future selves held fixed. Substituting a geometrically β -discounted equation changes that equilibrium problem, so its small residual would not affect. The preserved supplement supplies the full two-date algebra and the conventions. *Q.E.D.*

APPENDIX E: TRACE ESTIMATION AND LIMITATIONS OF LOCAL TRAINING

matrix A and $z \sim N(0, I)$, diagonalize A orthogonally. Rotational invariance of the coordinates imply $\mathbb{E} z' A z = \text{tr } A$ and $\text{Var}(z' A z) = 2\|A\|_F^2$. The mean

uniform residual enclosure over state space.

Q.E.D.

y point also cannot substitute for a global feasible action maximum.
+ .2a on $[-2, 2]$, the local maximum near $-.973994$ and the global
are both isolated stationary points, with objective gap about .399875.
and isolated actor stationary points do not eliminate this difference.
ould give the global supporting-hyperplane gap, but concavity in an
imply concavity in shared network weights.

counterexample has the same logical role. A small integrated strong
roximation of $|x| - 1$ does not select the viscosity solution $1 - |x|$ of
oundary problem. Missing viscosity inequalities cannot be replaced by
derivations, exact boundary specifications, and regression checks for
duced unchanged in the preservation supplement.

scale stochastic-approximation theorem requires bounded iterates, ap-
e fast evaluation dynamics, martingale-noise control, and step-size
nclusion is an invariant-set statement for the limiting slow dynamics.
usion needs an additional argument ruling out nonoptimal attractors.
ns do not rely on asserting that every observed gradient trajectory has

particular, reducing a joint scalar training objective need not improve and a small error in a value representation need not identify the policy at a

Neural Bellman Operators (NBO) as a separated evaluation-improvement
 NBO: RLP PRESERVATION SUPPLEMENT
 ems. The critic approximates the continuation value of a specified feasible
 proves that policy against a frozen continuation object. A third, independent
 the enacted policy and its feasible deviation gains. We study both the
 formulation and a positive finite-horizon operator formulation. The latter
 transition and the action maximization explicit and provides a reference
 d neural representation can be assessed.

stantive results. First, we give a quantitative connection between policy
 approximation, feasible improvement, and loss of economic value. The
 nd applies to a stopped controlled diffusion and requires uniform, rather
 , residual bounds. Its finite-horizon counterpart is directly computable
 on approximation. Second, we establish a comparative-static result for
 ce formation. When the adjustment cost enters as a price of a discounted
 i increase in that price reduces the optimal budget. This conclusion holds
 d does not require differentiability of the value function or a pointwise
 ference adjustment. Third, we implement the separated method, including
 ion of a continuous HJB benchmark and trained nonlinear actor and critic
 stic preference-portfolio problem. Independent policy evaluation separates
 greement of the full policy vector.

lication preserves the interior preference and wealth dynamics of the
 pletes the model at the boundary by an explicit early-liquidation contract.
 cient has nonzero diffusion, so a bounded drift cannot keep it in a closed
 lity one. Likewise, the original consumption floor is incompatible with a
 process constrained to the lower endpoint. Liquidation is an economically
 ondition from reflection or a capital injection; it is specified as such,
 ent fee and its sensitivity. No transition replenishes wealth and then
 on.

ence has distinct roles. A continuous-time Merton benchmark tests the HJB
 arated gradient graph. A two-state NDU problem tests a trained nonlinear
 against an independent exhaustive stochastic Bellman solver. A re-solved
 evaluates the economic comparative static. Recursive utility, sophisticated
 dynamic capacity game exercise additional operator interfaces. A coupled
 tratic problem checks dynamic cross-state dependence at dimensions 4, 8,
 nments retain the general economic scope without conflating an algebraic
 el equilibrium, and a trained solution of a continuous nonlinear economy.

1.1. Relation to the literature

ological antecedents already combine policy improvement and neural
 Kim, Kim, and Cho (2025) study physics-informed neural policy iteration

to these approaches is the particular combination of boundary-aware economically interpretable preference-adjustment result, and an executable plan, improvement, and policy-certification objects. The finite-operator distinction between errors of neural representation and errors of the bounds below are not presented as a priority claim for residual-based methods; they specify the conditions and constants used by this implementation.

Our approach connects portfolio choice (Merton, 1969), recursive preferences (Epstein and Zwick, 1992), and the motivating endogenous-preference methods (Epstein and Zwick, 1992). Projection methods remain important comparators in computational finance (Epstein and Zwick, 2017). In our two-state experiment, exhaustive search is faster than neural training. That result is reported, not converted into a claim of advantage. A neural representation can nevertheless be evaluated as a proxy for a controlled operator, and an independently measured value loss is a measure of that representation.

Deep BSDE methods (Han, Jentzen, and E, 2018) is about the equations governing whether an analyst can write a closed-form control. Semilinearity is the highest-order derivative. Eliminating a control analytically can reduce dependence. Conversely, an admissible BSDE representation need not be linear. Section 7 states the differentiation costs conditionally, including the accuracy demanded of a trace approximation.

THE ECONOMIC PROBLEM AND ITS BOUNDARY

Let x be in an open bounded domain $D \subset \mathbb{R}^d$, let $A(s)$ be a compact feasible set, and let π be an admissible feedback. Up to $\tau = T \wedge \inf\{r \geq t : S_r \notin D\}$,

$$dS_r = b(r, S_r, \pi_r) dr + \Sigma(r, S_r, \pi_r) dW_r. \quad (1)$$

The existence of the stopped process and integrability of the stated payoff. With discount rate ρ , define

$$\left[-\rho(r-t) \ell(r, S_r, \pi_r) dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right], \quad V^* = \sup_{\pi} J^{\pi}. \quad (2)$$

For the problem, not an output of a projection routine. At T , it equals the terminal payoff. The operator is

$$\nabla v + \frac{1}{2} \text{tr}(\Sigma^a (\Sigma^a)' D^2 v), \quad R_a(v) = v_t + \mathcal{L}^a v + \ell^a - \rho v.$$

If the representation is valid, fixed-policy evaluation has $R_{\pi}(V^{\pi}) = 0$, while optimality has $R_a(V^a) = 0$. Both have the same declared parabolic-boundary data. A reflected BSDE includes regulators and an appropriate derivative boundary condition; the BSDE replaces that condition with a value penalty.

ferences, fixed-policy utility is defined before optimization. On a domain of
 ere the driver is admissible, it solves

$$Y_r^\pi = f(r, S_r, \pi_r, Y_r^\pi) dr - Z_r^\pi dW_r, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

s, comparison, and integrability of this policy-wise equation are prereq-
 -supremum. The Markov state must contain any variable on which the
 rtunities depend. A general continuous martingale is not replaced by a
 ly because its quadratic variation is known; the executed models here use
 r finite Markov specifications. The recursive extension and its sign domain
 ion 6.1.

3. SEPARATED NEURAL BELLMAN OPERATORS

realization, a critic v_ξ is trained for a fixed enacted policy by its evaluation
 ropriate boundary loss. The actor parameters ψ are updated against a

$$L_E(\xi; \pi) = \mathbb{E}_\nu R_\pi(v_\xi)^2 + \lambda_B \mathbb{E}_{\nu_B} |v_\xi - g|^2, \quad (4)$$

$$L_I(\psi; \xi) = -\mathbb{E}_{\nu_I} \mathcal{H}^{\pi_\psi} [\text{stopgrad}(v_\xi, \nabla v_\xi, D^2 v_\xi)]. \quad (5)$$

by the actor parameterization or by explicit constrained maximization. A
 constraint is an approximation and must be accompanied by a feasibility
 rovement no parameter gradient reaches ξ . In a game no player- i gradient
 s of a frozen rival.

local improvement surrogate. It is not generally the lifetime-objective
 itrary sampling measure. Such an identification requires the appropri-
 int weighting and differentiability conditions. Likewise, minimizing a
 and (5) over both parameter blocks solves a different optimization prob-
 ins the composite-loss and viscosity counterexamples and states what the
 ent does establish.

3.1. A positive finite-operator realization

economy, let $\mathcal{S}_h$ be a finite state set and A_h a finite feasible action set.
 are included in r_n^a and surviving transitions in a substochastic kernel P_n^a .

$$T_n^a w = r_n^a + \delta P_n^a w, \quad T_n w = \max_{a \in A_h} T_n^a w. \quad (6)$$

s training loss. At deployment the actor chooses its hard argmax, evaluation targets are detached. This is a finite-horizon neural Bellman that one Adam update performs exact policy iteration. an independently coded NumPy backward induction with exhaustive bilinear interpolation. The neural operator is implemented separator-agreement test precedes training. The trained policy is then nce operator; this evaluation is different from reading the critic’s used for finite-model certificates range over the entire stated finite set.

Exact improvement and quantitative approximation

ct separated-operator limit: *Let the policy space Π be compact in the valuation E be continuous into a precompact subset of $C^{1,2}$, endowed n the function, its time derivative, gradient, and Hessian. Assume a ps Π into itself, is continuous at accumulation points, and satisfies the ion conditions of Assumption 1. If $\pi_{j+1} = I(\pi_j)$, every accumulation * , and the common limiting value solves the HJB boundary problem.*

quality of limiting values, not equality of successive limiting policies. ndix A. Closure under the exact selector is a real hypothesis and is not al class.

ry-aware policy-value certificate: *Under Assumption 1, let v be a test policy. Suppose, uniformly on the stopped domain,*

$$\sup_{\mathcal{A}(s)} \{R_a(v) - R_\pi(v)\} \leq \eta, \quad |v - g| \leq b \quad \text{on the parabolic boundary.}$$

$$e^{-\rho h})/\rho \text{ and } C_0(h) = h,$$

$$V^*(t, s) - J^\pi(t, s) \leq 2b + C_\rho(T - t)(2\epsilon + \eta). \quad (7)$$

I do not claim that an actor stationary point is a maximizer, is required for

not discounted at the terminal date: exit can occur earlier. Theorem 1 nningful errors into a value loss. A sampled mean-square loss does not eses. A verified residual Lipschitz constant and a covering radius can niform bound; such constants must be established, not inferred from a

horizon neural-policy certificate: *For the finite operators (6), let $\hat{v}_N$ st b . For an enacted policy π , define*

$$\|\hat{v}_n - T_n^\pi \hat{v}_{n+1}\|_\infty, \quad \eta_n = \|T_n^\pi \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1}\|_\infty.$$

Global action improvement versus parameter stationarity

$r(a) = -(a^2 - 1)^2 + .2a$ on $[-2, 2]$ has a stable local ascent attractor near al maximum near 1.024120. Their payoff gap is .3998746. Both stationary us an exact fast critic, diminishing two-timescale steps, and isolated actor ot imply a policy-improvement fixed point.

condition concerns the *action* maximization. If H is differentiable and compact feasible set, then

$$H^* - H(a_0) \leq \max_{a \in A} \langle \nabla H(a_0), a - a_0 \rangle. \quad (9)$$

a global supporting-hyperplane gap and handles boundary optima. Point-ot imply concavity in shared neural parameters. Our finite implementation maximizing finite action directly. The separate asymptotic stochastic-ent, with its invariant-set conclusion and additional optimality conditions, E.

ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

te u is the coefficient of relative risk aversion and X is financial wealth. umption c , preference adjustment θ , and the risky portfolio share π . Before

$$+ .05 dZ^u, \quad (10)$$

$$2 + .06\pi)X - c]dt + .2\pi X dZ^X, \quad d\langle Z^u, Z^X \rangle = -.25 dt. \quad (11)$$

$2.8) \times (.5, 2)$ and the control box is $[.05, .8] \times [-.2, .2] \times [-.5, .8]$. The and $\rho = .04$. Utility is cardinally fixed as

$$\theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad G(u, X) = -.02(u - 2)^2 + .1 \log X.$$

and this normalization are held fixed throughout the cost panel. An affine depends on the controlled preference state generally changes the economic

pend on k . No additional wealth is supplied after settlement because

than disguises the boundary economics. Constant preference diffusion
le-rectangle interpretation. At $X = .5$, even the largest wealth drift
box is $-.016$. Keeping the original interior process and the original
quires exit, reflection with an explicit regulator, or a different resource
Economic statements about this contract must not be interpreted as
y reflected self-financing economy.

$$\frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2 + \theta V_u + [(.02 + .06\pi)X - c]V_X - \rho V \\ + \frac{1}{2}(.05)^2 V_{uu} + \frac{1}{2}(.2\pi X)^2 V_{XX} - .25(.05)(.2)\pi X V_{uX} \Big\}, \quad (13)$$

t boundary. All three second-order contributions are retained.

4.2. Policies and economic interpretation

s interior and the indicated derivatives have the required signs,

$$\theta = V_u/k, \quad \pi = -\frac{.06V_X}{.2^2 X V_{XX}} - \frac{.05(-.25)V_{uX}}{.2X V_{XX}}. \quad (14)$$

substitutes for constrained maximization. Consumption and adjust-
n objectives. The portfolio objective is concave only when $V_{XX} \leq 0$;
arisons are essential. Appendix B gives the full selection rule, including
sitive marginal wealth value.

roduces a hedge against preference shocks correlated with returns. Its
er a negative shock correlation nor an assumed positive V_{uX} alone
ilio comparative static. Likewise, dividing V_u by k does not determine
tive changes with k . The following result gives a global economic
shortcut.

of preference adjustment: *Suppose the feasible policy set, stopping
act do not depend on k . Write*

$$-t) \frac{c_r^{1-u_r}}{1-u_r} dr + e^{-\rho(\tau-t)} g(\tau, S_\tau) \Big], \quad B(\pi) = \mathbb{E} \int_t^\tau e^{-\rho(r-t)} \frac{\theta_r^2}{2} dr.$$

$r) - kB(\pi)\}$ is convex and nonincreasing. If $k_1 < k_2$ and optimal
 $\pi_2) \leq B(\pi_1)$. At a differentiability point with an attained optimizer,
exible-preference value weakly exceeds the value under the feasible

stationary Merton economy with CRRA coefficient 2, discount .04, riskless return .06, and volatility .2. A homothetic critic is $v(x) = -\exp(w)/x$ and represents $m = c/x \in [.005, .07]$ and $\pi \in [.5, 1]$. First and second wealth derivatives are computed by automatic differentiation. Critic optimization minimizes the scaled dual. Actor optimization holds these derivatives fixed. Neither loss receives controls or value as a label.

Backpropagation gives $A^\pi = 1/(mD)$ for $J^\pi = -A^\pi/x$, where $D = .04 + .02 + .06$. The optimum has $m^* = .04125$, $\pi^* = .75$. Across seeds 0, 1, and 2 the maximum error covers m within 8.25×10^{-8} and π within 8.29×10^{-10} . The maximum scaled value loss on $[\cdot 5, 2]$ is 4.70×10^{-9} . The scaled HJB residual is at most 10^{-10} . An assertion verifies that actor backpropagation leaves the critic gradient at the homothetic neural benchmark with three trained parameters, not evidence of a functional approximation theorem. Its value is that the continuous generator, critic evaluation, and actor graph can all be checked independently.

2. Stochastic NDU transition and neural architecture

In the portfolio model, four equiprobable branches use independent signs $z_1, z_2 \in \{-1, 1\}$.

$$\Delta + .05\sqrt{\Delta}z_1, \quad (15)$$

$$[c + (.02 + .06\pi)X - c]\Delta + .2\pi X\sqrt{\Delta}[-.25z_1 + \sqrt{1 - .25^2}z_2]. \quad (16)$$

At each time step, the process exits, it settles at its first intersection and terminates; otherwise continue with bilinear interpolation. The running payoff is integrated with the exact solution over the elapsed fractional step while holding the current flow fixed. This defines a numerical Merton economy, not exact Brownian first-passage integration. Interior drift and diffusion are run before boundary corrections.

Actor and critic each have two hidden layers of width 48 with tanh activations. The actor outputs, representing five values for each of the three controls. The critic uses log distances to the four boundary faces; it is represented as $G + (1 - t)N_\xi$ with 48 nodes, with exact prescribed settlement at boundary nodes. The finite difference approximation is this boundary-switched representation to be a globally C^2 function. The numerical certificate applied to it is Theorem 2, not an unverified use of the theorem.

For training, training uses all interior nodes of a 25×31 lattice, 500 actor Adam steps followed by critic L-BFGS refinement. The learning rate is .004. Actor targets are frozen. This experiment is a lattice-trained finite-operator approximation-free sampling experiment. Seeds 0, 1, and 2 are reported without selecting

runs meet the stated finite-model date-zero value-loss target of .05. larger than the enacted policy loss, illustrating why a critic predic-
tion certificate. Boundary discrepancies are at most 4.80×10^{-7} in
. For seed 0 the maximum component differences in (c, θ, π) from
are (.375, .2, .975); the corresponding interior mean differences are
The portfolio policy is therefore not uniformly accurate even when
y seed’s complete component vector is retained in the ledger.
ackward induction takes about .60 seconds in the seed-0 comparison,
training and .65 seconds for independent policy evaluation. There is no
e in this two-state task. The result instead establishes that a genuinely
implement the stated stochastic operator with small independently
policy loss. Whether representation reuse or higher-dimensional nonlinear
omparison requires matched experiments on those objects.
failed: its maximum policy value loss was 4.10, its critic error 4.49,
tation was inaccurate. The failure and its executed source are retained.
te runs reached training output but timed out before completing their
led as incomplete, not successful runs. These records are not included
count.

5.3. Re-solved preference-adjustment costs

elve dates, the 25×31 state lattice, and the 125-action stochastic
ndently re-solved, along with the restricted $\theta = 0$ economy. Table II
(.95). The budget falls by approximately 86 percent between $k = .5$ and

TABLE II
REFERENCE FORMATION UNDER THE LIQUIDATION CONTRACT

	$k = .5$	$k = 2$	$k = 8$
	−1.351522	−1.372708	−1.404420
adjustment budget B	.016520	.011921	.002309
mitting adjustment	.064480	.043293	.011582
ustment θ	.2	.2	.1
ity	.248935	.249938	.249920
with nonzero θ	.896927	.764993	.475637

or $\theta^2/2$ and time discounting. Exit probabilities refer to this numerical liquidation
res average over all decision dates and interior lattice states, not over a population.

ts are deliberately not labeled solution-error certificates. At the same
or (N, n_u, n_X, n_a) are

$(6, 13, 16, 5)$	-1.423383	NBO: R11 PRESERVATION SUPPLEMENT
$(12, 13, 16, 5)$	-1.564151	
$(12, 25, 31, 5)$	-1.372708	
$(24, 25, 31, 5)$	-1.451486	
$(24, 49, 61, 5)$	-1.344458	
$(12, 25, 31, 7)$	-1.359628	

r of values per control. The changes are material. Refining time alone while
interpolation grid fixed need not improve a diffusion approximation: the inter-
o a generator can scale as h^2/Δ . Action refinement and boundary-crossing
atter. We therefore report a verified finite-model neural approximation and
del cost panel, but not a numerically certified continuous-diffusion value
comparison in Theorem 3 does apply to the continuous contract under its
tions.

ment fee from 8 to 6 or 10, with the twelve-date baseline discretization held
l value to -1.363909 or -1.379622 . These are different contracts rather
irical corrections to the same contract. The complete exit and adjustment
ned so that subsequent boundary refinements can be evaluated without
on feasibility as a zero-error result.

RECURSIVE UTILITY AND EQUILIBRIUM EXTENSIONS

6.1. Recursive utility

$\rho = 1 - \gamma$, and $bV > 0$. In the normalization used here, the Epstein–Zin
egator is

$$f(c, V) = \frac{\rho}{a} c^a (bV)^{1-a/b} - \frac{\rho b}{a} V, \quad (17)$$

limit when $a = 0$. The architecture $V = \exp(W)/b$ enforces the sign
A compact positive margin for bV , along with consumption bounds,
egularity; a sign assertion without executing the transform is not a domain

valuation equation is $V_t + \mathcal{L}^{c, \pi} V + f(c, V) = 0$. Formal stationary homoth-
 $(\mu, r, \sigma) = (.04, 5, 1.5, .08, .02, .2)$ are $\pi = .3$ and $m = .0455$. An algebraic
a verified infinite-horizon solution. We retain these anchors and add an
n implicit recursive-utility calculation. The latter has horizon .5, ten dates,
d 80 consumption fractions, and solves each fixed-action implicit value
ve-value branch. Its maximum implicit-equation residual is 2.46×10^{-13} ;
9625; the executed terminal sign transform differs from the terminal target

$\beta = e^{-\rho\Delta}$, a sophisticated acting self takes the continuation policy of

$$s) \in \arg \max_{c \in \mathcal{C}} \{u(c)\Delta + \beta \delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (18)$$

$$s) = u(c_n^*(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))]. \quad (19)$$

s the extra factor β . For two consumption dates followed by terminal and $\delta = 1$, ordinary continuation has coefficient 2 on log wealth at sophisticated initial consumption share is $1/(1 + 2\beta)$. At $\beta = .7$ this is metric- β evaluation gives .4566210. The corrected executed evaluation 10^{-16} and a dense feasible one-shot search finds no positive gain. Strict local first-order condition establish global optimality for this particular finite search alone would provide only a lower bound on exploitability. corrected but is not used to detect an error that vanishes at $\beta = 1$.

6.3. Dynamic capacity competition

ival policy defines both the evaluation operator and an independent problem. The static Cournot benchmark has Nash output 1/3 and joint-dynamic-differentiation test confirms that the rival gradient is absent from

ly dynamic finite game. Two firms have capacities in $\{0, .5, 1\}$, output up to capacity, and investment $i \in \{0, 1\}$ that adds .5 capacity when cost is $z \in \{2.5, 3.5\}$ with transition matrix having diagonal .8 and profit is $q_i(z - q_i - q_j) - .2i$, discount is .96, and the horizon is backward enumeration selects a pure Nash equilibrium in every state; solves multiplicity in three state-date problems. The maximum feasible is computed as zero in this finite game. The full profile and selection asymmetric equilibria. This is not a neural dynamic-game training unit calculation. The latter would require a market-size normalization limiting economy; the operator formulation and its economic questions

STOCHASTIC TRACES AND COUPLED DYNAMICS

Gaussian probes z_k , write $q_k = z_k' A z_k$ and $\bar{q}_K = K^{-1} \sum q_k$. The $\bar{q}_K$ satisfies

$$\mathbb{E}(a - \bar{q}_K/2)^2 = (a - \text{tr } A/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (20)$$

stead leaves a variance penalty independent of K . With $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ and station is $1 + 7.5/K$, not 8.5 for every K . The replication runs 30,000

s, and the probe budget needed for accuracy all enter total cost. There is
t crossover at state dimension 50 or 100.
dynamic check has $d \in \{4, 8, 16\}$, eight dates, $x_{n+1} = Ax_n + a_n + S\varepsilon_{n+1}$,
 $= I + .12111'/d$, $R = .3I$, and $S = .1I + .02111'/d$. A linear neural actor
a frozen quadratic policy evaluator, and compared with an independent
-diagonal transition, payoff, and shock terms are retained. Maximum
at the three dimensions are respectively 6.60×10^{-14} , 1.55×10^{-14} , and
an unconstrained linear-quadratic dynamic regression, not the earlier static
ccati evaluation is faster, and the known quadratic value family makes the
table. No constrained sparse-grid superiority or general high-dimensional
n follows from it.

8. CONCLUSION

evaluation, feasible improvement, and independent economic evaluation as
al objects. The distinction yields explicit value-loss bounds and prevents
n actor stationary point from being mistaken for a solution certificate.
portfolio application, an explicit liquidation contract makes the stochastic
ical transition agree; the re-solved adjustment-cost panel implements
static for the discounted preference-adjustment budget. Actual neural
ation, diffusion moments, temporal-self recursion, trace estimands, and
ions are now executable.
e includes recursive utility and strategic interaction without assigning them
l hypotheses. Further accuracy must be earned through the declared error
OU runs certify the finite stochastic economy, while continuous-diffusion
aterial. All historical models, derivations, figures, and prior reports remain
th a retention map distinguishing corrected equations, supplementary
ys, and executed evidence. This structure permits the next review to assess
he numerical economy rather than reconstruct their intended relationship.

APPENDIX A: PROOFS AND THE SCOPE OF THE VALUE CERTIFICATES

A.1. Proof of Proposition 1

The pointwise selector inequality is $\mathcal{H}^{I(\pi_j)}[V_j] \geq \mathcal{H}^{\pi_j}[V_j]$. Comparison
fication inequality imply $V_{j+1} \geq V_j$. Precompactness of the evaluation
edness. Therefore at each state-time point the monotone sequence has a

of V_j has a further convergent subsequence in the declared strong norm.
limit must be $\bar{V}$, by monotonicity and pointwise uniqueness of the limit.
ble sequence converges uniformly to $\bar{V}$; in fact it converges in the declared
otherwise a subsequence separated from its unique possible strong cluster
t precompactness. Derivatives of any strong cluster point are the derivatives

Hamiltonian maximum evaluated at $E(p) = V$. Its evaluation at this

$$-\bar{V}_t = \mathcal{H}^{I(p)}[\bar{V}] = \sup_{a \in A} \mathcal{H}^a[\bar{V}].$$

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passes to the limit in the declared topology. Comparison identifies a gives optimality. In particular $E(p) = V^*$, even if p and $I(p)$ differ successive limiting *policies* merely from a small difference of their

ns an exact operator on a compact, selector-closed policy class. Neither a finite-sample optimizer automatically meets those assumptions. The is instead assessed by Theorem 2. For recursive drivers the comparison supplied for that driver and value domain; the proof does not obtain d “recursive.”

A.2. Proof of Theorem 1

formula to the test function under any feasible policy a and stop at τ . s the identity

$$_{t,s} \left[e^{-\rho(\tau-t)}(g-v)(\tau, S_\tau) + \int_t^\tau e^{-\rho(r-t)} R_a(v)(r, S_r) dr \right]. \quad (21)$$

a followed by the integrability hypothesis justifies removal of the he enacted policy, $R_\pi(v) \geq -\epsilon$ and the boundary difference is at least $C_\rho(T-t)\epsilon$. For every feasible alternative, $R_a(v) \leq R_\pi(v) + \eta \leq \epsilon + \eta$ nce is at most b . Thus $J^a \leq v + b + C_\rho(T-t)(\epsilon + \eta)$. Taking a ves and subtracting proves (7). Attainment of that supremum is not

d differentiability of the true value or differentiability of the policy with It does need a legitimate test function, a legitimate stopped process, ver any competing policy can travel. A residual measured under the eed not control an alternative policy. This is why an off-policy uniform ved change-of-measure bound, is essential for the displayed worst-case

d domain qualifications. The theorem bounds the stopped payoff as ptimal value lacks a classical extension to every boundary face. The generate in wealth under $\pi = 0$. We do not assert that all boundary ry controlled diffusion, or that the continuous NDU value belongs to mscalar payoff at a boundary starting state is settlement by definition; e may require a generalized viscosity boundary interpretation. The d residual hypotheses would still have to be verified before numerical switched critic is not used to bypass this requirement.

These inequalities are deterministic covering statements, not consequences. No such global constants have been computed for the NDU neural critic, certificate is reported for it.

A.3. Proof of Theorem 2

sums at most one imply that T_n^π and T_n are monotone and δ -Lipschitz in $\|\hat{v}_n - V_n^\pi\|_\infty$. The evaluation defect gives $d_n^\pi \leq e_n + \delta d_{n+1}^\pi$, and hence

$$d_n^\pi \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} e_j.$$

the, the residual with respect to T_n is at most $e_n + \eta_n$ in absolute value, $=(\hat{v}_n - T_n^\pi \hat{v}_{n+1}) - (T_n \hat{v}_{n+1} - T_n^\pi \hat{v}_{n+1})$. The same backward estimate

$$\|\hat{v}_n - V_n^*\|_\infty \leq \delta^{N-n} b + \sum_{j=n}^{N-1} \delta^{j-n} (e_j + \eta_j).$$

proves the result. For exact re-evaluation, both e and b vanish; the code maximum feasible gain and accumulates it with the factor δ . creates no additional continuation coefficient: its payoff, including the ent, is already in r_n^a . The substochastic surviving row sums remain at most are replaced by their immediate settlement operator in both the optimizer The theorem therefore applies to the finite model even though its boundary economically enacted.

verified numerical-economy error. Suppose, for the same policies and dependent argument establishes $|J^\pi - J_h^\pi| \leq \zeta$ uniformly and $|V^* - V_h^*| \leq \zeta_*$. $(V_h^* - J_h^\pi) + \zeta$. This decomposition identifies the missing bridge without particular, a coarse-fine value difference is neither ζ nor ζ_* without a t.

A.4. Proof of Theorem 3

policy, $J_k(\pi) = A(\pi) - kB(\pi)$ is affine and nonincreasing in k , since m of these affine functions is convex; the pointwise supremum of nonin-nonincreasing. At an initial state where optimizers exist, their optimality

$$A(\pi_1) - k_1 B(\pi_1) \geq A(\pi_2) - k_1 B(\pi_2),$$

$$A(\pi_2) - k_2 B(\pi_2) \geq A(\pi_1) - k_2 B(\pi_1).$$

the action set, the utility normalization, or the law of preference shocks. pointwise ordering of θ , a portfolio-share ordering, or identification of data.

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AL ACTION SELECTION AND THE NDU STOCHASTIC OPERATOR

3.1. *Constrained Hamiltonian maximization*

NDU maximand separates across c, θ, π . Let their bounds be (c_-, c_+) , consumption maximizes $c^{1-u}/(1-u) - V_X c$, whose second derivative is $-u c^{-u-1}$. If $u > 0$, its optimum is the projection of $V_X^{-1/u}$ onto $[c_-, c_+]$. If $V_X \leq 0$, c_- is optimal and the optimum is c_+ . Adjustment maximizes $V_u \theta - k\theta^2/2$ and the optimum is V_u/k .

is concave as $a_1\pi + a_2\pi^2/2$, where

$$a_1 = -6XV_X + .2(.05)(-.25)XV_{uX}, \quad a_2 = .2^2X^2V_{XX}.$$

ected value $-a_1/a_2$. If $a_2 > 0$, compare the two endpoints. If $a_2 = 0$, the optimum is determined by the sign of a_1 , with any feasible action optimal when $a_1 = 0$. The portfolio slice need not create an inaccessible local search problem in this algorithm instead enumerates its declared action grid; it never labels an action as a global action optimum.

differentiable concave maximand, concavity gives $H(a) \leq H(a_0) + \nabla H(a_0)(a - a_0)$. Finding a supremum proves (9). If an interior maximizer and an m -move are known, the familiar gradient-square bound $H^* - H(a_0) \leq \frac{1}{2} \nabla^2 H(a_0)(a - a_0)^2$ follows by completing the square. The supporting-hyperplane formulation is also applicable. Neither argument transfers concavity from actions to shared

3.2. *Local consistency and preference reachability*

Let Δ be the correction (16) and let C denote the covariance matrix of the two-dimensional correction, (16) satisfies

$$\Delta = b\Delta, \quad \text{Cov}(\Delta S) = C\Delta, \quad C_{uX} = \sigma_u \sigma_X \pi X \rho_{uX}.$$

As $1, u, X, u^2, X^2, uX$ give the correct constant and linear generators, the generator errors $O(\Delta)$ from the drift outer product. At $(u, X) = (2, 1)$, the continuous generators are

$$(0, .13, -.341, .5225, -.6651, -.553625).$$

Step sizes .1, .05, .025, .0125. The covariance error is below 10^{-12} and the discrepancy declines at the predicted rate. These tests include, rather than just the drift, the generator contribution of u^2 is

stency limit requires a coupled choice such as $(h_u^2 + h_X^2)/\Delta \rightarrow 0$, together
 a and time errors. Near the killed boundary, additional exit-time consis-
 regularity are needed. The four-branch linear-segment rule is not an exact
 sion why the displayed square
 calculation. These qualifications explain why the displayed square
 alternative diagnostics but not a continuum error certificate.

B.3. Boundary cash flows and recorded exits

ement d , the exit fraction is the minimum positive fraction at which $s + \lambda d$
 , capped at one. At a crossing the continuation value is $g(t + \lambda\Delta, s + \lambda d)$,
 count integral $[1 - \exp(-\rho\lambda\Delta)]/\rho$, and there is no continuation policy
 gression starts at $X = .51$, chooses $c = .8$, $\pi = 0$, and verifies that every
 5 with fraction strictly below one and then terminates. The program never
 and permits another consumption date under the pre-exit contract.
 nguishes an exit event from a post-correction state. The adjustment budget
 until settlement. For a surviving branch interpolated to a boundary node, the
 ttlement is part of the finite kernel. This interpolation effect is a numerical
 l reflection. The settlement fee is a model primitive; comparing fees 6, 8,
 beled a comparison of contracts.

RECURSIVE AGGREGATION, TEMPORAL SELVES, AND STRATEGIC INTERACTION

C.1. The recursive aggregator and its domain

$= 1 - \gamma$, and $bV > 0$, expand the aggregator as in (17). Its consumption

$$f_c(c, V) = \rho c^{a-1} (bV)^{1-a/b} > 0.$$

$$f_V(c, V) = \frac{\rho}{a} (b - a) c^a (bV)^{-a/b} - \frac{\rho b}{a}.$$

with consumption bounded away from zero and bV bounded away from
 s are bounded. This is a local Lipschitz conclusion on the specified range,
 existence theorem for all values. The transform $V = e^W/b$ enforces $bV > 0$,
 erify integrability, terminal compatibility, comparison, or transversality.
 sive evaluation solves (3); in the Markov setting its equation is $v_t + \mathcal{L}^\pi v +$
 f is Lipschitz in v with constant L , standard comparison of a perturbed
 fied evaluation gives a finite-horizon error envelope of the form

$$e^{L(T-t)} b + \int_t^T e^{L(r-t)} \epsilon dr$$

ulation fixes $\gamma = 5$, $\psi = 1.5$, and $\rho = .04$ on a deterministic saving
length .05. At each wealth node $x \in [.5, 2]$, consumption is $c = mx$
.6], and the next wealth is the stated saving transition in the code. For
valuation equation is

$$y = I_h V_{n+1}(x') + \Delta f(c, y), \quad by > 0.$$

s bracketed and solved numerically before maximizing over the action
is x^b/b . Outside the wealth lattice, continuation uses the specified
as numerical boundary data at that date. It is not a solution on an
in. At the central node the recorded value is $-.29658656$ and the
action is the lower bound .01. This active bound is reported rather than
stationary ratio.

mplicit-equation residual tests are actual evaluated quantities. Neural
use this experiment solves scalar implicit reference equations rather
neural policy. The general recursive portfolio model remains part of the
on and analytical derivations; its stationary ratios .3 and .0455 remain
eir full infinite-horizon verification is supplied.

3. Ordinary continuation and acting-self utility

on policy, ordinary log-wealth utility in the two-consumption-date
 $= A_n \log w + B_n$. Starting from $A_N = 1, B_N = 0$, the sophisticated

$$\alpha_n w, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log \alpha_n + \delta[A_{n+1} \log(1 - \alpha_n) + B_{n+1}].$$

e improvement uses $\beta\delta$. With $\delta = 1$ and two decisions, $A_1 = 2$ and
uting β into the evaluation coefficient instead produces the different
 2). At $\beta = 1$ the two formulas coincide, so this special case cannot

evaluated against the same frozen continuation policy whose equilib-
placing its continuation with a different grid's optimized value changes
ral finite sophisticated-self computation, all feasible actions can be
using the correctly evaluated future policy. For continuous actions a
feasible deviation lower bound unless concavity, a global oracle, or an
ulation closes the search gap.

4. Dynamic games and equilibrium selection

player i 's payoff with its own objective and transition law. A Nash

$$\sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}) - J_i(\pi_i, \pi_{-i}),$$

tion values from the already selected future equilibria. A pure equilibrium is a best response for each player against the other current action and the profile. Every enumerated state has at least one pure equilibrium in this strategic form game, and the full profile includes a best response for each player. In the finite-horizon game, this backward best-response construction establishes subgame perfect equilibrium in the state space; no uniqueness follows from it. In the infinite-horizon game, on the other hand, the demand and capacity scaling, investment, and equilibrium concept must be fixed before taking a limit. Symmetric games do not admit a proof of equilibrium nor an equilibrium-selection theorem. Those games are best thought of as model extensions rather than filled with the historical illustrative examples. A trained high-dimensional game or a solved competitive limit is not an exact finite game.

DIX D: TRACE ESTIMANDS AND COMPUTATIONAL COST

Let z be a Gaussian z and symmetric A , diagonalization and the independent squared-residuals estimator gives $\mathbb{E}z'Az = \text{tr } A$ and $\text{Var}(z'Az) = 2\|A\|_F^2$. Independence across K probes gives $\text{Var}(z'Az)/K$ for their mean. Expanding the squared residual then proves (20). The diagonal contribution to variance vanishes, so the formula must be used uncritically.

The cross-product correction

$$U_K = a^2 - a\bar{q}_K + \frac{1}{4K(K-1)} \sum_{i \neq j} q_i q_j. \quad (22)$$

Exact probes gives $\mathbb{E}U_K = (a - \text{tr } A/2)^2$. The statistic is unbiased but can be batched and may have a different variance and optimization behavior. The additional estimand, not a blanket replacement justified by unbiasedness, reports its empirical mean and standard error separately. When A depends on parameters, the uncorrected variance term itself varies with those parameters; a squared residual can consequently change the expected minimizer.

The actual cost of a directional second derivative and its parameter gradient for a matrix Σ has m columns, exact contraction can be organized as m directional contractions with loading and batching costs. A K -probe contraction costs roughly $K C_{\text{jet}}$ rather than an unconditional $O(d)$ solver. Total work additionally multiplies by the number of training iterations, and any restarts. Peak memory, warm-up, and accelerator costs can be measured for a hardware comparison. The current CPU timers do not account for the overhead, so that metric remains null.

Quadratic dynamic regression. For the finite-horizon quadratic-cost economy, the cost is a quadratic value $x'Px + c$. Given a frozen next evaluator, the actor's cost is the quadratic stage-plus-continuation cost. Gaussian shocks contribute

ing law nor compare against adaptive sparse grids. The original static n is retained in the R3 archive as a small linear-algebra check, separate ment.

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OXIMATION, COUNTEREXAMPLES, AND OPTIMIZATION LIMITS

1. *A composite objective can bias evaluation*

iminal evaluation problem whose true value is zero and whose critic is ation residual is v . A joint surrogate $v^2 - \lambda v$ has minimizer $v = \lambda/2$, for every $\lambda \neq 0$. The example isolates the logical issue: optimizing ion over the same value parameter need not preserve the evaluation e residual weight changes the bias but does not prove exact evaluation. o gradient into that value parameter, so it does not create this particular

ng a residual is not a stability theorem. The derivative of $R^2/2$ is onditioning, sampling, and step sizes govern a finite optimizer. Value driver remains inside this derivative. A zero target does not make the the value level.

ull squared residual does not select a viscosity solution

ary values zero, consider $|v'| = 1$. The nonnegative distance function ution. The function $|x| - 1$ has the same boundary values and satisfies re, but it fails the viscosity supersolution test at zero: the constant test om below and has $|\phi'| - 1 = -1 < 0$.

s $v_\varepsilon(x) = \sqrt{x^2 + \varepsilon^2} - \sqrt{1 + \varepsilon^2}$ have exactly the boundary data and -1 . Their squared residual $\int_{-1}^1 (|v'_\varepsilon| - 1)^2 dx$ tends to zero by domi-ugh the limit is not the viscosity solution. Thus boundary agreement, L^2 residual do not establish viscosity selection. Monotone, stable, appropriate comparison can provide a different route (Barles and ivity of an interior interpolation alone does not verify all of those undary.

l neural networks can approximate a sufficiently smooth function and subsets under the relevant approximation hypotheses. This does not k, ensure global boundary regularity, close a finite policy class under a prove that stochastic optimization finds the approximant. The retained motivation is used only in that conditional sense.

wo-timescale statements and suboptimal attractors

ic approximation, a conventional ODE analysis requires bounded y and stability, martingale-difference noise with bounded conditional

IMPLEMENTATION, EVIDENCE CLASSES, AND REPRODUCIBILITY

F.1. Executed parameterizations and data flow

erton critic has one learned log-amplitude and its actor has two learned
automatic differentiation through the value with respect to wealth, including
d actor optimizers are separate. Its independently evaluated policy has
decay D . Analytical controls are used only after training to measure error.
ferent: time-specific nonlinear networks represent a finite Bellman operator.
ormalized states to 125 logits through two 48-unit tanh layers. Its critic
cluding log distances to boundary faces, through two 48-unit tanh layers to
terminal utility. Next-date critics are frozen, the actor target uses only their
ons, and the current critic is fitted to the enacted actor's policy-evaluation
nt NumPy optimizer and evaluator are not imported as training labels.
are not out-of-sample certificates because all interior finite states are
e independent evaluator changes the computational path, not the economic
ains separate NumPy and PyTorch implementations of the transition and
equality is tested on deterministic data. Cross-grid reference calculations
gnostics; they have not been relabeled a held-out neural accuracy result.

F.2. Interpreting metrics

st the finite optimal reference, a critic error against the evaluated policy, a
ference, and a unilateral gain are different estimands. Policy component
d for consumption, adjustment, and portfolio share. Boundary-node actor
ed after settlement; full-grid maxima and interior summaries must be
interpreting them. A finite exhaustive best-response maximum is a finite-
feasible search over only some continuous deviations is merely a lower
ty.
d tolerance status are separate fields. A program that completes can miss
audit has no final solution certificate. A null numerical error has an explicit
measured continuous-diffusion error or a recursive reference that did not
A numerical zero is used only for a quantity actually calculated, such as
ion maximum or a tested monotonicity violation.
neural source and its weights, metrics, and raw arrays are retained in the
The successful three-seed sources and weights are separate. The attempted
marked incomplete. Wall-clock times are metadata and are excluded from

alone supplement, embedded arrays, earlier reports, and R3 numerical
d. Core model formulations, constrained policy derivations, recursive
lf and game extensions, trace analysis, and theoretical qualifications
t and the102 appendices. Historical unexecuted plots are retained as
ented again as learned current outcomes. The retention map identifies
the corresponding corrected discussion. This preserves the scientific
incompatible equations as simultaneous current specifications.

TABLE I
RETAINED R5 NDU PANEL

NBO: R11 PRESERVATION SUPPLEMENT

uration	<i>n</i>	Raw	Safeguarded	Max. bound	Seconds
e	10	10	10	0.0496	18.19
horizon	3	0	3	0.0064	34.64
2	2	1	2	0.0471	16.86
4	2	2	2	0.0461	20.03
	2	0	2	0.0050	18.52
	2	2	2	0.0339	17.95
eps 300	2	0	2	0.0065	13.66
eps 400	2	1	2	0.0440	15.70

number of runs, not the number of time steps. Pass counts use the historical finite-candidate historical recorded training measure; this is not the new full safeguard re-audit timing.

TABLE II
RETAINED R5 QUADRATIC-CONTROL COMPARISON

d	Mean loss	Sampling upper	Train (s)	Audit (s)	MiB
	0.0010	0.0018	14.93	3.45	377.5
-PI	-0.0001	0.0007	12.59	6.61	376.9
	0.0037	0.0055	14.29	3.84	378.6
-PI	0.0013	0.0032	11.67	5.69	377.7
	0.0102	0.0145	16.29	3.72	381.3
-PI	0.0075	0.0117	13.34	5.64	382.1

ole-Hopf/QMC benchmark with an analytic greedy control and an in-house PINN-PI
ative estimated loss reflects reference or simulation uncertainty, not performance exceeding
has higher mean loss in each row pair. This result is retained despite the favorable R8
nonconvex-control comparison.

0.0436	0.0040	0.5502	0.0106
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al solver has an analytically manufactured verification pair and no trained actor. It is earned NDU-geometry test, which also explicitly modifies the running payoff.

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TABLE IV

RETAINED R5 COUPLED NDU REFINEMENTS					
grid	n_a	h^2/Δ	V_0	B_0	Exit
$\times 16$	3	0.2133	-1.624355	0.012266	0.1543
$\times 31$	3	0.1067	-1.510132	0.012248	0.1558
$\times 61$	5	0.0533	-1.391646	0.011969	0.1024
$\times 121$	5	0.0267	-1.353683	0.012234	0.0987

change in this historical panel. The table supplies numerical differences, not validated continuous error bounds or a separated convergence order.

TABLE V

RETAINED R5 RECURSIVE-UTILITY POLICIES				
raw log loss	Final bound	Final log loss	m_0	p_0
0.0425	0.0000	0.0000	2.506	0.955
0.0502	0.0000	0.0000	2.442	1.093

ass the historical .01 log-value target. The final candidate-set bounds use repeated safeguards; the raw failures are not relabeled as neural successes.

TABLE VI

RETAINED R5 NEURAL GAME			
Player 1, $t = 0$	Player 2, $t = 0$	All subgames	Seconds
0.00245	0.00673	0.00673	13.87
0.00312	0.00305	0.00368	14.04

ree-date game with exact continuous best-response auditing for the declared finite ator, not a continuous-state equilibrium scalability result.

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